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Acronyms

Acronym	Meaning
ABC	Accelerated Bridge Construction
ARTBA	American Road Transportation and Builders Association
BCA	Benefit-Cost Analysis
BCA-U/Regret	Benefit-Cost Analysis under Uncertainty/Regret Analysis
BPR	Bureau of Public Roads
CMIP	Coupled Model Intercomparison Project
DOT	Department of Transportation
EMAT	Exploratory Modeling and Analysis Tool
FEMA	Federal Emergency Management Agency
FHWA	Federal Highway Administration
GIS	Geographic Information System
GMNS	General Modeling Network Specification
GTFS	General Transit Feed Specification
GUI	Graphical User Interface
H-GAC	Houston-Galveston Area Commission
HERS	Highway Economic Requirements System
HRPDC	Hampton Roads Planning District Commission
HRTPO	Hampton Roads Transportation Planning Organization
IRI	International Roughness Index
LHS	Latin Hypercube Sampling
MPO	Metropolitan Planning Organization
OMX	OpenMatrix
OST	Office of the Secretary of Transportation
PHT	Person Hours Traveled
PMT	Person Miles Traveled
RDR	Resilience and Disaster Recovery
RDRM	Resilience and Disaster Recovery Metamodel
ROI	Return on Investment
STRAHNET	Strategic Highway Network
TAZ	Traffic Analysis Zones
TDM	Travel Demand Model(ing)
TMIP	Travel Model Improvement Program
USDOT	United States Department of Transportation
VAST	Vulnerability Assessment Scoring Tool
VDOT	Virginia Department of Transportation
VHT	Vehicle Hours Traveled
VMT	Vehicle Miles Traveled
XLRM	EXternal factors, policy Levers, Relationships, and Metrics

Executive Summary

The **Resilience and Disaster Recovery (RDR) Tool Suite** was developed to help transportation agencies explore a large scenario space and evaluate the performance of resilience investments in the context of long-range transportation planning. The tool suite utilizes established Robust Decision-Making concepts^{1,2} to build on current Travel Demand Model (TDM) analyses and address deeply uncertain future scenarios. Robust Decision-Making has been used under a similar modeling context by the Federal Highway Administration (FHWA) Travel Model Improvement Program – Exploratory Modeling and Analysis Tool (TMIP-EMAT)³, a scenario-based decision-making tool that can be integrated with existing travel demand forecasting models. The RDR Tool Suite enables transportation agencies to assess transportation resilience return on investment (ROI) for specific transportation assets over a range of potential future conditions and hazard scenarios, which can then be used as a consideration in existing project prioritization processes.

Motivation

Transportation planning decision-makers need to make decisions today about investing in the transportation systems of the future, decisions whose consequences will last many decades. Investments may take several forms, such as adding capacity via a new highway lane or transit line, making improvements aimed at safety, or making resilience investments (e.g., designing a road or bridge to withstand an earthquake or flood).

The classic paradigm for transportation planning is to first, **forecast** what will happen in the future (e.g., trips in a region will increase 20%), and then **act** on that forecast (e.g., add transportation capacity). This paradigm breaks down when the future is highly uncertain, such as trying to predict storms, earthquakes, or other hazards. Under these conditions, the prediction of a single future is unlikely to be correct, and the resulting decisions may be grossly sub-optimal.

An alternative approach is to focus on performance across a **range** of potential futures rather than selecting specific forecasts. With robust decision-making, the objective is not to predict the future, but rather, to make decisions today that produce good outcomes under a wide range of plausible futures. This alternative approach is especially appropriate for prioritizing which projects will be included in long-range investment plans, as long-range investment planning tends to focus on which assets will be deployed or improved to provide the best return.

The objective of the RDR Tool Suite is to help state Departments of Transportation (DOTs), Metropolitan/Regional Planning Organizations (MPOs) and others make informed infrastructure investment decisions by evaluating the performance of potential resilience investments across the set of uncertain future events of interest. It supports long-range investment analyses where agencies need to decide which assets to improve using general information about the options and future conditions. The RDR Tool Suite can be used whether agencies already have proposed projects or are simply exploring

¹ RAND. Robust Decision Making. *RAND*. [Online] [Cited: July 20, 2022.]

<https://www.rand.org/pubs/tools/TL320/tool/robust-decision-making.html>

² Lempert, R. (2019). Robust Decision Making (RDM). *Decision Making under Deep Uncertainty: From Theory to Practice*. V. A. W. J. Marchau, W. E. Walker, P. J. T. M. Bloemen and S. W. E. Popper, Springer: 329.

³ TMIP-EMAT Development Team. TMIP-EMAT. [Online] <https://tmip-emat.github.io/>.

what potential assets they could improve. The outputs of the RDR Tool Suite are focused on total and net benefits of the project in terms of investment cost, repair cost, and network performance.

Approach

Transportation agencies prioritize investments in transportation infrastructure based on their ROI with regard to reducing congestion (vehicle miles and hours traveled) and other locally important factors. Many state DOTs and MPOs currently use Travel Demand Models (TDMs) to forecast traffic flows on the transportation system and evaluate changes in performance resulting from future conditions and transportation infrastructure projects. However, there is currently no standard tool or approach used by state DOTs and/or MPOs to analyze resilience investment ROI. Additionally, the frequency of high-cost hazard events is increasing,^{4, 5} making resilience a critical element of long-range transportation planning.⁶ MPOs and state DOTs need tools and resources to evaluate resilience ROI across a range of future scenarios and hazard conditions as part of their project prioritization processes.

The RDR Tool Suite provides modeling tools to estimate performance of resilience investments across a range of uncertain hazard-related disruptions, recovery patterns, and ROI analysis periods, leveraging existing TDM analysis approaches and performance metrics. U.S. DOT has developed the RDR Tool Suite to be usable in any geographic setting and to enable state DOTs and MPOs to incorporate the costs and benefits of resilience into the project prioritization process. The RDR Tool Suite is intended to be location and hazard agnostic, so that any agency can utilize it for any hazard that results in geospatially-predictable impacts on network capacity.

A resilience-focused ROI analysis must take into account:

- (1) the range of potential hazard events;
- (2) the range of hazard impacts on transportation assets and operations, including travel disruption during hazard events, asset damage, and the travel disruption due to subsequent repairs;
- (3) the costs associated with travel disruptions, measured by changes in system performance and asset damage repair and cleanup costs due to hazard events; and
- (4) the investment required to mitigate the impact of the range of potential hazard events.

Measures of system performance include changes in

- Person hours traveled (PHT), monetized using value-of-time for the person traveling;
- Person miles traveled (PMT);
- Vehicle hours traveled (VHT);
- Vehicle miles traveled (VMT), monetized using per-mile vehicle operating cost;
- Trips (a disruption may prevent some trips from being made).

⁴ National Oceanic and Atmospheric Administration. (2020) "National Centers for Environmental Information (NCEI) U.S. Billion-Dollar Weather and Climate Disasters." DOI: 10.25921/stkw-7w73. Accessed on November 13, 2020 from <https://www.ncdc.noaa.gov/billions/>

⁵ FHWA. (June 21, 2017). "Texas Resilience and Planning Workshop: Summary Report." FHWA-HEP-17-095. Accessed on November 13, 2020 from https://www.fhwa.dot.gov/environment/sustainability/resilience/workshops_and_peer_exchanges/texas_06_2017/index.cfm

⁶ For the purposes of this project, resilience refers to the ability of an asset to tolerate or recover from a given hazard. A resilience-related investment intends to mitigate the impacts of a hazard on transportation assets by reducing damage and the resulting disruption of travel.

The RDR Tool Suite allows the user to determine priority assets for resilience investment by assessing the disruption and repair costs associated with different levels of hazard events. The RDR Tool Suite also calculates the benefits of deploying a resilience investment (including up front/deployment investment versus performance and repair costs) against what would have occurred under the same hazard events if the resilience investment had not been deployed.

This technical documentation describes the structure and functions of the components of the RDR Tool Suite and their application for a Resilience ROI Analysis. The RDR Tool Suite (see Figure 0-1) includes the RDR Metamodel (RDRM), the RDR ROI Analysis Tool, the RDR Exposure Analysis Tool, and the RDR Benefits Analysis Tool.

The RDRM takes the performance metrics results from a TDM and performs complementary disruption analyses using an open-source routing model called Aequilibræ.⁷ The RDRM then performs regressions and iterative scenario expansion to estimate the change in VHT and VMT associated with a range of hazard conditions and investment scenarios. The RDRM models travel behavior response to disruptions in link capacity and availability by assigning a value of trip demand elasticity,⁸ which determines how many trips will be made, their mileage, and the time of those trips in an equilibrium state. The RDRM calculates the impact of a resilience investment (represented as mitigation of the hazard-related disruption at that location) as changes in system performance when the investment is deployed versus a baseline disruption scenario in which the project alternatives are not deployed, including estimates of the impact of hazard duration and repair recovery times and costs.

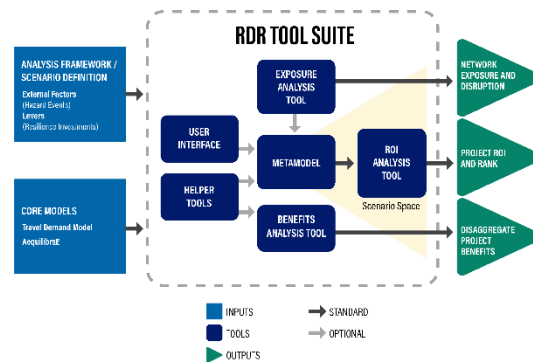


Figure 0-1: Structure of the RDR Tool Suite. The scenario space increases as more potential hazard severities, durations, recovery periods, and resilience investments are assessed.

The RDR ROI Analysis Tool uses the RDRM disruption and network performance outputs to estimate economic net benefits of avoiding disruption over the economic analysis period for a range of resilience investments and recovery patterns. It then collates the performance of different resilience investments into a visualization dashboard and ranks the projects based on their performance across the full range of uncertainties analyzed.

Section 1 of this document presents an overview of the approach and the analytical framework. Section 2 lists the software components that comprise the tool suite. The following sections 3-9 provide detailed documentation of each component of the RDR Tool Suite, and the processes included in each, as follows:

- Scenario Definition: Taking in information on hazard events, travel demand, and resilience investments as defined by the analyst (Section 3).

⁷ Aequilibræ. [Online] [Cited: 07 26, 2022.] <http://aequilibrae.com/python/latest/>. RDR 2026.1 was developed using Aequilibræ 1.6.2.

⁸ See Section 6.2.1.1

- RDR Exposure Analysis Tool: Overlay of hazard severity grids (e.g., inundation depth grids) and network links, to create network exposure severity scenarios (Section 4).
- RDR Disruption Module: Calculation of disruption based on exposure extent and default or user-defined relationship between exposure severity and roadway link capacity (RDR Exposure Analysis Tool for visualization, RDRM for full ROI analysis; Section 5.1).
- RDRM Parameterization: Identification of core model sample runs using Latin Hypercube Sampling, initial scenario runs in the core models, and use of a fast user equilibrium routing model, the open-source AequilibraE software package, to supplement runs from the MPO's more time-intensive TDM (Section 5.2).
- RDRM Regression: Regression on these results to generate performance measure estimates for the entire scenario space (Section 5.3). Performance measures include trips completed, miles traveled, and hours of travel.
- RDR ROI Analysis Tool: Assignment of economic value, economic performance analysis (benefit, cost, and regret), project prioritization (Section 6).
- Reporting and Data Visualization: Enable results exploration and comparison of resilience investments (Section 7).
- RDR Benefits Analysis Tool: Disaggregation of travel performance changes due to a resilience investment during a particular hazard by traffic analysis zone (TAZ) categories (Section 8).
- RDR Graphical User Interface: Provide guided walkthrough of setting up a resilience return on investment scenario in RDR (Section 9).

Section 10 acknowledges our pilot partner organizations, and Section 11 provides a list of known limitations and caveats for the current RDR Tool Suite.

A companion document, the RDR User Guide, enables a user to install the tool suite and execute analyses. The RDR Quick Start Guide provides step-by-step instructions for installing RDR and running a first scenario on a small, hypothetical transportation network. The RDR Reference Scenario Library provides example analyses in the RDR Tool Suite on a series of hypothetical and real-world transportation networks to demonstrate functionality and analysis options. The RDR Run Checklist provides a short reference on the required input file and parameter changes needed to execute a custom analysis. The RDR Tool Suite and documentation are available on GitHub at <https://volpeusdot.github.io/RDR-Public>, along with test data for executing the Quick Start and Reference Scenario examples.

Agencies vary in their level of data availability, analytical tools and resources, and knowledge of potential hazards and disruptions in their region. The RDR Tool Suite is set up to allow for different levels of analysis depending on the information the analyst can bring to the model. If an analyst does not have access to a travel demand model but can estimate trip tables for AequilibraE, all parameterization of the RDRM can be performed using the open source AequilibraE. And if an agency does not have specific project plans for investing in resilience and therefore does not have an estimate of the total cost of an investment, the RDR Tool Suite can still be used to estimate a breakeven point (total benefits) for potential project cost. Thus, the RDR Tool Suite is intended to be used by agencies with a range of information and capabilities. The RDR Tool Suite can also use data from publicly available sources, if converted to the correct format compatible with the tool suite. The analyst can use these sources to complement their specific data following guidance in the RDR documentation.

It should be noted that the RDR Tool Suite is not a “one-stop shop” for all resilience planning needs. It does not provide or incorporate engineering evaluation considerations, except to the extent that these

define the capacity and availability of links in the transportation network and the costs associated with resilience investment actions. It also does not help agencies determine what kind(s) of project(s) would be appropriate for a given asset based on predicted vulnerability to exposure. The outputs are based on investment cost, repair cost, and network performance; it does not calculate economic impact analysis at the broader societal level (e.g., impact on employment, transfers between parties, etc.).

In summary, the RDR Tool Suite provides a transportation agency with a set of tools for evaluating the ROI provided by a set of resilience investments across a range of transportation assets and uncertain future hazard conditions, and for ranking those projects based on performance. The RDR ROI Analysis Tool project rankings and thorough visualizations can be used as a factor in overall transportation infrastructure project prioritization for long-range transportation planning in combination with other prioritization factors such as congestion reduction, safety, engineering considerations, and budgets.

1 Introduction

1.1 Background and Overview

Transportation agencies, including state Departments of Transportation (DOTs) and Metropolitan Planning Organizations (MPOs), prioritize investments in transportation infrastructure based on their return on investment (ROI) with regard to reducing congestion and improving flow (e.g., reducing vehicle hours or miles traveled) as well as other locally important factors. Objective analytical processes such as Benefit-Cost Analysis (BCA) can assist agencies in justifying project investments both for their constituents and to obtain federal funding for their projects. Many state DOTs and MPOs currently use Travel Demand Models (TDMs), which forecast traffic flows on the transportation system, to evaluate changes in performance resulting from future conditions and transportation infrastructure projects.

TDMs can provide the necessary inputs to execute a BCA for transportation investments. However, standard TDMs do not incorporate the potential for hazard events (such as storms, earthquakes, etc.) to disrupt and damage transportation assets. Even where hazards could be modeled mechanically by turning off individual links, standard TDMs lack the temporal resolution to model hazard events that occur over multiple days, weeks, or months. Nor do TDMs have a method for evaluating the benefit of mitigating those impacts by investing in resilience to specific hazards. Thus, there is currently no standard tool or approach used by state DOTs and MPOs to analyze how investments in resilience can contribute to ROI, particularly in the face of uncertainty of both the frequency and magnitude of future hazard events. Yet external hazards such as storms and flooding can lead to massive transportation disruptions and billions of dollars in repair costs. Additionally, the frequency of high-cost hazard events is increasing,^{9,10} making resilience a critical element of long-range transportation planning.¹¹ There is a need for tools and resources to enable MPOs and state DOTs to evaluate the ROI of resilience investments across a range of future scenarios and hazard conditions as part of their project prioritization processes.

To fill this gap, the Federal Highway Administration (FHWA) and the Office of the Secretary of Transportation (OST) of the U.S. Department of Transportation (USDOT) worked with the Volpe National Transportation Systems Center to develop the Resilience and Disaster Recovery (RDR) Tool Suite, which includes the RDR Exposure Analysis Tool, the RDR Metamodel (RDRM), the RDR ROI Analysis Tool, and the RDR Benefits Analysis Tool. The RDR Tool Suite extends standard TDM analyses to address resilience investments in roadway and transit infrastructure (roads, bridges, and on-road or rail transit) using established Robust Decision Making concepts^{12,13} developed to address future scenarios in an environment of deep uncertainty. The RDR Exposure Analysis Tool provides a method to link exposure

⁹ National Oceanic and Atmospheric Administration (NOAA). (2020) “National Centers for Environmental Information (NCEI) U.S. Billion-Dollar Weather and Climate Disasters.” DOI: 10.25921/stkw-7w73. Accessed on November 13, 2020 from <https://www.ncdc.noaa.gov/billions/>

¹⁰ FHWA. (June 21, 2017). “Texas Resilience and Planning Workshop: Summary Report.” FHWA-HEP-17-095. Accessed on November 13, 2020 from

https://www.fhwa.dot.gov/environment/sustainability/resilience/workshops_and_peer_exchanges/texas_06_2017/index.cfm

¹¹ For the purposes of this project, resilience refers to the ability of an asset to tolerate or recover from a given hazard. A resilience-related investment intends to mitigate the impacts of a hazard on transportation assets by reducing damage and the resulting disruption of travel.

¹² RAND. Robust Decision Making. *RAND*. [Online] [Cited: July 20, 2022.] <https://www.rand.org/pubs/tools/TL320/tool/robust-decision-making.html>.

¹³ “RDM” is typically used as an acronym for Robust Decision Making in the literature, but this document does not use it to avoid confusion with the Resilience and Disaster Recovery (RDR) and similar acronyms.

data to a transportation network and test various disruption scenarios to generate impacts for input to the RDRM. The RDRM uses an initial sampling of scenarios produced in the TDM along with other travel demand model tools and the impacts associated with a specific subset of hazards and mitigation options. It computes travel impacts for all of the scenarios and resilience investment alternatives and compares the total outcome against the hazard event baseline (i.e., without resilience investment). The RDR ROI Analysis Tool analyzes these scenario outcomes using a benefit cost analysis (BCA), benefit cost analysis under uncertainty/regret analysis (BCA-U/Regret), or breakeven approach to inform agencies' exploration of a wide range of future conditions and scenarios. This approach enables transportation agencies to assess transportation asset-specific resilience ROI over a range of potential future conditions and hazard scenarios, which can then be used as a consideration in existing project prioritization processes. The RDR Benefits Analysis Tool focuses on a single resilience investment and a single hazard, separating out how travel impacts are distributed across different users of the transportation network.

Resilience investment planning is particularly challenging because hazard events are rare and unique. The impacts of a hazard on travel inherently involve uncertainties regarding the probability, severity, timing, location, and duration of a hazard. That challenge is compounded by the fact that different resilience investments may perform differently under the wide variety of potential hazard events. In addition to uncertainty about hazard events and the suitability of resilience investments, there is also uncertainty about how land-use patterns, demographics, trip demand, and traveler behavior will impact system performance.

The RDR Tool Suite allows the user to determine priority assets for resilience investment by calculating the difference between impacts of exposure of the transportation network at different hazard levels without and with an additional resilience investment and aggregates this information across the range of potential hazard scenarios. This analysis approach complements existing federal and other resources.¹⁴ The RDR Tool Suite is intended to be location and hazard agnostic, so that any state DOT or MPO can utilize the methods to assess how a range of physical hazards may impact the transportation network and how resilience projects can provide economic benefits.

This technical documentation describes the components of the RDR Tool Suite (see Figure 1-1), their purpose, structure, and functions.

- The RDR Exposure Analysis Tool helps the analyst overlay agency-supplied hazard severity information onto their transportation network.
- The RDR Metamodel rapidly estimates network performance under a range of uncertain future hazard scenarios and resilience investments using travel demand model (TDM) results to provide initial inputs and an open-source routing model called AequilibraE to produce disruption scenario results.
- The RDR ROI Analysis Tool monetizes investment, repair, and change in performance under disruption with and without resilience investments that mitigate the disruption at a given location.
- The RDR ROI Analysis Tool evaluates return on investment using Benefit-Cost Analysis (BCA), Benefit-Cost Analysis under Uncertainty/Regret Analysis, or Breakeven Analysis, depending on the user's data.

¹⁴ Including the National Cooperative Highway Research Program (NCHRP) 20-101: Extreme Weather and Climate Change: Guidelines to Incorporate Costs and Benefits of Adaptation Measures, FHWA tools and resources include the Vulnerability Assessment and Adaptation Framework and U.S. DOT BUILD Discretionary Grant Program guidance.

- The outputs of the tool suite include rank ordering of project performance over all scenarios and future conditions. Results can be explored using a series of graphical dashboards.
- The RDR Benefits Analysis Tool helps the analyst quantify the benefits of a resilience investment during a particular hazard, as measured in changes in travel performance measures, by a categorical or continuous attribute of interest at the TAZ level.

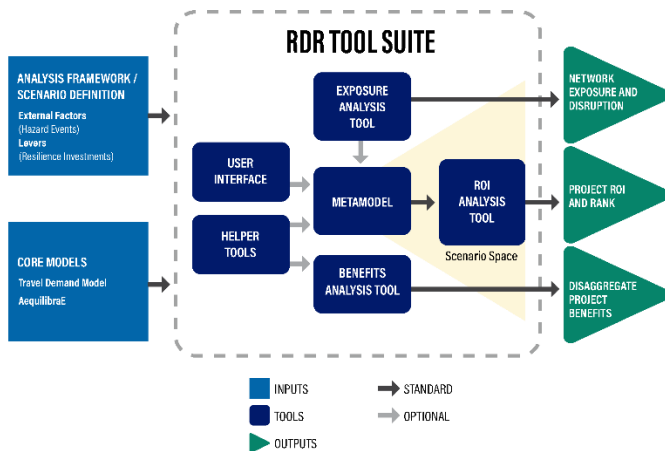


Figure 1-1: Structure of the RDR Tool Suite. The scenario space increases as more potential hazard severities, durations, recovery periods, and resilience investments are assessed.

A companion document, the RDR User Guide, provides more instruction on preparing data, executing analyses, and interpreting results as well as test data for running sample scenarios. The RDR Quick Start Guide provides step-by-step instructions for installing RDR and running a first scenario on a model transportation network. The RDR Reference Scenario Library provides example analyses in the RDR Tool Suite on a series of hypothetical and real-world transportation networks to demonstrate the Tool Suite's options and functionality. The RDR Run Checklist provides a short reference on the inputs to check before executing an analysis. The RDR Tool Suite is available at <https://volpeusdot.github.io/RDR-Public>.

1.2 Approach

A resilience-focused ROI analysis must take into account:

- the range of potential hazard events;
- the range of impacts of those hazard events on transportation assets and operations including travel disruption during hazard events, the asset damage from those events, and the travel disruption due to the subsequent necessary repairs;
- the costs associated with travel disruptions measured by changes in system performance (PHT, VMT, and trips) and asset damage repair and cleanup costs due to hazard events; and
- the investment required to mitigate the impact of the range of potential hazard events.

The RDR Tool Suite is designed to overcome these challenges of calculating the impact of resilience investments under a wide range of possible futures. It accomplishes this using the established Robust Decision Making framework,^{15,16} which has been used under a similar modeling context by the Travel

¹⁵ RAND. Robust Decision Making. RAND. [Online] [Cited: July 20, 2022.] <https://www.rand.org/pubs/tools/TL320/tool/robust-decision-making.html>.

¹⁶ Lempert, R. (2019). Robust Decision Making (RDM). *Decision Making under Deep Uncertainty: From Theory to Practice*. V. A. W. J. Marchau, W. E. Walker, P. J. T. M. Bloemen and S. W. E. Popper, Springer: 329.

Model Improvement Program – Exploratory Analysis and Modeling Tool (TMIP-EMAT),¹⁷ a scenario-based decision-making tool that can be integrated with existing travel demand forecasting models of all types.

The Robust Decision Making approach provides a method for comparing alternatives under deep uncertainty by comparing alternative policy approaches under a wide range of scenarios. It includes the following steps:

1. **Define the analysis framework:** Apply an “XLRM” framework to guide stakeholder engagement, data assembly, and model development.¹⁸ The X refers to external factors, or uncertainties (e.g., hazard scenarios); the L refers to possible policy levers (e.g., resilience investments); the R refers to relationships between the other elements that are reflected in the TDMs and RDR Tool Suite to estimate performance; and the M refers to performance metrics. The scenario space is the set of all possible scenarios that the analysis will consider and is defined by the unique combinations of external factors and policy levers.
2. **Run a core model:** The underlying travel demand model is called the “core model,” and reflects the causal relationships of the XLRM framework. In the transportation context, the core model represents the transportation network and its system performance, which is typically computed using a TDM. The RDR Tool Suite uses two core models to populate the metamodel:
 - a. **A regional travel demand model (TDM)** which is commonly used by State DOTs and MPOs. It typically provides
 - i. Tables describing baseline movements;
 - ii. Mode choice;
 - iii. Network routing of the trips.
 - b. **The RDR core model leveraging the open-source AequilibraE,**¹⁹ which provides a fast, simplified representation of network routing for a representative day. The RDR core model is run many times, using the trip tables from the regional travel demand model, to compute the outcomes for many scenarios in the full scenario space. The RDR Tool Suite uses AequilibraE to enhance initial TDM core model sampling to populate the metamodel.
3. **Use a metamodel to analyze the full scenario space:** If the core model has a longer run time (e.g., several hours to days for an MPO TDM), then in most cases the scenario space will be too large to use the core model to calculate all results. The metamodel works by running regressions on a limited set of runs from the core model that are then used to fill in the gaps where the core model has not been used to produce scenario results. A metamodel typically involves the use of regression models that are parameterized using a subset of results produced by the core model.

¹⁷ TMIP-EMAT Development Team. TMIP-EMAT. [Online] <https://tmip-emat.github.io/>.

¹⁸ Lempert, R. J., D. G. Groves, S. W. Popper and S. C. Bankes (2006). "A General, Analytic Method for Generating Robust Strategies and Narrative Scenarios." *Management Science* 52(4): 514-528.

¹⁹ AequilibraE. [Online] [Cited: 07 26, 2022.] <http://aequilibrae.com/python/latest/>. RDR 2024.2 was developed using AequilibraE 1.0.0.

4. **Simulate and analyze:** Experiments are run using the metamodel to build a range of outcomes across multiple axes of variation. Analysis is then performed to determine the best course of action based on the range of potential futures.

The RDR Tool Suite calculates results for different time units at different steps in the overall analysis. At the initial stage, the scenario space is defined at the single day level in the configuration file. Successive steps within the RDR Tool Suite build the scenarios from single-day “steady state” outcomes of network performance up to complete hazard events with defined hazard duration and recovery processes that occur as a series or stream of annual evaluations of hazard events occurring with some probability over the economic analysis period as follows:

- The core models produce single-day estimates of network performance based on the variables of hazard severity, economic scenarios, and trip demand elasticity for a subset of the scenario space. The analyst must provide core model results for the initial year for each hazard event severity (see Sections 5.2.2.3 through 5.2.2.6).
- The RDRM regression computes single-day estimates of network performance for the remaining scenarios in the scenario space that were not estimated in the core models (see Section 5.3.2.1).
- The Recovery step of the RDRM analysis incorporates the hazard duration and hazard recovery process into the scenario space, by extending the single-day event estimations into multi-day events over which the hazard event recedes (see Sections 5.3.2.2 through 5.3.2.3).
- The RDR ROI Analysis Tool extends the scenarios from single-event outcomes to a stream of future events over the economic analysis period of interest using hazard event probabilities and economic parameters. The start and end years of the period of analysis can be different from the initial and future years of the scenario (see Sections 5.3.2.4 and 6.2.2).

Figure 1-2 shows how the RDR Tool Suite develops the results for the full scenario space in successive steps of the tool suite where each circle represents a scenario within the scenario space.

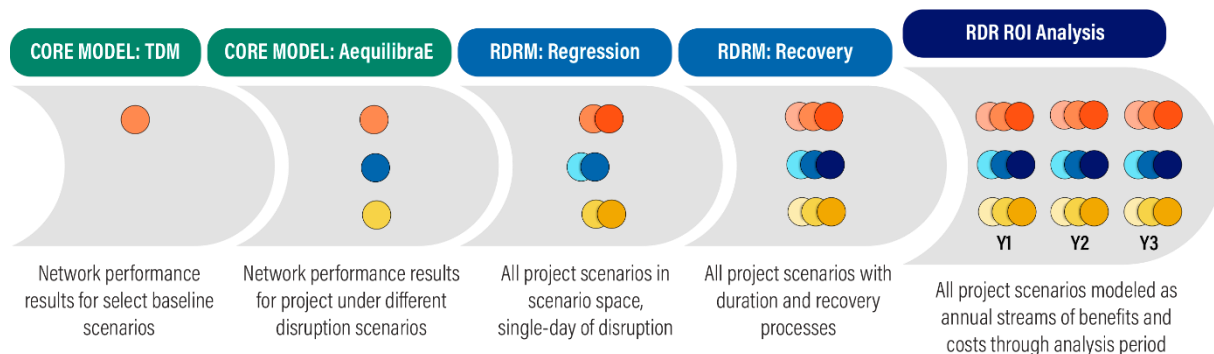


Figure 1-2: Diagram of how the RDR Tool Suite develops the scenario space results through the successive steps of the RDR Tool Suite analysis process. Different color groups represent different hazard severities, different color intensities within color groups represent different durations and recovery processes.

1.3 Return on Investment Analysis

Hazard events affect travel (transportation network performance) through the loss of network links or reduction in level of service (capacity). This disruption persists at least until the hazard ends and may also necessitate repair for travel to return to normal levels. Lost network performance can be quantified in terms of costs of repair/replacement and/or cleanup of the asset (e.g., debris removal) as well as the

cost assigned to the increase in vehicle hours traveled (VHT), vehicle miles traveled (VMT), and lost trips resulting from the disruption.

To maintain and enhance the reliability of the transportation network, state DOTs and MPOs seek to make specific infrastructure investments that will increase the resilience of transportation assets (i.e., reduce impacts and/or speed recovery time). In practice, there may be some resilience investments that would provide network performance or cost saving benefits in the absence of hazards, but the focus of the RDR Tool Suite is on investments that mitigate the impact of a hazard (reduce disruption and associated costs), but do not necessarily provide improvements in network performance under non-hazard scenarios.

Figure 1-3 shows a hypothetical comparison of the loss of network performance and period of recovery for a hazard event without resilience investment (yellow-shaded area labeled 7) and with resilience investment (blue-shaded area labeled 6). The impact on system performance is lessened both during the event and during the repair period, as the asset was less damaged due to the resilience investment.

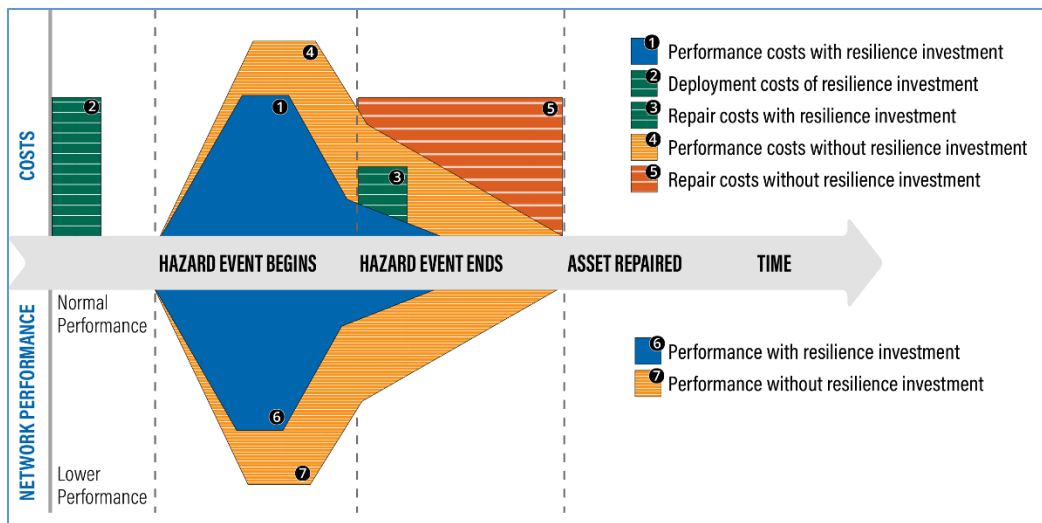


Figure 1-3: Resilience investment analysis concept: network performance during exposure and repair with and without a resilience investment (lower half) and components of the associated benefit-cost analysis (upper half). In the area below the timeline axis, the yellow-lined area shows a hypothetical disruption to network performance during a hazard event and the disruption after the hazard event due to damage of a transportation asset. The disruption to the network ends when the asset is repaired to working order as it was before the hazard event. The blue-shaded area shows the disruption that occurs in the case of the same hazard event when a resilience investment is deployed. The area above the timeline axis shows components of the benefit-cost analysis of a single hazard event, including monetized loss of network performance and repair costs in the absence of resilience investments (yellow and orange areas) and the cost of resilience investments and associated reduced network performance and repair costs (green and blue areas).

This hypothetical resilience investment example can be extended to an ROI analysis appropriate for long-range decision making like benefit cost analysis, as shown in Figure 1-3. In Figure 1-3, network performance has been monetized as costs, and the costs of damage repair and the cost of the resilience investments have been included. The benefit of the resilience investment is the difference between the costs when the investment is deployed (green and blue areas labeled 1, 2, and 3) versus when it is not deployed (yellow and orange areas labeled 4 and 5).

Hazard events can vary in their severity, frequency, and duration in significant ways that would have a differential impact on transportation. Standard methods of infrastructure investment analysis are not designed to accommodate significant uncertainty, and therefore new methods and tools are needed to support the ROI analysis of resilience investments, such as the RDR Tool Suite Robust Decision-Making approach.

1.3.1 Benefit Cost Analysis

The Return on Investment (ROI) metrics can be used to support a Benefit-Cost Analysis (BCA). In standard BCA, analysts use a “predict then act” (or “agree-on-assumptions”) approach that relies on a singular best prediction of future conditions under which the proposed policy alternatives are tested.²⁰ In such analyses, a best estimate of the conditions of the future world is predicted (or assumptions about the future are agreed upon) using the expected utility approach in which agencies assign their best estimate of the probabilities of the mutually exclusive possible future conditions of the world, and then the probability of each future state of the world is applied to the value of the outcomes of those futures. The sum product of all outcomes by its probability represents the best estimate of the expected outcome of the future. Analysts are then able to compare the impact of each of the investment alternatives against one another using a baseline scenario as the basis of comparison.

The NCHRP 20-101 report provides guidance for how to compute a BCA for transportation investments in resilience. It is a great resource for agencies who are still developing an understanding of the fundamentals of BCA. The NCHRP 20-101 guidebook chapters 7 and 8 describe a pen and paper method for building out a BCA analysis for a single flood resilience project that considers a range of future flood levels. This technical document expands on the NCHRP guidebook in that it broadens the analysis to include more assets and more uncertainties.

The RDR Tool Suite BCA approach uses all of the standard assumptions of BCA including an exhaustive list of the benefits and costs of the project, ignores transfers between parties, avoids double counting, and applies discounting of future costs and benefits to reflect the time value of money.

1.3.2 BCA-U/Regret

The Return on Investment metrics also allows an alternative ranking approach called BCA under Uncertainty (BCA-U) or Regret analysis which uses an “agree on decisions” approach in which the investment alternatives are determined first, and then those alternatives are tested under all of the potential future conditions.²¹ The project prioritization approach within the BCA-U/Regret analysis approach addresses the need to appropriately balance the project costs and benefits across the wide range of potential hazard exposures with uncertain probabilities. The BCA-U/Regret approach is a way of adapting the expected utility functions for situations in which there is low confidence in the estimated probabilities of future events.²² These methods are designed to produce better decisions under inherent

²⁰ Lempert, R.J., Steven W. Popper, and Steven C. Bankes. (2003). “Shaping the Next One Hundred Years: New Methods for Quantitative Long-Term Policy Analysis.” Rand Pardee Center. Accessed on 12/19/2019 from https://www.rand.org/content/dam/rand/pubs/monograph_reports/2007/MR1626.pdf

²¹ Lempert, R.J., Steven W. Popper, and Steven C. Bankes. (2003). “Shaping the Next One Hundred Years: New Methods for Quantitative Long-Term Policy Analysis.” Rand Pardee Center. Accessed on 12/19/2019 from https://www.rand.org/content/dam/rand/pubs/monograph_reports/2007/MR1626.pdf

²² Society of Decision Making under Deep Uncertainty: <http://www.deepuncertainty.org/>. Accessed on 15 July 2019.

and unresolvable uncertainty through exploratory analysis, rather than to improve predictions of future conditions.

In BCA-U/Regret, the expected net benefits criterion for prioritizing projects cannot be used because there are no probabilities associated with the future scenarios and therefore the results of the scenarios cannot be risk-weighted. Instead of expected net benefits, an alternative concept is used called regret, which is defined as “the difference between the performance of the option in a specific state of the world and the performance of the best possible option in that state of the world.”²³ In this context, regret is measured by comparing the outcome of each alternative project scenario with the outcome of the scenario that would have been the ideal course of action under an assumed future state of the world, and this process is repeated under each possible future state. The project scenario outcome is compared against what the agency would have elected to do instead if it had known what would actually occur. In some cases, the ideal course of action under a given scenario is a project alternative, and in others it may be the do-nothing (no project) alternative.

The regret metric is calculated for every project under every scenario, and then the regret scores from each scenario for each project are summed to provide a cumulative regret metric for each project. Project cumulative regret scores can then be compared using the decision rule that ranks projects where lower regret scores are preferred. The approach implemented by the RDR Tool Suite uses the Laplacian decision rule which assumes that all scenarios have the same probability of occurrence. In many contexts in which the regret methodology is used, the scenarios are mutually exclusive so that aggregation of the regret scores across scenarios has a different meaning. Mutually exclusive outcomes would allow different kinds of decision rules such as Minimax, in which the best outcome is the one that has the least bad outcome. However, the scenarios considered in the RDR Tool Suite are not mutually exclusive. The hazard events are stochastically independent so that any combination of them could occur in the same year (e.g., a 10-year flood and a 100-year flood could occur separately in the same year; two 10-year storms could occur in the same year, etc.). Therefore, the appropriate regret decision rule for the RDR Tool Suite is a low-to-high ranking of the cumulative regret metric that can account for the multiple realizable hazard events and treat the other future conditions such as the socioeconomic outcome or the elasticity value as having equal probability.

The RDR Tool Suite calculates the BCA-U/Regret in cases where the user does not have probabilities for each uncertainty which allows the user to prioritize projects by their cumulative regret value summed over all possible future scenarios or a subset of future scenarios that are most relevant for the user.

1.3.3 Breakeven

Finally, the RDR Tool Suite can also be used to calculate a breakeven analysis in which the maximum possible cost of the project is estimated by calculating the sum of the expected social benefits (the benefits as in a BCA).²⁴ The breakeven value can be used to determine whether an investment is

²³ Kwakkel J.H., Haasnoot M. (2019) Supporting DMDU: A Taxonomy of Approaches and Tools. In: Marchau V., Walker W., Bloemen P., Popper S. (eds) Decision Making under Deep Uncertainty. Springer, Cham. Accessed on 12/19/2019 from https://link.springer.com/chapter/10.1007/978-3-030-05252-2_15

²⁴ Breakeven analysis is a special case of BCA in which the net benefits are \$0, by construction. Breakeven analysis is a partial BCA approach that calculates the benefits of notional resilience projects and interprets the potential benefits as defining the upper bound of costs that such a project could have and still produce a positive economic outcome. This approach works under the assumption that agencies would be indifferent between project alternatives that produce zero net positive benefits compared to the baseline but can also be applied “in reverse”

potentially viable based on the benefits it could produce, and it can also be used to rank projects based on their benefits. The breakeven calculation is flexible and can be used whether the user has associated probabilities for each uncertainty (as in the expected utility approach of BCA) or when they do not (as in the BCA-U/Regret approach).

Breakeven analysis is useful for agencies that have not yet identified candidate investments and want to explore how hypothetical investments would impact their transportation network in order to help them select candidate projects. It is also useful for agencies that already have candidate investments but do not yet have cost estimates and want to better understand what their constraints are in terms of their projects' deployment costs.

1.4 RDR Tool Suite Use Cases

The RDR Tool Suite is designed to help transportation agencies address questions regarding their potential resilience investments, including:

- 1) Which of my road, bridge, and transit assets are vulnerable under a given hazard condition or range of conditions?
- 2) What will link-level capacity loss be under a given hazard condition?
- 3) Which resilience projects will give the most benefit across the range of hazards of concern?
- 4) Under which hazard conditions does a project perform well? How is the performance of a given project distributed across the range of hazards?
- 5) How are the benefits of a resilience project distributed across TAZ categories, in terms of impact of a project during a particular hazard on travel performance metrics?

Agencies vary in their level of data availability, analytical tools and resources, and knowledge of potential hazards and disruptions in their region. The RDR Tool Suite is set up to allow for different levels of analysis depending on the information the analyst can bring to the model. If an analyst does not have access to a travel demand model but can estimate trip tables for AequilibraE, all parameterization of the RDRM can be performed using the open source AequilibraE. And if an agency does not have specific project plans for investing in resilience and therefore does not have an estimate of the total cost of an investment, the RDR Tool Suite can still be used to estimate a breakeven point (total benefits) for potential project cost. The RDR Tool Suite can also use data from publicly available sources, if converted to the correct format compatible with the tool suite. The analyst can use these sources to complement their specific data, following guidance in the RDR documentation.

It should be noted that the RDR Tool Suite is not a “one-stop shop” for all resilience planning needs. It does not incorporate engineering evaluation considerations, except to the extent that these define the capacity and availability of links in the transportation network and the costs associated with resilience investment actions. It also does not help agencies determine what kind(s) of project(s) would be appropriate for a given asset based on predicted vulnerability to exposure. The outputs are based on investment cost, repair cost, and network performance; it does not calculate economic impact analysis at the broader societal level (e.g., impact on employment, transfers between parties, etc.).

In summary, the RDR Tool Suite provides a transportation agency with a set of tools for evaluating the ROI provided by a set of resilience investments across a range of transportation assets and uncertain

to define the range of net benefits that a project would need to provide under a given scenario in order to justify its initial investment and recurring costs.

future hazard conditions, and for ranking those projects based on performance. The RDR ROI Analysis Tool project rankings and visualizations can be used as a factor in overall transportation infrastructure project prioritization for long-range transportation planning in combination with other prioritization factors such as congestion reduction, safety, engineering considerations, and budgets.

2 Software Components

The RDR Tool Suite is written in Python and R. Both programs are required software and will be downloaded and installed during the RDR Tool Suite setup process. The required Python packages for installation, specifically the Python package AequilibraE, are managed using the conda dependency management system. Current installation instructions are available in the RDR Quick Start Guide and RDR User Guide.

The following programs are used by the RDR Tool Suite to run an analysis:

1. Conda dependency management system²⁵ is used for software package management (version 23.10.0 or newer). Installation through Anaconda or Miniconda²⁶ is recommended for conda functionalities.
2. Python²⁷ is the primary coding language of the RDR Tool Suite.
3. AequilibraE²⁸ is an open-source Python-based routing model used as the core model to compute network flows and summary statistics for quantifying the impact of disruption on the transportation network.
4. R²⁹ is used to run the Latin hypercube sampling and regression modules. R will be installed when setting up the conda environment and does not need to be installed separately by the user.
5. SQLite is used to store core model inputs and results. If the user would like to fully explore SQLite-generated outputs, they are encouraged to install a SQLite database browser, such as DB Browser for SQLite.
6. Tableau Reader³⁰ or Tableau Desktop is used for visualizing RDR ROI Analysis Tool outputs.
7. Spatial analysis, such as the RDR Exposure Analysis Tool and Transit Network Helper Tool, use open-source approaches based on geopandas, and do not require ESRI ArcGIS Pro.
8. The RDR Tool Suite Version 2026.1 is the tool itself and can be downloaded from the public GitHub repository at <https://volpeusdot.github.io/RDR-Public>.

See the RDR User Guide for more information on installation of RDR Tool Suite required software.

²⁵ Conda. [Online]. User Guide: Installation. Accessed 26 July 2022 from <https://docs.conda.io/projects/conda/en/latest/user-guide/install/>.

²⁶ <https://docs.anaconda.com/anaconda/install/>

²⁷ Python. [Online]. Download. Accessed 26 July 2022 from <https://www.python.org/downloads/>

²⁸ AequilibraE. [Online] [Cited: 07 26, 2022.] <http://aequilibrae.com/python/latest/>. RDR 2022.1 was initially developed using AequilibraE 0.7.2. The current version, RDR 2026.1, was developed using AequilibraE 1.6.2

²⁹ R. [Online]. The R Project for Statistical Computing. Accessed 26 July 2022 from <https://www.r-project.org/>

³⁰ Tableau Reader. [Online]. Accessed 26 July 2022 from <https://www.tableau.com/products/reader>

3 RDR Scenario Definition Inputs

3.1 Purpose: Defining the Scenario Space

The first step in the analytical process is to define the analysis framework, which includes the hazard scenarios, resilience investments, and other uncertainties. The hazard scenario(s) will determine which assets are affected and where a resilience investment could be implemented to reduce impacts.³¹ For the RDR Tool Suite, hazard characteristics such as magnitude and duration will influence the extent of impacts on travel behavior, while the probability of the hazard scenario occurring influences the anticipated costs and benefits associated with investing in an asset to improve resilience.

In the “XLRM” framework used in Robust Decision Making (as described in the Introduction), the scenarios comprise the external factors (“X”) that have a causal role in the overall outcome of a given scenario. In the RDRM context, these uncertainties can be grouped into the hazard event uncertainties and the travel demand uncertainties, listed below:

- Hazard uncertainties:
 - Hazard severity
 - Hazard duration
 - Hazard event annual probability (probability of occurring each year)
 - Future hazard event frequency, i.e., the change in a hazard probability over time
- Travel demand uncertainties:
 - Future demographics
 - Future land use decisions
 - Future economic activity
 - Trip demand elasticity with respect to travel time (the hazard event may lead to longer travel times for some trips)

The policy levers of the “XLRM” framework in this context are the resilience investments, or “project alternatives,” that will mitigate the impact of a hazard event either through reducing travel disruption, damage, or both (e.g., project alternatives could include raising a roadway or scour prevention for a bridge).

The total scenario space is defined by the unique combinations of each possible value for each uncertainty. Each project alternative is tested in each scenario in the scenario space. Figure 3-1 shows a notional example of this: each arrow represents a different value of the given uncertainty. Each path following the arrows from top to bottom represents one scenario in the scenario space and is a unique combination of values of the uncertainties that represents one possible future. The diagram highlights one possible path with red enlarged arrows. The RDR Tool Suite approach tests the impact of each project alternative in each of these different possible futures.³²

³¹ In this document, a “hazard” will refer to the hazard type such as flooding or earthquake, and a “hazard scenario” will refer to a particular hazard type event where characteristics of the hazard scenario are fully defined (e.g., 500-year flood, 6.1 magnitude earthquake).

³² The RDR Tool Suite uses different approaches for estimating the outcomes of these scenarios. Some of the scenarios are estimated in the core model using the TDM and AequilibraE, while the remaining are estimated in the RDR Metamodel. The process for estimating each of these scenarios is describe in section 5.

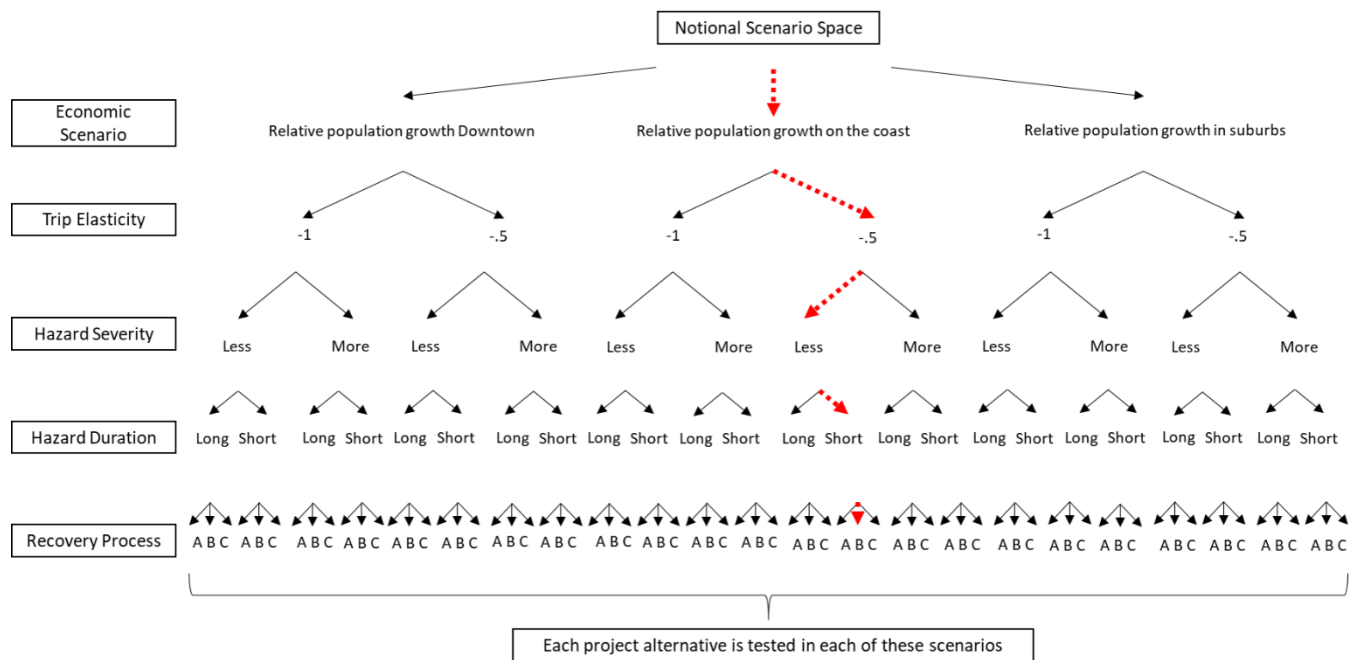


Figure 3-1: Tree diagram of notional scenario space in which each resilience project alternative is tested. Red arrows highlight one scenario in the scenario space, which is a unique combination of uncertainty values, leading to one possible future.

It is important that the hazard scenarios are defined to accurately capture, to the best available knowledge, the future potential hazards, and how those hazards may impact the transportation network. Understanding the full range of potential hazards ensures the resilience investments being deployed will appropriately mitigate future hazard impacts. The agency will need to identify the potential hazards that they face, as well as develop reasonable ranges of values for the variables of hazard event severity, hazard event probability, and future hazard event frequency.

Two other constraints may limit the ability to define hazards in the most robust way: data quality and study area definition. For example, in the case of flooding, the data available may only specify whether a roadway section is inundated, but not the depth of inundation. Similarly, the data available may only be capable of specifying the conditions of certain flooding events, such as a storm surge/sea level rise, but not for others such as extreme precipitation events.

The RDR Tool Suite is agnostic about both the hazards and the resilience investments to mitigate those hazards. However, the RDR User Guide provides background information on developing hazard scenarios for input into the RDR Tool Suite and provides a few potential data sources that can help an agency estimate hazard extent and severity conditions. Hazard event duration and hazard event recovery variables can be informed by agency knowledge of the probability of certain kinds of events or can be estimated as assumptions for analysis purposes.

Note that the RDR Tool Suite is a region-specific analysis and will not capture the interaction with regions that are not included in the analysis, even though they may affect travel into and out of the area.

3.2 Structure, Functions, and Parameters

3.2.1 Hazards Event Uncertainties

The RDR Tool Suite analyzes a single hazard at a time (e.g., flooding OR earthquake OR wildfire). The hazard event uncertainty parameters include:

- Hazard severity: The hazard event severity in terms of how much exposure it causes is operationalized in the RDR Exposure Analysis Tool, but the hazard event parameter here supplies the label that the RDR Tool Suite will use associated with different hazard exposure impacts. The hazard severity and hazard frequency are typically linked. At least two separate hazard severities should be included to take advantage of the RDR Metamodel approach.
- Hazard duration: This parameter defines how long the hazard event lasts in days.³³ The impact of a hazard event is aggregated to an annual impact for project prioritization regardless of hazard event probability. If the hazard is a single event like an earthquake, the hazard duration would be set to 1 day. All hazard severities in a given analysis are assigned the same range of duration values.
- Hazard event recovery: This set of parameters define how hazard events will subside over time. All hazard severities in a given analysis are assigned the same recovery pattern.
- Hazard event annual probability: This parameter defines the hazard event's annual probability of occurrence, which the user should be able to justify based on existing data. Hazard severity and hazard frequency are typically linked. In the case of flooding, annual flood risk is defined by its "return period" which essentially is the annual probability of occurrence of a flooding event of a given level of flooding (e.g., in feet). This parameter can be set to more than 1 for events that are expected to occur more than once per year. This parameter must be appropriately set if running a benefit cost analysis or breakeven analysis.
- Hazard event frequency factor: This parameter defines how the risk of hazard events will change over time. This parameter is a fixed annual increase in probability which is applied as a multiplier to the costs and benefits incurred in each successive year. The parameter applies to all hazard events uniformly. The relationship between severity and probability also impacts the future event probability value. For example, in the context of flooding, changes in climate may increase the frequency and severity of storms and thus the depth and duration of flooding events.³⁴ With a changing climate, the likelihood of what is currently considered a 100-year storm may be greater than the estimated one percent probability per year, and the storms with a 100-year return period in the future may be stronger than the current 100-year storms.³⁵

The user is responsible for ensuring the scope of hazards and their parameters are appropriate for the decision-making task at hand and should use best practices and expert knowledge to determine hazard parameters. It is possible for users to bias their results by selecting an unrepresentative or unrealistic set of hazard events. Agencies should avoid selecting hazard scenarios to obtain a predetermined, desired result, or that do not encompass the range of likely scenarios. For example, if the analysis considers only the most severe hazard events, the analysis is more likely to identify a significant need for investment, while other investments may actually perform better under more likely hazard severities.

³³ The RDR Core Models estimate travel at the daily level so hourly level assessments cannot be made.

³⁴ <https://www.fhwa.dot.gov/engineering/hydraulics/pubs/hif16018.pdf>

³⁵ Holmes, R.R., Jr., Dinicola, K., 2010, 100-Year flood—it's all about chance: U.S. Geological Survey General Information Product 106, 1 p. Accessed on 1/2/2020 from <https://pubs.usgs.gov/gip/106/>

3.2.2 Travel Demand Uncertainties

In addition to the hazard event uncertainties, more standard uncertainties associated with human travel behavior could impact the outcome of a hazard event and/or the potential of a resilience investment to mitigate a hazard. The two primary behavioral choices in the RDR Tool Suite are captured in the following mechanisms:

- **Elasticity of Travel Demand:** Travel demand elasticity is the change in travel (driver decision to make their trip) in response to a change in generalized cost. For example, if the elasticity is -0.5, a 10% increase in generalized cost will result in a 5% decrease in travel. There is empirical uncertainty about the specific value of this parameter, and it has substantial impact on the travel behavior as modeled by the core models. The RDR Tool Suite allows the user to use multiple elasticity of demand values to capture this uncertainty.
- **Economic Scenario:** The economic scenario reflects uncertainty about long-term changes in land-use, demographics, and trip demand patterns in the region over the period of analysis. This is operationalized in the trip demand tables that come from the core model, which reflect the pattern of trip demand to and from each travel analysis zone (TAZ). Users must create a future year trip table to represent the expected pattern of trip demand for the region, which itself reflects land-use and demographics of the future year. Users can use a single future economic scenario to represent their best guess as to how their region will grow leading up to the future year. Or, if there is uncertainty about how the region will grow over the period of analysis, the user can generate multiple economic scenarios to reflect the different potential future patterns of trip demand.

The variables of economic scenario and trip demand elasticity are already part of agency scenario planning, and most agencies will already have a range of possible values that they want to consider. Agencies should be careful not to be too narrow in their assumptions about future conditions.

4 RDR Exposure Analysis Tool

4.1 Purpose: Applying Hazards to Networks

An asset hazard exposure analysis is used to determine how a transportation asset is impacted by a hazard. The RDR Exposure Analysis Tool is a tool that provides a method to link exposure data to a transportation network. Agencies that have already assessed exposure using other techniques (e.g., have overlaid flooding depth grids on their transportation network or linked earthquake vulnerability estimates for transportation assets) can proceed directly to using the RDRM. The RDR Exposure Analysis Tool uses open-source geospatial tools; outputs can be visualized using GIS software such as QGIS or ArcGIS Pro.

Hazards do not impact all transportation assets uniformly, as hazard severity varies across geography, e.g., flood levels depend on land elevation and soil composition. Hazard exposure also depends on the asset location, type, and characteristics such as elevation, construction materials, condition, and context (e.g., surrounding topography, etc.).

To conduct an exposure analysis, a user must obtain geospatial data on the transportation assets (roadway and/or transit network), identify a specific hazard type to be analyzed (e.g., flooding), and identify the intersection of a particular hazard type and event with transportation assets in the region. The RDR Exposure Analysis Tool combines individual asset information with hazard severity to evaluate the potential asset exposure (e.g., flood inundation depth on individual network links). The output of the RDR Exposure Analysis Tool is used in the Disruption Analysis Submodule to assess the impact of exposure on travel.

A resilience investment in the context of this analysis is considered to be a unique project or project expansion that is intended to reduce the impact of a hazard on the transportation system. Resilience projects can reduce hazard exposure by moving or redesigning a transportation asset (e.g., moving a roadway out of a floodplain or raising it to reduce flooding exposure, or enhancing structural elements to resist earthquake). Resilience projects could also involve adding infrastructure away from the transportation asset (e.g., adding a sea wall or berm to reduce exposure of the asset to flooding). The effect of these projects on exposure can be represented in an additional resilience investment exposure table that reflects reduced exposure on the modified links where the resilience investment is planned.

4.2 Structure, Functions, and Parameters

The RDR Exposure Analysis Tool uses the following inputs:

- Routable transportation network:** A routable transportation network can be acquired from a TDM or publicly available sources such as OpenStreetMap (see the User Guide for more detail). For the RDRM process, using a TDM network is the simplest approach if available since results are directly linked to the TDM assets. Most TDMs can export their network in a GIS format to combine with other geospatial data, such as existing geospatial hazard severity datasets. However, if the TDM network is highly schematized (i.e., not representing links in real-world geography), it may be better to use a true shapes file that is more geospatially-explicit and then match up the links to a TDM network after calculating exposure. A routable transit network (e.g., bus, tram, subway links) can be incorporated into the transportation network input files for the RDR Tool Suite as well; see the User Guide for more detail on transit network specifications.

- Exposure data: These data are generated for each hazard scenario. More information on defining exposure data, identifying relevant individual asset data, and surveying potential data sources can be found in the User Guide. Users may have already conducted vulnerability assessments that may be appropriate as input data for this analysis. To incorporate data from such analyses, the data must link hazards to assets of interest (links in the transportation network), including the magnitude of exposure.
- Resilience investment-related exposure data: Mitigation of exposure associated with a given resilience investment.

The RDR Exposure Analysis Tool associates the links in the network with the geospatially-coincident exposure grid cells to estimate exposure severity on each link in the network and assigns the most severe exposure on a link as its exposure level.

The RDR Exposure Analysis Tool outputs are a file containing a list of the transportation network links and the exposure of those links under different hazard scenarios and resilience investment scenarios.

5 RDR Metamodel (RDRM)

5.1 RDR Disruption Analysis Submodule

5.1.1 Purpose: Translating Exposure Severity to Network Capacity

To evaluate how a hazard impacts travel as well as how the resilience investment mitigates that impact, the hazard exposure for assets in the transportation network is translated into reduction in capacity (disruption). The RDRM approach handles hazards as changes in network link capacity. While potentially limiting, this allows the RDRM tool to be completely hazard-agnostic – exposure from any hazard can be modeled. The RDR Disruption Analysis Submodule provides several options to define how exposure translates to capacity loss. Agencies that have estimates of asset disruption for each hazard event can proceed directly to the core model runs that inform the metamodeling.

5.1.2 Structure, Functions and Parameters

5.1.2.1 Disruption Calculations

Either the RDR Exposure Analysis Tool or the main RDR Metamodel can be used to calculate disruption of the network based on exposure using the same methodology options and parameters that are defined by the user. The RDRM disruption analysis converts exposure data into disruption levels to be used downstream in the RDR Tool Suite.

The same five options for the disruption analysis are built into both the RDR Exposure Analysis Tool and the RDRM:

- **Default (flood-specific):** The default flood exposure function utilizes a depth-disruption function adapted from an existing function from the literature (Pregolato et al. 2017), where the availability of a roadway decreases from 100% to 0% at a linear rate between 0 and 300 millimeters of flood depth.³⁶ This function is only applicable to flooding depth grids, not other exposure grids, and is relevant only for road network links. This method can be used when a simple linear function converting flood depth to disruption is desired, without the complexity of a manual approach or custom exposure-disruption function.
- **Binary:** If the binary approach is selected, any network segment with an exposure above 0 is assigned a link availability of 0. Any network segment without any hazard exposure is assigned a link availability of 1 (fully available). This option is useful when data limitations in the underlying exposure analysis do not allow fine-grained assessment.
- **Manual:** The manual method allows users to define their own bins (ranges of severity) for converting exposure values to disruption (e.g., severity of level x to y corresponds to reduction in capacity of z). This method is used when the user does not have a specific exposure-disruption function but can still estimate categories of disruption based on exposure.
- **Facility Type Manual:** The facility type manual method allows user-defined bins representing the conversion of exposure into link availability by facility type. This method allows for more nuance compared to Manual in estimating categories of disruption based on exposure.

³⁶ Pregolato, M., Ford, A., Wilkinson, S., & Dawson, R. (2017). "The impact of flooding on road transport: A depth-disruption function." *Transportation Research Part D: Transport and Environment*, Vol. 55, pp.67-81.
<https://www.sciencedirect.com/science/article/pii/S1361920916308367>

- **Beta Distribution:** This option allows the user to define a custom curve representing the relationship between hazard severity and transportation capacity change.³⁷ This gives the user flexibility in determining how link availability could change in a non-linear fashion due to varying levels of exposure.

5.1.2.2 Resilience Investment Disruption Mitigation

Different types of resilience investments can mitigate the impact of a hazard at a given location (see Table 5-1).

Table 5-1 Resilience Investments in the RDR Tool Suite

Resilience investment	How this is implemented in the RDR Tool Suite
New alternate route	New links in the network
Modify existing route to make its disruption less likely (e.g., a road subject to flooding is raised)	Greater link availability in the network at those resilience project links
Ability to recover faster	Reflected in the recovery times and damage recovery path, model will reflect a less disrupted network earlier in the recovery process
Lower cost of repair	Reflected in the cost analysis

5.1.2.3 Calculating Emergency Response, Evacuation Disruption, and Other Analyses

In addition to supporting project prioritization, the RDR Tool Suite disruption module can be used to estimate which roads would still remain accessible to emergency vehicles or other high clearance vehicles. Exposure and disruption data can also be used to determine the disruptions to emergency response and evacuation routes.

Emergency response vehicles usually have higher clearance than passenger vehicles; therefore, emergency vehicle access limitations during a hazard may differ from those for passenger vehicles when clearance affects passage (e.g., flooding). The emergency response disruption analysis allows users to identify impacted routes based on ground clearance. For example, in a flood-based scenario, a user could specify that inundation levels between one and two feet are only passable by emergency vehicles and inundation levels above two feet are impassable to everyone. However, it should also be noted that the resulting damage and therefore disruption will also impact the ability of emergency response vehicles to respond and the corresponding increases in travel time should be accounted for.

The emergency response disruption analysis feature of the RDR Exposure Analysis Tool can be further used to assess viability of evacuation routes under various hazard conditions. An analyst can use the map feature in GIS to find critical failure points in an evacuation route and identify alternative routes for affected segments. The analyst can also evaluate the changes to travel times on the degraded evacuation or alternative route using either the TDM or the AequilibraE core models tools.

³⁷ There are examples of predictive relationships such as depth velocity functions defined by Pregolato et. al (see Appendix G in the User Guide) that can be used for a more nuanced analysis estimating level of change in capacity or speed based on level of exposure.

The results of the disruption analysis can also be exported for use outside of the RDR Tool Suite to inform regional analyses of access disruption to other transportation modes (e.g., ports, airports) as well as key regional assets, such as military bases, hospitals, and utilities.

5.2 RDR Metamodel Parameterization Submodule

5.2.1 Purpose: Calculating Network Performance Under Select Hazard Conditions

RDRM performs a regression to estimate performance across the range of scenario conditions at the daily level. To do this, the RDRM needs a starting set of analyses to parameterize the regressions. The RDRM leverages a sample of core model runs (travel demand model, if user has one, and open-source AequilibraE) to calculate daily network performance under the baseline and select hazard and scenario conditions.

5.2.2 Structure, Functions and Parameters

The metamodel parameterization submodule has two components:

- **Latin Hypercube Sampling:** The Latin Hypercube Sampling selects the array of scenarios from across the scenario space that will provide sufficient data to parameterize the regression model.
- **Core Models:** The core model component runs the scenarios selected by the Latin Hypercube Sampling module so that the outputs can be used in the regression module to estimate network statistics for all scenarios in the scenario space. The core model used to model network effects of scenario uncertainties is AequilibraE.

The following subsections detail the functions and parameters for each of these components.

5.2.2.1 Latin Hypercube Sampling

The Latin hypercube sampling (LHS) module is used to select the set of core model runs that populate the regression model in the RDRM. The module chooses scenarios randomly and runs a series of coverage tests to make sure the selected set provides adequate data to fully fit a regression model for the entire scenario space. The module outputs a set of core model runs to pass to the RDRM based on the scenario space and the regression model to be used.

The full universe of parameter combinations depends on the agency's decisions about which uncertainties to include, which subsequently define the scenario space. For example, a set of inputs to the RDRM defined on the given uncertainties might include:

- Socio economic – 1 level
- Project group – 3 levels
- Resilience investment – 2 levels
- Elasticity – 3 levels
- Hazard – 3 levels
- Recovery – 6 levels

$$1 \times 3 \times 2 \times 3 \times 3 \times 6 = 324 \text{ combinations of 'scenario factors'}$$

The LHS module enumerates the full scenario space specified by the analyst and randomly selects a number of combinations of 'scenario factors' (a parameter chosen by the analyst) such that all levels of each uncertainty are represented by a core model run. Based on the regression model selected,

additional coverage checks are run to ensure the regression can be expanded to cover the full scenario space (e.g., full coverage of hazard-recovery interactions if the regression model with interactions is chosen). While the LHS module will confirm the RDRM can construct a full regression model based on selected set of core model runs, the performance and model fit of the RDRM improves as the analyst opts to select more core model runs. Practically, the number of core model runs is limited by the computationally intensive nature of those model runs. In the above example, an analyst may only be able to conduct 75 core model runs in a reasonable amount of time using AequilibraE. The RDRM regression can then use those as ‘training data’ and provide interpolated outputs for the remaining 249 combinations of scenario factors.

The LHS module provides additional functionality to the analyst for supplementing an existing set of selected core model runs with additional samples for use in the modified scenario space. This allows for reuse of previously-run core model runs, as long as core model inputs have not been changed from what was used in the previous run (e.g., routable network, trip tables). To supplement an existing set of core model runs, the LHS module follows these steps:

1. The modified scenario space is constructed.
2. Existing core model runs are filtered to only those that still fall within the modified scenario space.
3. A new random set of core model runs is chosen to reach the number of core model samples the analyst wishes to select for the modified scenario space. A buffer of additional runs is included in this initial selection of new core model runs to account for potential overlap with the existing set.
4. The existing and new sets are combined, with duplicate selections removed. The final set is down-selected to only the number of combinations indicated by the analyst.
5. The full selection is checked for coverage of the scenario space based on the regression model to be used.

5.2.2.2 *Travel Demand Model (TDM) Core Model*

Many transportation planning organizations use travel demand models (TDM) to analyze flows within the planning region. The RDR Tool Suite can take in information from standard TDM analyses as the primary core model that forms the basis for the metamodeling.

A classic “four-step” TDM uses the following four steps: trip generation, trip distribution, mode choice, and vehicle routing.

Normally, a TDM is calibrated and validated for current conditions by comparing TDM outputs with observed conditions. The TDM is then applied to future year scenarios. These scenarios may include changes in:

- Population, employment and the resulting changes in trip generation and distribution.
- Infrastructure changes (e.g., planned highway or transit expansions).

These are steady-state futures, as distinguished from hazard scenarios, which disrupt the network for a specific time period as described in Section 5.2.2.5.

Trip generation primarily depends on land use data at the TAZ level. Land use data includes population, employment, and other socio-economic data, including household car ownership (which acts as a proxy

for vehicle access) and income. Trips are categorized by various purposes (home-based work, home-based shop, home-based social/recreation, home-based other, nonhome-based work, and nonhome-based other). Trips are then produced (origin) or attracted (destination) depending on established trip production and attraction rates. Home-based work attractions can be adjusted based on an accessibility factor for each zone.³⁸ That is, if a zone is highly accessible (high employment combined with low travel times to other zones), then the attractions are adjusted upwards. There are separate procedures for dealing with external trips and with truck trips. Outputs of the trip generation step include trip-ends for both peak and off-peak periods.

Trip distribution estimates the number of trips that will travel between each TAZ pair. A gravity model uses a generalized cost (i.e., travel time plus toll/value-of-time) between zone pairs to estimate the distribution of trips. These costs are estimated based on congested highway travel times from a previous model run. The output of trip distribution is a set of person trip tables, typically organized as productions (e.g., home) and attractions (e.g., work, shopping). They may be further organized by peak and off-peak periods, divided by trip purpose and auto availability. Table 5-2 lists the trip tables for the peak period set for an example based on an existing sample TDM.

RDR typically obtains trips in one of two ways. First, if an MPO has a trip-based demand model, trips are obtained from the output of the trip distribution step. If the MPO has an activity-based model, trips are usually available as an intermediate output. Second, if there is no MPO model (e.g., RDR is being run for a rural area), RDR provides helper tools to synthesize a set of trips based on population and employment data within Census block groups in the study area, plus traffic counts on major roads entering the study area.

To be used by the RDR Tool Suite, these trips tables must, first, be converted from production-attraction to origin-destination, and second, be aggregated to daily person trips. The RDR Tool Suite accepts as input origin-destination trip tables disaggregated by auto availability (households with cars vs. households without cars), though still at the level of daily person trips aggregated across trip purpose.

Table 5-2 Trip distribution outputs

Table name	Trip purpose	Auto Availability
hbo_0pk	Home based other (includes social/recreation)	0
hbo_1pk	Home based other (includes social/recreation)	1 or more
hbs_0pk	Home based shopping	0
hbs_1pk	Home based shopping	1 or more
hbw_0pk	Home based work	0
hbw_1pk	Home based work	1 or more
nhb_pk	Non home based	any

Mode choice allocates trips between auto and transit modes. It allocates the trips to time periods. Four periods each day (AM, PM, midday, and night) are often used.

³⁸ One example of a TAZ accessibility factor calculation is on page 41 of the Hampton Roads Model Methodology Report Version 1.0. $Accessibility(zone\ i) = \sum_j \frac{Employment(zone\ j)}{travel\ time_{ij}^2}$

Finally, **vehicle routing** puts the passenger vehicle and freight truck trips onto the road network taking speed and capacity into account, and finding a user equilibrium routing where no user can improve their travel time by changing routes. The TDM outputs include PHT and PMT, VHT and VMT, and the number of trips for the scenario and future analyzed. Inputs and outputs to the core model module of the RDR Metamodel are summarized in Table 5-3, below.

The vehicle routing component of a core model requires a routable network (nodes and links) as an input. The RDR Tool Suite is designed to work with the nodes and links in a General Modeling Network Specification (GMNS) network.³⁹ GMNS is an open-source network specification, designed to facilitate sharing of networks. The roadway network in RDR includes the following types of links:

- Road centroid connectors, artificial links with high capacity and low speed,⁴⁰ connecting the centroids to the rest of the road network.
- Road links, including streets of various functional classes, typically including freeways, arterials, and collectors.

For RDR scenarios that use transit, the following types of links are required in the network:

- Transit centroid connectors, artificial links with high capacity and low (typically, walking) speed, connecting the centroids to the rest of the transit network.
- Transit boarding links, placed at each transit stop for boarding and de-boarding transit vehicles.
- Transit service links, the segments comprising bus, tram/light rail, and subway/heavy rail routes.
- Transit transfer links, for connecting between routes.

The open-source program, GTFS2GMNS,⁴¹ converts a set of static General Transit Feed Specification (GTFS) files to the appropriate boarding, service, and transfer links. GTFS has emerged as a frequently used (de-facto) standard for sharing transit data in the United States. The RDR Format Network helper tools create transit centroid connectors given a set of TAZ centroids and transit network nodes in GMNS format.

Table 5-3 RDR Metamodel Core Model Inputs and Outputs

Network	Origin-Destination Trips	Origin-Destination Impedances
Baseline network (TDM or AequilibraE)	Trip tables listing number of daily trips between each origin and destination (O-D), optionally broken out by auto availability	Tables (skims) listing impedances (e.g., travel time, generalized travel cost) for each O-D pair
Network with some level of disruption and (optionally) resilience investment (AequilibraE)	Trip tables listing daily trips (with a reduction in trips for those O-D pairs that are affected by the disruption)	Tables (skims) listing impedances (with an increase in impedances for those O-D pairs that are affected by the disruption), which are

³⁹ <https://github.com/zephyr-data-specs/GMNS>

⁴⁰ The combination of travel time and toll on a centroid connector should be set high enough so that the centroid connectors are generally used only where they must be used: at the start or end of a trip. This is why centroid connectors are generally assigned very low speeds.

⁴¹ <https://github.com/VolpeUSDOT/GTFS2GMNS>, forked repository from original code at <https://github.com/asu-trans-ai-lab/GTFS2GMNS> by Arizona State University Transportation AI Lab

Network	Origin-Destination Trips	Origin-Destination Impedances generated in the disruption analysis
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The baseline network and trips should come from the TDM. Since the AequilibraE core model described in the following sections is used as a proxy for the TDM in generating disruption scenario origin-destination impedances, it is recommended that the analyst validate the AequilibraE core model against the TDM for the baseline network. The RDR Tool Suite includes a baseline network run helper tool for this purpose. In addition, if the RDR scenario involves a transit network, the analyst should validate the AequilibraE core model mode share outputs against the TDM to ensure travel time and generalized travel cost inputs have been accurately set. The next few sections describe how TDM files are used as input for AequilibraE, and also the mechanics of how AequilibraE approximates the TDM.

5.2.2.3 *AequilibraE Core Model*

The RDR Tool Suite includes a link to AequilibraE (version 1.6.2), an externally provided open-source shortest path and routing model that may be used to compute network flows and summary statistics for baseline and disrupted networks. In the RDR, AequilibraE is used to quickly determine daily link flows and skims (times and distances between each origin and destination) for both base and disrupted networks. AequilibraE can be used within an RDR analysis to supplement or replace the TDM as a core model.⁴²

While most MPO core routing models route vehicles, the AequilibraE model in RDR routes persons. This is so that person flows on roads and person flows in transit can be routed in the same model.

AequilibraE requires the following inputs:

- An origin-destination trip table file, stored in open matrix (OMX) format. The OMX file must contain a trip table, named 'matrix', for daily person trips. The OMX file may optionally contain two trip tables, named 'matrix' and 'nocar', to distinguish between trips by 1+ car households and 0 car households, respectively. The RDR Tool Suite will automatically create OMX files from provided comma-separated values (CSV) files if provided in the correct format.
- A routable network consisting of node and link tables, which are input as CSV text files, and then stored in an SQLite database.

Up to three sets of core model runs are performed, to provide the needed summary outputs.

First, a baseline run is completed on a non-disrupted network. This model run may be based on either a shortest path or user equilibrium routing calculation. Summary outputs for a day include:

- Total trips
- Total PMT
- Total PHT

⁴² See Appendix B for additional validation of AequilibraE as a core model and Appendix C for use of a general modeling network specification (GMNS) network in AequilibraE.

Second, disrupted runs are completed, on a disrupted network. This run includes adjustments of the number of trips with the same summary outputs.

Finally, disrupted runs with resilience investments may be completed, on a modified disrupted network. For both disrupted runs, additional summary outputs (trips by mode, passenger miles by mode, transit wait time, and time enroute by mode) disaggregating transit and car trips may be generated if specified by the analyst.

5.2.2.4 Base Runs

Baseline network outputs (with no disruption) are drawn from either TDM core model outputs, or from an AequilibraE base run. Figure 5-1 illustrates the data flows. The four step TDM is shown on the left, while AequilibraE is shown on the right. Both produce link flows and times, and O-D flows and times.

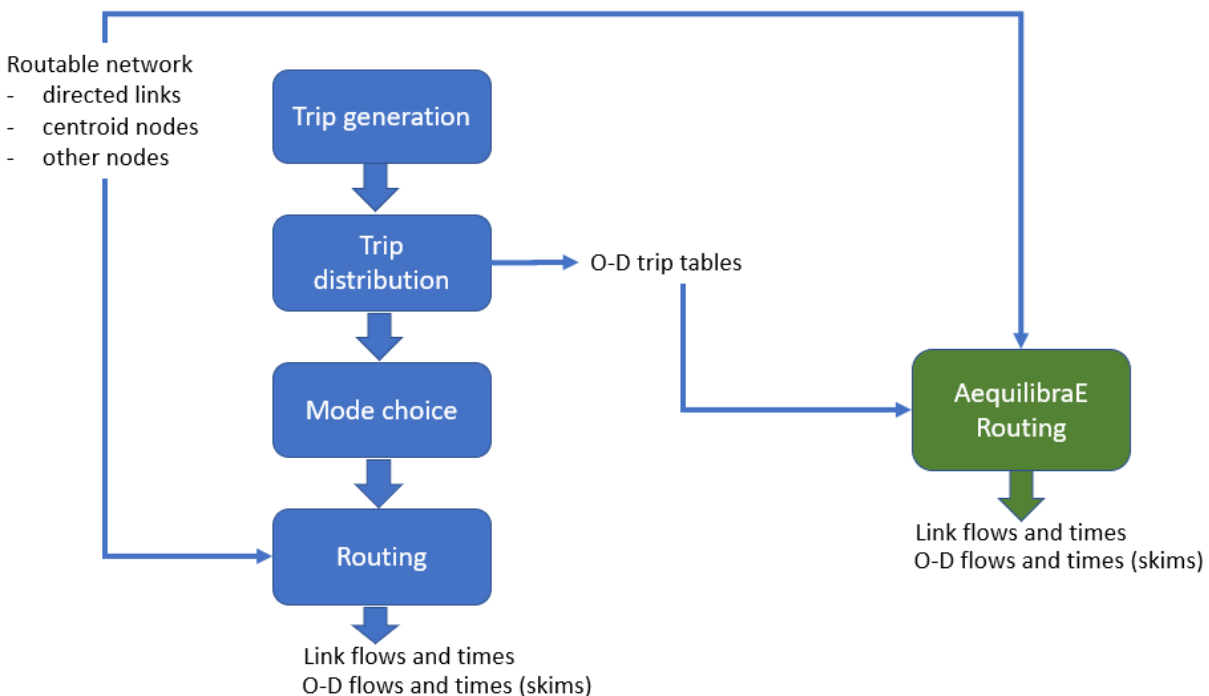


Figure 5-1 Base run data flows

5.2.2.5 Disruption Scenarios using AequilibraE

Once the trip tables and network have been assembled, the hazard disruptions can be applied to produce model outputs for a range of hazard event disruptions and resilience investments. These modifications lead to capacity and trip routing changes in the baseline hazard scenario that alter the generalized cost of travel for particular links in the network. These changes in the generalized cost of travel then impact travel behavior, as modeled in the core model (i.e., AequilibraE), and changes the PHT, PMT, and trips, along with resulting changes to VHT and VMT. The core model (i.e., AequilibraE) models how trip-makers will respond to the change in generalized cost of travel by adjusting whether or not they make trips, and by changing how those trips are routed, which could mean increases in travel time or travel mileage depending on the circuitry of the route under hazard exposure or asset damage. The impact of the resilience investment is quantified as the difference between the PHT, VMT, and trip levels in the baseline hazard scenario compared to the PHT, VMT, and trip levels in the same hazard

scenario when the resilience investment is deployed. The difference in PHT and VMT are monetized using standard DOT values for value of time and vehicle operating costs or using values supplied by the agency (see Section 6.2.2). The trip values are monetized using the difference in trip time multiplied by the generalized cost of those trips in the baseline.

A permanent hazard condition may make certain links in the network and certain TAZs permanently unusable. With these links and TAZs being removed, the transportation system will reach a new equilibrium. Therefore, for these scenarios, it is appropriate to run the entire TDM analysis, considering the broader effects of the permanent hazard on trip-making. Steps are as follows (Table 5-4):

Table 5-4: Modeling the Effects of a Permanent Hazard

What happens:	
1	Identify the scenarios of interest, for example: - Long-term hazard severity (e.g., land subsidence). - Level and type of infrastructure investment.
2	Translate the exposure level at various times in the recovery process to the amount of disruption on previously identified TAZs and links (see Sections 4 and 5).
3	Review and approve the proposed list of TAZ and network changes from step 2 for the model run.
4	Determine how (or whether) to adjust productions/attractions for the other TAZs. If a permanent disruption makes a TAZ much less accessible, it may be worthwhile to reassess the number of generated trips to or from that TAZ.
5	Run the TDM under the new conditions. Outputs from the TDM include - origin-destination trips and impedances (travel times) - total number of trips - Passenger and Vehicle Miles traveled (PMT and VMT) - Passenger and Vehicle Hours traveled (PHT and VHT).
6	Use the results of the core model runs in the RDRM and RDR ROI Analysis Tool.

A temporary event, such as intermittent flooding due to storm surge or precipitation, temporarily removes links and TAZs from the network, but does not lead to a permanent change in trip making. Note that in many cases, the effects of temporary events (e.g., a 100-year flood), will be exacerbated by changes in permanent hazard conditions (e.g., a 3 foot land subsidence).

Recall that a disrupted network state under a hazard condition is characterized by:

- Some set of links disabled (i.e., due to a hazard)
- Some set of origin and destination zones disabled, or have reduced trip-making
- Some set of links with reduced capacity

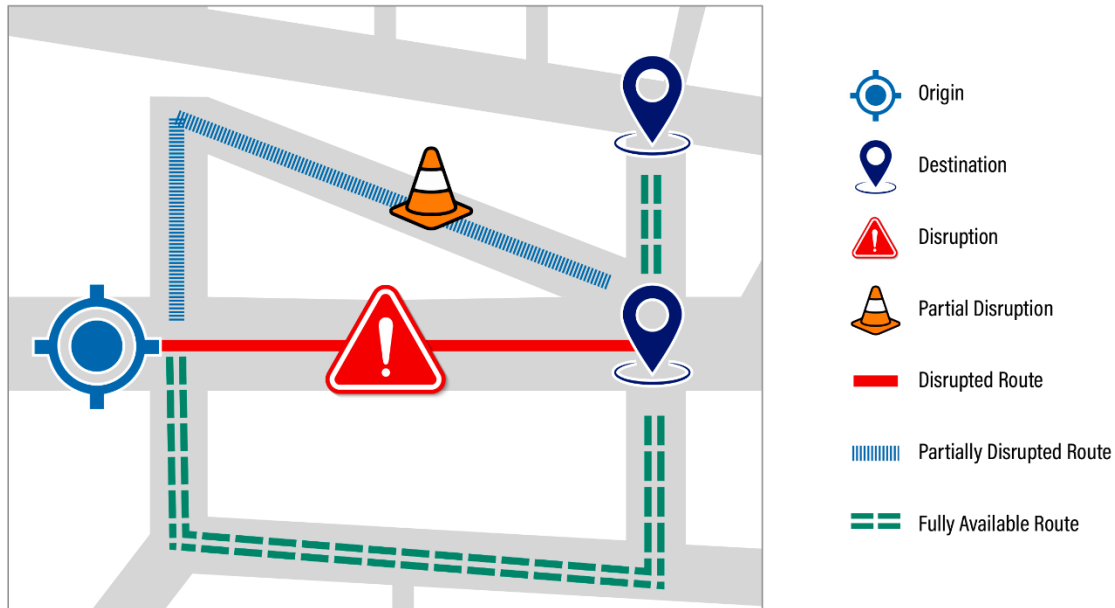


Figure 5-2 A hazard disrupts certain links in the network, leading the core model to identify alternate route(s) around a disruption and calculate the change in trip-making and performance.

The base core model run (TDM or AequilibraE) returns origin-destination impedances, while the disrupted run returns impedances that consider out-of-route travel.

Table 5-5: Modeling the Effects of a Temporary Hazard

What happens:	
1	Identify the scenarios of interest, for example: - Hazard event (see Section 3.2.1) – e.g., storm damage, including recovery steps (e.g., the flooding receding). - Resilience investment options (i.e., infrastructure modifications). - Elasticity of trip demand.
2	Translate the exposure level at various times in the recovery process to amount of disruption on previously identified TAZs and links (see Sections 4 and 5).
3	Adjust trip tables in accordance with the removed / disrupted TAZs as input to the TDM.
4	Recalculate generalized travel costs for the remaining origin-destination pairs. Two approaches: (1) Rerun the user equilibrium routing portion of the TDM. (2) AequilibraE module (See Section 5.2.2.3).
5	Adjust trip tables in accordance with more circuitous travel: - This makes use of the elasticity of trip demand.⁴³ - Alternatively, use a simple cutoff.
6	Rerun traffic assignment to calculate the trips that are lost, as well as the generalized cost of the remaining trips. Again, there are two approaches: (1) Run the assignment portion of the TDM to produce congested travel time and VMT.

⁴³ $\text{new_demand} = \text{old_demand} \times (\text{new_travel_time} / \text{old_travel_time})^{\text{elasticity}}$ where elasticity ≤ 0 .

What happens:

Outputs are reported for each time period (typically AM, PM, midday, night), and will include:

- Trips
- PMT and VMT
- PHT and VHT

(2) Run the AequilibraE routing module on daily person trips. Outputs are reported for the entire day, and include trips, PMT and PHT.

7 Using the congested travel times from the results of step 6, adjust the trip tables again. To lessen the likelihood of an unstable result (that reduces trips too much), the value of elasticity is set to one-half of the original value.

8 Rerun traffic assignment again.

9 Using the results of the core model runs, perform the meta-analysis.

5.2.2.6 AequilibraE outputs: Origin-Destination impedances

Key outputs from AequilibraE include origin-destination impedances, or skims. They are determined either via a routing algorithm (which considers link capacity and congestion) or via shortest path (which only considers link impedance).

After the trips have been routed on the network, the origin-destination impedances are then multiplied by the origin-to-destination flows for each baseline and disruption scenario to calculate overall performance measures, including number of trips, person miles traveled (PMT), and person hours traveled (PHT). Performance measures disaggregated by car and transit mode can also be calculated based on link impedances on different components of the network. Disaggregate performance measures include unlinked trips by mode (car, bus, tram/light rail, subway/heavy rail), passenger miles by mode, transit wait time, and passenger hours in transit.

Finally, the trips, PMT (which are eventually converted to VMT using user-provided occupancy rates computed within the TDM or using federal or state vehicle default occupancy values), and PHT are scaled up from a generic hour to overall daily values in the scenario state.

5.3 RDR Scenario Expansion Submodule

5.3.1 Purpose: Estimating Network Performance Across Hazard Conditions and Resilience Investments

The RDRM uses core model runs to estimate network performance under disruption and estimate performance during hazard recession and repair recovery stages. It then concatenates outputs in terms of daily values into annualized performance based on hazard recovery and repair recovery. The annualized values are then passed to the RDR ROI Analysis Tool. The RDR Metamodel has four modules:

- **Regression:** The RDRM regression module is used to produce outcomes for each scenario in the scenario space. The regression model circumvents needing to run the core model, which is time intensive, for every scenario. Outputs of the regression module are single-day estimates of network performance under each uncertainty scenario.
- **Exposure recovery modeling:** The RDRM Recovery process transforms the single-day estimates of the impact of hazard exposure calculated in the core models and RDRM regression module into realistic multi-day hazard events.

- Repair recovery modeling: The RDRM Recovery process also extends the estimates of the impact of hazard exposure to include impacts on network performance during the repair period incurred by damage on the asset after the end of a hazard event. Benefits of less severe and shorter repair recovery due to a resilience investment mitigation compared to a no-action baseline are quantified to incorporate in the ROI analysis.
- Annualization: The annualization submodule translates the scenario performance outcomes from single hazard events to annual values.

5.3.2 Structure, Functions, and Parameters

5.3.2.1 Regression

The RDRM regression model takes the parameters used in the core model runs and uses the output from those models to estimate the target network performance variables (i.e., number of trips, PHT, PMT) for combinations of parameters not run through the core model. The regression uses the core model scenario performance measures and the scenario space parameters to statistically estimate network performance measures for the full range of the scenario space across hazard severities, recovery stages, economic futures, elasticities, resilience investments, and resilience project groups. Only scenario space parameters (e.g., hazard events, economic futures, elasticities, etc.) with more than one distinct value provided by the analyst are used in the regression model.

The default model is a multitarget Gaussian regression. Four additional models can be selected using the “metamodel_type” parameter in the configuration file. These additional models enable the use of a simple and explainable linear regression, the addition of interaction terms, distinct linear models for project groupings, and mixed-effects modeling. Each of these has different advantages depending on needs and the input data properties:

- ‘multitarget’ (default model) – This regression model jointly infers all dependent variables (trips, PMT, PHT overall and disaggregated by car or transit mode when specified) using the same Gaussian process model.⁴⁴ In other words, the model estimates multiple performance metrics at the same time to take into account their dependence on the same underlying factors. The regression model outputs the inferred mean function value for each scenario in the scenario space. While the model has no explicit functional form (compared to the linear regression models) and relies entirely on fit to sampled data, it has the benefit of inferring all network performance measures together. Testing has shown this approach to outperform the other four models at predicting trips, miles, and hours, when given the same number of LHS samples. The disadvantage of this model is its significantly longer runtime, though the ‘multitarget’ R package used by the RDR Tool Suite enables parallelization so some savings may occur for hardware with more computing power.
 - In the event when there are insufficient core model runs to fit a Gaussian multitarget regression model, or the library finds the variance-covariance matrix is singular, the regression module will revert to the ‘linear’ model described below for inferring each of the dependent variables separately.

⁴⁴ Section 2.2 of <http://gaussianprocess.org/gpml/chapters/RW2.pdf>.

- ‘linear’ – This regression model is the simplest and requires the fewest samples of all of the models, making it applicable in the greatest number of situations. The linear regression model uses each scenario space parameter independently as a decision variable.
- ‘interact’ – This regression model incorporates interaction terms between (a) hazard severities and recovery stages, (b) project groups and resilience investments. The model improves on the ‘linear’ model in cases where the relationship between hazard severities and how each recedes across the network is not uniform, and in cases with several project groups with uneven impacts on network performance. Use of this model requires significantly more core model runs to be sampled by the LHS module than the default linear regression, as every interaction term must be represented in the sample.
- ‘projgroupLM’ – This regression model fits a separate (independent) linear regression model for each project group subset of the scenario space. This model also incorporates interaction terms between hazard severities and recovery stages. As a result, use of this model requires significantly more core model runs to be sampled by the LHS module than the ‘linear’ or ‘interact’ model.
- ‘mixedeffects’ – This model combines fixed effects (project group designation) and random effects (all other scenario space parameters). This model works best in cases where variation between distinct project groups is large compared to variation of resilience projects with a project group. In testing, this model performs second only to ‘multitarget’, but requires significantly more core model samples than the other models (e.g., generally cannot be fit with fewer than 50% sample coverage).

5.3.2.2 Recovery Modeling

In order to assess the full cost of a temporary hazard, and the benefit of resilience investments to mitigate it, it is necessary to model the recovery process. The RDRM Recovery process transforms the single-day estimates of the impact of hazard exposure calculated in the core models and RDRM regression modules into realistic multi-day hazard events. The post-hazard event recovery process is comprised of two stages: exposure recovery (i.e., cessation of hazard condition) and repair recovery. For each scenario identified, the recovery module models the progression of exposure stages from initial hazard exposure to no exposure as well as the progression of asset repair from maximum damage to fully repaired, keeping track of damage repair costs and times throughout. The RDRM uses a set of user-provided input parameters in the configuration file to define the recovery processes that can occur under each scenario:

- Each scenario defines a network state for a single day given by the initial exposure of the hazard event.
- The RDR *exposure recovery* modeling constructs exposure recovery paths that describe how network states progress from initial exposure to the end of the hazard event, the period described as exposure recovery. The set of exposure recovery paths is generated from six parameters in the [recovery] section of the configuration file.
- The RDR *repair recovery* modeling constructs a repair recovery path to describe how assets progress from a state of maximum damage immediately after the end of a hazard event to a fully-repaired state.

The recovery process uses the scenario states defined through the core model runs and the first stage of the metamodel and further defines the scenarios by a recovery profile. The recovery profile defines how long each scenario state lasts and defines the progressing of states from the initial exposure state to the final no-exposure state.

5.3.2.2.1 Metamodel Exposure Recovery

The exposure recovery process works in the following steps. A scenario is chosen that defines the initial conditions of the scenario, such as initial flooding depth of X feet from a 100-year flood. The metamodel generates the network state for the initial condition under X feet of flooding, as well as a series of network states with incrementally less flooding. The exposure recovery path then defines the series of network states that occur after the initial scenario state, along with the number of days each network state lasts. The exposure recovery path is defined by the initial hazard exposure, the duration of the entire hazard event, and the severity and duration of each network state in the series of hazard recession stages. The recovery module generates a range of exposure recovery paths to allow the evaluation of the impact of possible recoveries on potential resilience investments.

The exposure recovery path represents a further uncertainty that is captured by the metamodel which is how the exposure will occur over time. An analyst can define many exposure recovery paths to represent the uncertainty about how recovery will occur.

The four-link example shown above in Figure 5-2 illustrates the framework for modeling of a recovery process. The example includes three zones: A, B, and C, and three hazard scenarios: Large, Medium, and Small disruption, the impacts from which are listed in Table 5-6. The example provides a description of a single exposure recovery path for each of the three hazard scenarios and demonstrates how the hazard severity and exposure recovery interact. The RDRM builds out several potential exposure recovery paths for each hazard scenario in order to evaluate resilience investments across both hazard events and potential recoveries.

Table 5-6 Four-link disruption impacts

	<i>Link 1</i>	<i>Link 2</i>	<i>Link 3</i>	<i>Link 4</i>
<i>Large disruption</i>	<i>4 weeks</i>	<i>1 week</i>	<i>No effect</i>	<i>3 weeks</i>
<i>Medium disruption</i>	<i>3 weeks</i>	<i>No effect</i>	<i>No effect</i>	<i>1 week</i>
<i>Small disruption</i>	<i>1 week</i>	<i>No effect</i>	<i>No effect</i>	<i>No effect</i>

Figure 5-3 illustrates these impacts at weeks 1, 2, 3, 4, and later. Note that there are only 4 unique network states: the baseline with all links available, 1 link disrupted, 2 links disrupted, and 3 links disrupted. The red lines are disrupted links, the brown line is the primary alternate route, and the blue lines represent other links in the network. These unique network states are generated using the core model (full TDM run or shortest path) and/or the first stage of the metamodel (regression) which associates trips, PMT (which are eventually converted to VMT using user-provided occupancy rates computed within the TDM or using federal or state default vehicle occupancy values), and PHT with these four network states. As stated above, the recovery process further defines scenarios by defining the progression of states from an initial exposure condition.

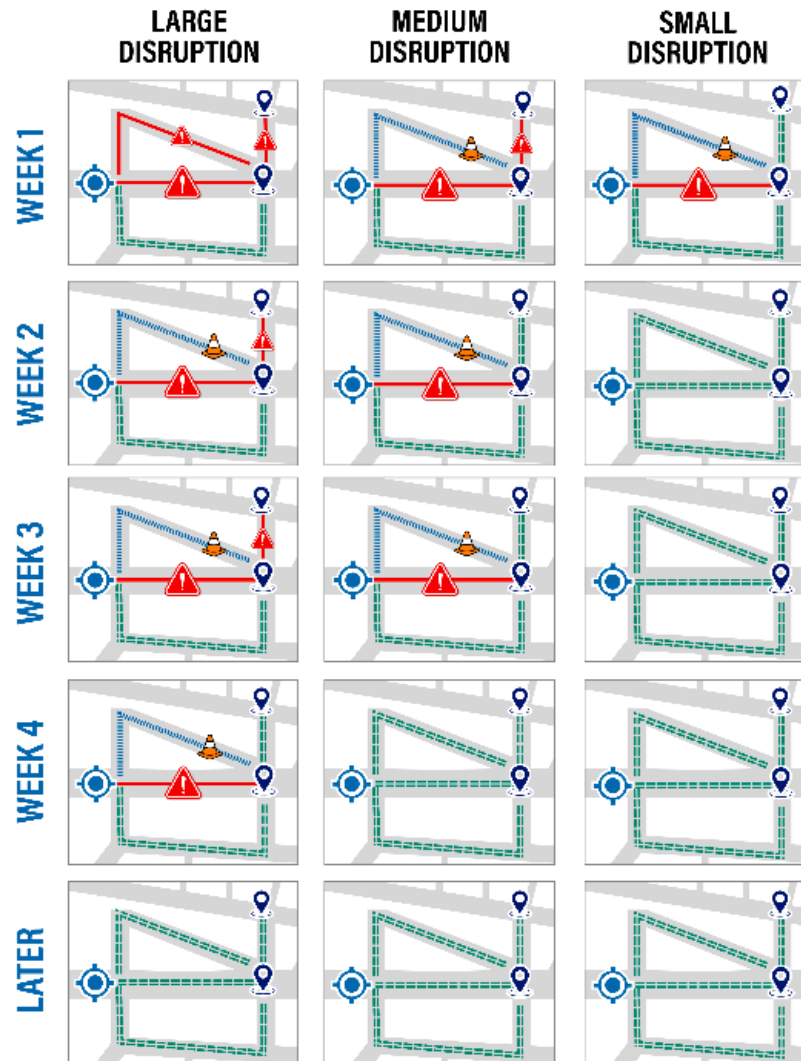


Figure 5-3 Four-link model, impacts, and network states

5.3.2.2.2 Metamodel Damage and Repair Recovery

If the asset damage from a hazard requires repair, a post-hazard damage estimate and repair period disruptions must be included to effectively estimate resilience ROI. The repair recovery process is modeled separately from the exposure recovery process in the metamodel, because repair is an agency decision, unlike exposure recovery which cannot be controlled by the agency. The asset damage costs include the potential disruption due to asset damage after the hazard ends and during the repair/construction period in addition to the material and labor repair/replacement costs. The RDRM currently relies on a fixed model for repair recovery, where the asset is repaired such that capacity increases linearly from maximum damage at the start of the repair recovery period back to full capacity. This model approximates partial repair of asset damage across the repair recovery period; the benefits accrued for the resilience investment compared to the baseline factor in both the reduced damage incurred during the hazard and the shorter repair period. The RDRM Recovery module uses repair time

to expand the number of days of additional disruption, as the core models and RDRM regression module only estimate a single day of travel.

Repair recovery is modeled to occur after the exposure recovery and encompasses the gradual repair of the asset from maximum asset damage to full recovery. For a specific resilience project investment, repair recovery is compared to a baseline with no resilience project investment. The impact of the resilience project investment on repair costs during the repair recovery period is quantified for the affected asset through its lessened damage and shorter repair time, while the state of the rest of the network is assumed to be the same as the baseline.

This process is similar to exposure recovery, although the repair recovery model is not configurable through configuration file parameters—a damage recovery path describes how the asset is repaired through a linear progression of network states until it is fully repaired (see Figure 5-4). Repair recovery is defined by:

- Damage to the asset
 - The level of functionality at given level of damage
 - The minimum time required to restore the asset to complete functionality (with and without resilience investment)
 - The intermediate stages of partial repair with reduced functionality
- Debris cleanup level of effort
 - Minimum time to remove debris
 - May overlap with damage repair

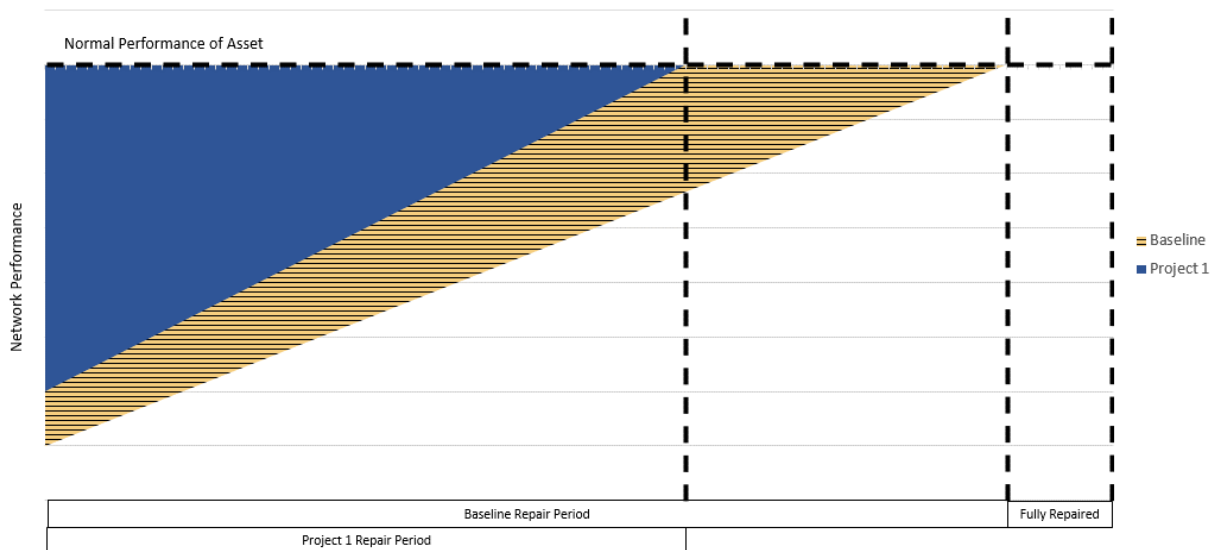


Figure 5-4: Example of Repair Recovery – One resilience project vs. no action baseline showing linear recovery for both scenarios starting at different maximum performance loss and recovering across differing repair recovery periods.

Similarly to the disruption analysis process, the RDR Tool Suite allows for multiple methods of asset damage estimation. For some hazards, standardized exposure-damage relationships exist which can inform damage estimates when faced with a hypothetical future hazard such as:

- Temperature to pavement damage^{45,46}
- Earthquake road-damage scale⁴⁷
- Flooding/inundation depth and damage⁴⁸

The RDRM provides flexibility in calculating asset damage by allowing the analyst to provide their own exposure-damage functions or rely on a set of default options such as the binary damage option or default depth-damage function. The RDRM includes default values for roadway, bridge, and transit assets, based on damage to asphalt concrete roadways from flood inundation, bridge stage-damage curves from a London city-wide analysis, and engineering characterization of rail rapid transit fragility to saltwater flood exposure.^{49, 50} The analyst can adjust the inputs in the configuration file and look-up tables to better model the repair process of their particular use case. Users that select the binned exposure-damage approach can define their own interval bins and mapping relationship to asset damage and should ensure that the values selected correspond to the hazard scenarios that are defined. Similarly, users that select the piecewise linear exposure-damage approach can define their own piecewise linear segments, ensuring that the values selected correspond to the hazard scenarios that are defined.

Once the percent damage is estimated, it is used to set the “no project” baseline repair recovery network state and is monetized using standard per mile costs for regular maintenance such as resurfacing, rehabilitation, and reconstruction (see following subsections and Section 6.2.2.2).

Repair duration is related to the extent of damage and depends on multiple local factors, including the number of repairs needed in the network in response to a given event, asset priority, equipment, personnel and materials availability, and weather conditions (e.g., heavy rains can delay construction if the ground is saturated for several days after the precipitation event).⁵¹ However, a minimum time for repair and/or rebuilding can be used as a lower bound for the exploration of recovery times in the RDRM. A table of the minimum time of repair can be entered as an optional input into the disruption analysis of the RDRM.

⁴⁵ Lu, Wei, Kayser Sascha, and Wellner Frohmut. 2013. Impact of Surface Temperature on Fatigue Damage in Asphalt Pavement. *Journal of Highway and Transportation Research and Development*: 7(3).

<https://doi.org/10.1061/JHTRCQ.0000324>

⁴⁶Maaty, Ahmed. 2017. Temperature Change Implications for Flexible Pavement Performance and Life. *International Journal of Transportation Engineering and Technology*: 3(1):1. DOI:10.11648/j.ijtet.20170301.11.

⁴⁷ Panjamani, Anbazhagan, Sushma Srinivas, and Deepu Chandran. 2011. Classification of road damage due to earthquakes. *Natural Hazards*: 60(2). DOI:10.1007/s11069-011-0025-0

⁴⁸ Huizinga, J., De Moel, H. and Szweczyk, W., *Global flood depth-damage functions: Methodology and the database with guidelines*, EUR 28552 EN, Publications Office of the European Union, Luxembourg, 2017, ISBN 978-92-79-67781-6, doi:10.2760/16510, JRC105688.

⁴⁹ Department of Civil and Environmental Engineering University of Western Ontario (2011). *The City of London: Vulnerability of Infrastructure to Climate Change*. https://www.london.ca/residents/Environment/Climate-Change/Documents/Final%20Report%20Vulnerability%20of%20City%20of%20London%20Climate%20Change%20August%202011_with%20Appendices_Dec%201%202011.pdf

⁵⁰ Martello, M.V., Whittle, A.J., Lyons-Galante, H.R. Depth-damage curves for rail rapid transit infrastructure. *Journal of Flood Risk Management*. <https://doi.org/10.1111/jfr3.12856>

⁵¹ VDOT. 2019. Road Building: Frequently Asked Questions. Accessed on April 20, 2020 from <https://www.virginiadot.org/projects/faq-road-built.asp>.

For a specific resilience project investment, damage recovery is compared to a baseline with no resilience project investment. The impact of the resilience project investment on repair costs during the damage recovery period is quantified for the affected asset, while the state of the rest of the network is assumed to be the same as the baseline. Damage to the asset associated with the resilience project is calculated for both the baseline case and the resilience investment case using the following steps:

1. Network links comprising the asset are identified using the `project_table.csv` input file.
2. Exposure to the hazard event is identified for each network link using the Exposure Analysis Tool or externally developed exposure analyses.
3. Damage to each network link is calculated according to a user-specified depth-damage approach in the configuration file: 'Binary', 'Default_Damage_Table', 'Manual_Bins', and 'Manual_Linear'.
4. The resilience investment hazard disruption mitigation, either complete or link-by-link partial mitigation, is considered in the asset damage calculation step.
5. Minimum repair duration for each network link is calculated. A default look-up table is provided, and the analyst may also provide their own minimum repair duration by setting the configuration file parameter `repair_time_approach` to 'User-Defined'. The default look-up table is based on input from Virginia DOT (VDOT) and a Cambridge Systematics report on Hurricane Sandy among other sources, and the values for each asset type can be found in the sections below. Repair durations are scaled by damage percent.

Once minimum repair durations have been calculated for both the baseline and resilience project investment cases at the network link level, the damage recovery path is constructed to approximate the gradual repair of the asset. The baseline case without resilience investment will have greater maximum damage and a longer time of repair for the specified asset. The resilience investment case has lower maximum damage, thus starting the repair period at a better network performance, and also requiring a shorter repair recovery time, i.e., reaching normal performance earlier. For both cases, the asset associated with the resilience project starts at its maximum damage at the beginning of the repair recovery period and is repaired such that capacity recovers linearly to normal performance across the repair recovery time. The difference in the network performance curves between the baseline case and the resilience investment case across the total repair recovery period is attributed to benefits from the resilience investment and passed to the ROI analysis.

5.3.2.2.3 Roadway Damage Cost and Time to Repair

The parameter `repair_cost_approach` in the configuration file can take two values: 'Default' and 'User-Defined'. Regardless of approach taken, repair costs are scaled by the damage percent incurred on the asset based on level of hazard exposure.

Existing roadway build cost estimates provide a starting point for calculations if local/regional estimates are not readily available. The RDR Tool Suite includes a default look-up table of with roadway damage costs based a schedule of typical costs per lane mile for repair and reconstruction actions for the Highway Economic Requirements System (HERS) model used for the 25th edition of the Conditions and Performance report, which is maintained and updated regularly by FHWA. HERS is the agency's BCA tool used in long-range funding decision making and congressionally mandated reporting.⁵² The HERS model

⁵² U.S. DOT, FHWA and FTA. (2024). "Status of the Nation's Highways, Bridges, and Transit: Conditions & Performance Report to Congress, 25th Edition." Accessed on December 16, 2024 from https://www.fhwa.dot.gov/policy/25cpr/pdf/CP25_Full_Report.pdf

uses standard values for reconstructing and widening a lane, reconstructing an existing lane, resurfacing and widening a lane, resurfacing existing shoulder, improving shoulder, adding a lane, and new alignment for different categories of roadway including functional class, population area size for urban areas, and geographic terrain (e.g., flat, rolling, and mountainous) for rural areas.

The default roadway damage costs (Figure 5-5) are measured per lane-mile and are estimated as the cost to reconstruct an existing lane. Damage costs are broken down by functional class of the roadway, the RDR default facility type for road network links. The costs in the report are given in units of 2018 dollars, which were converted to 2024 dollars for use in the RDRM using the National Highway Construction Cost Index (NHCCI).

Asset Type	Network Type	Facility Name	Facility Type	Unit	Damage Repair Cost	Total Repair Cost
Highway	Rural Flat	Interstate (HERS)		1 Lane-mile	2270656.963	2270656.963
Highway	Rural Flat	Other Freeway and Expressway (HERS)		2 Lane-mile	2141992.442	2141992.442
Highway	Rural Flat	Other Principal Arterial (HERS)		3 Lane-mile	2016952.274	2016952.274
Highway	Rural Flat	Minor Arterial (HERS)		4 Lane-mile	1837546.816	1837546.816
Highway	Rural Flat	Major Collector (HERS)		5 Lane-mile	1643643.947	1643643.947
Highway	Rural Flat	Minor Collector (HERS)		6 Lane-mile	1471487.194	1471487.194
Highway	Rural Flat	Local (HERS)		7 Lane-mile	1346447.026	1346447.026
Highway	Rural Rolling	Interstate (HERS)		1 Lane-mile	2685645.346	2685645.346
Highway	Rural Rolling	Other Freeway and Expressway (HERS)		2 Lane-mile	2529798.181	2529798.181
Highway	Rural Rolling	Other Principal Arterial (HERS)		3 Lane-mile	2381199.72	2381199.72
Highway	Rural Rolling	Minor Arterial (HERS)		4 Lane-mile	2169175.087	2169175.087
Highway	Rural Rolling	Major Collector (HERS)		5 Lane-mile	1940840.868	1940840.868
Highway	Rural Rolling	Minor Collector (HERS)		6 Lane-mile	1736064.941	1736064.941
Highway	Rural Rolling	Local (HERS)		7 Lane-mile	1585654.304	1585654.304
Highway	Rural Mountainous	Interstate (HERS)		1 Lane-mile	3405079.356	3405079.356
Highway	Rural Mountainous	Other Freeway and Expressway (HERS)		2 Lane-mile	3211176.487	3211176.487
Highway	Rural Mountainous	Other Principal Arterial (HERS)		3 Lane-mile	3022710.147	3022710.147
Highway	Rural Mountainous	Minor Arterial (HERS)		4 Lane-mile	2749071.519	2749071.519
Highway	Rural Mountainous	Major Collector (HERS)		5 Lane-mile	2459123.303	2459123.303
Highway	Rural Mountainous	Minor Collector (HERS)		6 Lane-mile	2199982.085	2199982.085
Highway	Rural Mountainous	Local (HERS)		7 Lane-mile	2002454.864	2002454.864
Highway	Small Urban	Interstate (HERS)		1 Lane-mile	3609855.283	3609855.283
Highway	Small Urban	Other Freeway and Expressway (HERS)		2 Lane-mile	3405079.356	3405079.356
Highway	Small Urban	Other Principal Arterial (HERS)		3 Lane-mile	3211176.487	3211176.487
Highway	Small Urban	Minor Arterial (HERS)		4 Lane-mile	2875923.863	2875923.863
Highway	Small Urban	Major Collector (HERS)		5 Lane-mile	2511676.417	2511676.417
Highway	Small Urban	Minor Collector (HERS)		6 Lane-mile	2228976.907	2228976.907
Highway	Small Urban	Local (HERS)		7 Lane-mile	2038698.39	2038698.39
Highway	Small Urbanized	Interstate (HERS)		1 Lane-mile	4124513.366	4124513.366
Highway	Small Urbanized	Other Freeway and Expressway (HERS)		2 Lane-mile	3890742.617	3890742.617
Highway	Small Urbanized	Other Principal Arterial (HERS)		3 Lane-mile	3664220.574	3664220.574
Highway	Small Urbanized	Minor Arterial (HERS)		4 Lane-mile	3274602.659	3274602.659
Highway	Small Urbanized	Major Collector (HERS)		5 Lane-mile	2857802.099	2857802.099
Highway	Small Urbanized	Minor Collector (HERS)		6 Lane-mile	2524361.652	2524361.652

Figure 5-5: Default repair damage cost table (from FHWA HERS model)

The RDRM also allows the analyst to provide their own look-up table for damage costs across roadways and bridges, as there are a number of resources that estimate these values nationally and regionally. The American Road Transportation and Builders Association (ARTBA)⁵³ estimates \$2-3 million per mile to construct a new 2-lane undivided road in a rural area, which goes up to \$3-5 million in urban areas, and

⁵³ <https://www.artba.org/about/fag/>. Cost to build a mile of road is included under “Funding, Financing & Costs”.

costs go up as the number of lanes increase. ARTBA estimates milling and resurfacing a 4-lane road to cost \$1.25 million per mile. Arkansas,⁵⁴ Florida,⁵⁵ and Virginia have published cost per mile estimates for road work. Both Arkansas and Florida provide the costs per mile of roadway construction for roads and bridges across a variety of functional classes (e.g., highways, non-highways, etc.) and a variety of construction or improvement types (e.g., reconstruction, new roads, etc.), whereas Virginia provides roadway construction cost for a range of functional classes. These examples can be used by an analyst as a reasonable default if they do not have locally relevant data. For agencies to provide their own damage cost look-up table, the parameter `repair_cost_approach` should be set to 'User-Defined' in the configuration file, and the structure of the provided table must mirror the default, breaking out the damage repair cost by facility type and in units of lane-mile.

The parameter `repair_time_approach` can take two values: 'Default' and 'User-Defined'. Minimum repair duration for road debris removal and repair can be estimated from past experience or from existing datasets and may also be classified by roadway functional class and improvement type. The RDRM default values refer to data provided by Virginia DOT, for which repair times for a rural secondary project can take between 1 to 8 weeks, repair times for a primary 4 lane road can take up to 4 months, and repair times for an interstate can take a day or weeks. If agencies provide their own repair time look-up table, the repair durations should be broken out by facility type. Repair times are also scaled by damage percent.

5.3.2.2.4 Bridge Damage Cost and Time to Repair

As with roadway assets, the RDRM allows an analyst to use default values for bridge damage cost and time to repair or provide their own values. Bridge costs and times are stored in the same look-up tables as roadway costs and times. The configuration file is used to specify whether the 'Default' or 'User-Defined' look-up table is used by the tool. The RDRM uses the FHWA estimate for bridge replacement costs⁵⁶ in the default look-up table for bridge damage cost, in 2024 dollars, and the VDOT estimates for bridge repair times.

In the FHWA-maintained annual collation of bridge replacement costs,⁵⁷ in 2024 the average national cost of bridge replacement was \$429 per square foot for structurally deficient National Highway System bridges, and \$383 per square foot for structurally deficient non-national highway system bridges.⁵⁸ FHWA regularly updates these values, and they can provide a basis for an initial evaluation of potential cost of repair related to hazard disruption. However, these values vary by location and also depend on what elements are included in the estimate. Based on VDOT estimates, bridge replacement averaged \$850 per square foot in 2019 for new bridge deck area including all phases of the development and construction, as well as associated non-bridge construction.⁵⁹ For analysts to replace the default repair cost with their own values, the parameter `repair_cost_approach` should be set to 'User-Defined' in the configuration file, and the structure of the provided table must mirror the default, breaking out the damage repair cost by "Bridge" asset type and in units of per square foot.

⁵⁴ ARDOT. 2020. Estimated Costs per Mile. <https://www.ardot.gov/wp-content/uploads/2021/11/2020-CPM.pdf>

⁵⁵ FDOT. Costs Per Mile Models for Long Range Estimating. Accessed July 26, 2020 from <https://www.fdot.gov/programmanagement/estimates/lre/costpermilemodels/cpmsummary.shtm>.

⁵⁶ FHWA. 2024. Bridge Replacement Unit Costs. <https://www.fhwa.dot.gov/bridge/nbi/sd2024.cfm>

⁵⁷ FHWA. 2024. Bridge Replacement Unit Costs. <https://www.fhwa.dot.gov/bridge/nbi/sd2024.cfm>

⁵⁸ FHWA. 2024. Bridge Replacement Unit Costs. <https://www.fhwa.dot.gov/bridge/nbi/sd2024.cfm>

⁵⁹ Eric Stringfield, VDOT, personal communication, 30 March 2020 to Dale Stith, Hampton Roads TPO.

The actual physical replacement of a bridge can take six months or more. For example, VDOT provided a rough estimate for new/replacement bridge construction on an emergency basis at:

- Bridges shorter than 20': 4 to 6 months
- Bridges greater than 20' and less than 50': 6 to 8 months
- Bridges greater than 50' and less than 100': 8 to 12 months
- Bridges greater than 100': 12 months to 18 months⁶⁰

VDOT indicated that these would take longer on a non-emergency basis. Accelerated Bridge Construction (ABC)⁶¹ components (prefabrication of specific bridge elements) and techniques for installing those prefabricated components have reduced the time period of bridge replacement, sometimes by up to 50 percent,⁶² and in some cases compressing the actual replacement down to 24 hours to 2 weeks.⁶³

If agencies use their own cost estimates, the bridge repair cost must be provided in units of dollars per square foot, and the bridge repair times broken out by length of the bridge.

5.3.2.2.5 Transit Damage Cost and Time to Repair

The RDRM allows an analyst to use default values for transit damage cost and time to repair or provide their own values. Transit costs and times are stored in the same look-up tables as roadway and bridge costs and times. The configuration file is used to specify whether the 'Default' or 'User-Defined' look-up table is used by the tool. The tool provides a general default approach for continuous estimates of the exposure-damage relationship for flooding inundation on rail transit infrastructure based on synthetic curves generated with engineering expertise and validated in the literature.⁶⁴ For repair cost and time, bus transit links are assumed to run on existing roadway links; no additional repair cost is included in the default repair cost table, and the repair time is assumed to be the same as higher functional class roadway segments. Transit assets have default repair cost values broken down by light rail, heavy rail,

⁶⁰ Eric Stringfield, VDOT, personal communication, 30 March 2020 to Dale Stith, Hampton Roads TPO.

⁶¹ FHWA. 2011. Accelerated Bridge Construction. Publication no. HIF-12-013.

<https://www.fhwa.dot.gov/bridge/abc/docs/abcmanual.pdf>

⁶² Salem, Ossama, Baris Salman, Sudipta Ghorai. Accelerating construction of roadway bridges using alternative techniques and procurement methods. Transport 33(2): 567-579.

<https://journals.vgtu.lt/index.php/Transport/article/view/160/129>

⁶³ FHWA. 2011. Accelerated Bridge Construction. Publication no. HIF-12-013.

<https://www.fhwa.dot.gov/bridge/abc/docs/abcmanual.pdf>

⁶⁴ Existing studies of the depth-damage relationship for flood-related damages to transit rail infrastructure are limited. One study developed depth-damage functions that may be useful for users to consider, but users should be careful about applying these in practice without additional engineering consideration.

Martello, Michael V., Andrew J. Whittle, and Hannah R. Lyons-Galante. September 12, 2022. Depth-damage curves for rail rapid transit infrastructure. Journal of Flood Risk Management. Volume 16. Issue 1, e12856. Accessed from: <https://doi.org/10.1111/jfr3.12856/>

and subway, derived from cost estimate reports on reconstruction efforts on railway assets around New York City in the aftermath of Hurricane Sandy.⁶⁵

There are many other sources for estimates of transit damage cost. A discrete threshold approach can be used for estimating transit repair costs based on a recent analysis that suggests rail can continue operation as long as the floodwater is below the rail head. Once the water reaches the bottom of the rail head (assumed 4 inches), then the train speed reduces to 5 mph. If the water is above the rail head (assumed 5 inches) or if the water is moving with any velocity (such as storm surge), rail service can be assumed to be halted because the ballast may have been removed or the rail embankment affected, placing the safety of the track in question. The user can incorporate this data into the RDRM through a user-specified exposure-damage look-up table. To date, a threshold of flood damage has not been located that would suggest full replacement. However, if the user wants to estimate the cost of reconstruction,⁶⁶ incremental damage costs are estimated from the following sources and can be incorporated into the repair cost table (converted to per lane-mile costs and inflated to FY2023 dollars):

- The default for the unit costs of a ballast repair is estimated to be \$13.25 (FY2011)/ft.
- A U.S. Army Corps of Engineers study⁶⁷ suggests a repair cost of \$2,487 (FY2011)/ft of track to replace a section of track and embankment (that is assumed to be needed when the track experiences embankment scour). This repair cost estimate is used in the default repair cost table, converted to FY2024 dollars using the NHCCI. This estimate is based on the cost of new materials, the cost for labor and machines at overtime rates, and mobilization costs. The actual cost depends on many factors such as width and depth of erosion.
- Using track replacement costs for approximately ¾ mile of inundated Amtrak tunnel in the aftermath of Hurricane Sandy,⁶⁸ a default cost per subway track-mile was developed. Taking the lowest cost estimate for one of the two fully inundated East River tubes, a conservative figure of \$59.333 million in 2014 was calculated, which is \$75.353 million in 2023, only adjusting for inflation.

Based on these cost estimates, it is assumed that 4 days are required to repair scour of the embankment and 1 day to repair ballast removal, and that rail operations even under very extreme flood, landslide, and washout conditions are operational within 14 days, though repair may continue for months.^{69,70} The estimated repair time of 5 days to repair ballast and scour the embankment is incorporated into the default repair time table used by the RDRM. Default repair time for heavy rail and subway was

⁶⁵ HNTB (2014). [NYC HNTB Tunnels Assessment Report](https://ded2pkg5zsyd4.cloudfront.net/images/article-0105/NYC-HNTB-tunnels-assessment-report.pdf?v=a502f3f167b6719a21025e61ccec80822b910c72). Retrieved from <https://ded2pkg5zsyd4.cloudfront.net/images/article-0105/NYC-HNTB-tunnels-assessment-report.pdf?v=a502f3f167b6719a21025e61ccec80822b910c72>

⁶⁶ FEMA Hazus-MH, 2010.

⁶⁷ USACE (2011). San Clemente Shoreline Feasibility Study Orange County, California; Cost Engineering Appendix

⁶⁸ HNTB (2014). [NYC HNTB Tunnels Assessment Report](https://ded2pkg5zsyd4.cloudfront.net/images/article-0105/NYC-HNTB-tunnels-assessment-report.pdf?v=a502f3f167b6719a21025e61ccec80822b910c72). Retrieved from <https://ded2pkg5zsyd4.cloudfront.net/images/article-0105/NYC-HNTB-tunnels-assessment-report.pdf?v=a502f3f167b6719a21025e61ccec80822b910c72>

⁶⁹ CSX Customer Advisory- CSX Rail Line in South Carolina Impacted by Flooding. (2015, October 7). Retrieved from <https://www.csx.com/index.cfm/customers/news/service-bulletins1/csx-rail-line-in-south-carolina-impacted-by-flooding/>

⁷⁰ Changnon (2006). From fogs to floods and heat to hurricanes, the impacts of weather and climate on American railroading. AMS.

developed from a report on repair time on rail assets in the New York City area after Hurricane Sandy. Hurricane Sandy restoration of service on transit facilities took between one month to over 14 months, with the 14-month figure not including initial out-of-service time of about one month. For the purposes of the default time values, the lower bound is assumed to be 40 days while the upper bound is assumed to be 460 days, with the average repair time being 250 days.⁷¹

5.3.2.3 *Annualization*

Once the exposure recovery and repair recovery periods are built out for each single-day scenario snapshot in the scenario space, network performance metrics have been computed for each scenario in the full scenario space. The disruption and recovery days are concatenated into a series of days modeling the full hazard impact in a hypothetical year, with the remaining days of the year assumed to be at the baseline performance. Depending on the level of data provided by the user, each scenario (e.g., hazard event) is associated with an annual probability of occurrence from which an expected impact of a resilience investment can be calculated. These annual values are used in the ROI Analysis Tool to estimate performance across the full period of analysis.

⁷¹ Siethoff, et. al. (2017). Post Hurricane Sandy Transportation Resilience Study in NY, NJ, and CT (bts.gov). Retrieved from: <https://rosap.ntl.bts.gov/view/dot/50860>

6 RDR ROI Analysis Tool

6.1 Purpose: Calculating Aggregated Economic Benefits Across Hazard Conditions

The RDR ROI Analysis Tool applies variations of the benefit cost analysis framework to the RDRM outputs to estimate ROI across the range of hazard conditions and other uncertainties. The BCA analysis approaches included in the RDR ROI Analysis Tool are BCA, BCA under Uncertainty with a regret measure (BCA-U/Regret), and break-even analysis. The different approaches allow agencies to conduct an analysis that is consistent with their own decision-making needs and available data. The user specifies which approach is used by the RDR ROI Analysis Tool. A standard BCA is recommended when the probabilities associated with hazards are well known; BCA-U defined by a regret measure is recommended when there is uncertainty associated with the hazard probability; and a break-even analysis is recommended when the designs or costs of potential resilience projects are unknown. This is an advancement from existing approaches to resilience investment planning; the RDRM can facilitate analysis even when data are less reliable or unavailable. Additional information about the ROI approaches can be found in section 1.3.

6.2 Structure, Functions, and Parameters

This section describes how the RDR ROI Analysis Tool including the tool features, inputs, outputs are generated and how to interpret the results. The analytical steps in the RDR ROI Analysis Tool are:

- ROI Analysis
 - Calculate the impact of the project relative to the baseline (e.g., PHT in project scenario minus PHT in the no-project baseline scenario).
 - Apply monetization to the performance metrics (including corresponding safety, noise, and emissions benefits).
 - Apply monetization to trips.
 - Apply discounting.
 - Incorporate estimated cost of repair of damaged infrastructure.
 - Compute resilience project costs and residual value across the period of analysis.
 - Compute net benefits (for BCA and BCA-U/Regret) and total benefits (for break-even analysis).
 - Compute regret.
 - Compute project rankings by regret.
- Export results for visualization.

6.2.1 Economic Parameters

6.2.1.1 Defining Economic Parameters

In order to conduct any of the ROI analyses (BCA, BCA-U/Regret, or breakeven analysis), the analyst will need to select the following economic parameters for inclusion in the input files.

Analysis Period: The period of analysis is the time period over which the project benefits accrue. Transportation resilience projects by their nature typically involve large capital investments whose benefits accrue over the service life of the investment.

The user will select the start year of the project; the project is assumed to be completed and begins accruing benefits in that year. The RDR Tool Suite is not a financial analysis tool and does not consider impacts of budget, deployment, or financing alternatives.

The user will also select the analysis period end year. Ideally, all of the potential resilience investments being considered would have the same service life. In practice, this will be rare. Therefore, the analyst should choose a period of analysis that is the least common multiple of the potential projects.⁷² This approach is preferred because it avoids the issue of having to calculate residual value of the projects (though these are calculated by the ROI Analysis Tool if necessary). For example, when considering three different resilience investments with service lives of 5, 25 and 50 years, the least common multiple, and therefore the period of analysis, would be 50 years. However, in practice this may be difficult given the variety of resilience investments that could be made and the challenge of predicting patterns over a long period of analysis. In those cases, the analyst should use their best judgement regarding period of analysis and specify the project lifespan for each of the resilience projects in their analysis. The RDR ROI Analysis Tool uses the project costs and project lifespan to compute the residual value for each project for the specified period of analysis, in addition to the total (discounted) project cost for that period. If redeployment costs are enabled, the project cost is replaced with the specified redeployment cost after the initial project deployment when calculating the residual value. Another optional aspect of the project cost stream, which is not used to calculate residual value, is the annual maintenance cost. If annual maintenance costs are enabled, the cost will be applied annually as part of the project cost stream.

Initial and Future Year Inputs: The steady-state conditions for the initial year and the future year are used to linearly interpolate the performance metrics for the entire period so that there are performance metrics associated with every year of the period of analysis. Given the linear interpolation/extrapolation method, the period of analysis (defined by the start year and end year) need not be between the initial year and the future year; the analyst can use any period starting after the initial year. There will be a initial year result for each hazard event severity and recovery stage, as economic futures and resilience projects will not apply before the period of analysis. There will be a future year result for each scenario in the scenario space, generated by the RDRM.

Inflation Adjustments: The analyst must ensure that all monetized cost and benefit values are expressed in the same dollar year so that a meaningful comparison between benefits and costs is possible. In most cases, analysts will need to convert at least some monetary values from their current dollar year. The US DOT recommends using the Gross Domestic Product (GDP) deflator as the general method of converting nominal dollars into real dollars, based on the recommendations in the OMB Circular A-94⁷³ and OMB Circular A-4.⁷⁴

Time Value of Money/Discounting: Resources at one's disposal today are worth more than having comparable resources available in the future. To account for this, future benefits and costs are discounted to their present values so that nominal levels of costs and benefits can be compared across years and across alternatives that generate different future time patterns of costs and benefits. The concept of discounting is demonstrated in Figure 6-1. The user specifies one discount factor in the

⁷² Wolfram Mathworld. 2022. Least Common Multiple. [online]. Accessed July 26, 2022 from <https://mathworld.wolfram.com/LeastCommonMultiple.html>.

⁷³ OMB Circular A-94 "Guidelines and Discount Rates for Benefit-Cost Analysis of Federal Programs": <https://obamawhitehouse.archives.gov/sites/default/files/omb/assets/a94/a094.pdf>

⁷⁴ OMB Circular A-4 "Regulatory Impact Analysis: A Primer": https://www.reginfo.gov/public/jsp/Utilities/circular-a-4_regulatory-impact-analysis-a-primer.pdf

configuration file for RDRM to use during the linear interpolation process for general discounting of benefits and costs.

Based on the December 2025 update to the US DOT BCA guidance, US DOT recommends analysts use a general discount rate of 7 percent per year. Per US DOT BCA guidance, the discount rate reflects the principle that benefits and costs that occur sooner in time are more highly valued than those that occur in the more distant future, and that there is thus a cost associated with diverting the resources needed for an investment from other productive uses in the future.

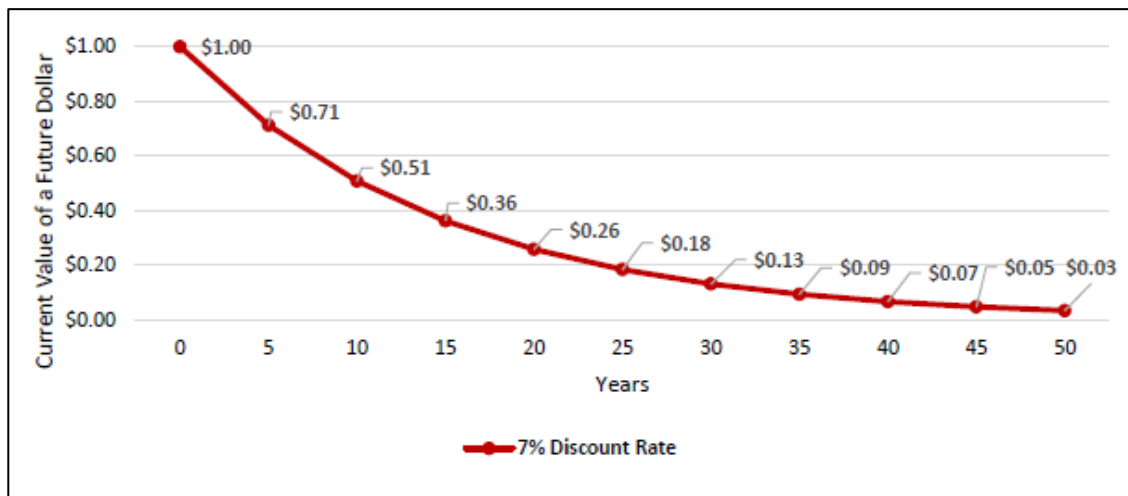


Figure 6-1: Demonstration of discounting⁷⁵

6.2.2 Monetization

The RDR ROI Analysis Tool computes the benefits of each project alternative in each scenario by monetizing the changes in transportation performance metrics.

6.2.2.1 Network Performance Benefits

In order to compute the performance metrics benefits, the user will need to assign monetary values for value of travel time, vehicle miles traveled for passenger vehicles and transit vehicles, safety, noise, air pollution emissions, lost trips, asset damage repair, and asset cleanup.

The RDR Tool Suite uses a set of vehicle occupancy rates for passenger vehicles and transit vehicles to convert RDRM outputs from passenger miles to vehicle miles. The default value for passenger vehicles is the US DOT-specified value, and the default values for buses, light rail, and heavy rail come from the FTA 2024 National Transit Summaries and Trends report Exhibit 16.1 column “Average Occupancy (PMT/VRM)”.⁷⁶

⁷⁵ FHWA. (January 6, 2023). “Benefit-Cost Analysis Guidance for Discretionary Grant Programs.” Accessed on January 6, 2023 from <https://www.transportation.gov/office-policy/transportation-policy/benefit-cost-analysis-guidance-discretionary-grant-programs-0>

⁷⁶ FTA. (2026). “National Transit Summaries and Trends: 2024 Edition” Accessed on September 30 2026 from https://www.transit.dot.gov/sites/fta.dot.gov/files/2026-04/2024%20National%20Transit%20Summaries%20and%20Trends_1.2.pdf

As a default, the RDR Tool Suite uses US DOT-specified values for travel time measured as vehicle hours traveled (VHT) for passengers using passenger vehicles or transit vehicles, value of time for waiting for transit users, operating costs for passenger vehicles miles traveled (VMT), and safety, noise, and emissions per mile costs. The operating costs per mile for bus, light rail, and heavy rail come from the FTA 2024 National Transit Summaries and Trends report Exhibit 16.1, using “Operating Expenses (Millions)” divided by “VRM (Millions)”, adjusted for inflation. The user can adjust these to alternative monetary values if preferred (e.g., to use local values).

US DOT does not have a standard method for monetizing lost trips, trips that are foregone because the path is not traversable or if the trip time makes the trip infeasible. Instead of a fixed value, the RDR ROI Analysis Tool uses the economic theory concept of consumer surplus to value lost trips. The trip values are computed using the implied value of trips based on the trip elasticity value and the network equilibrium generated in the core model. Essentially, the value of the trip is assumed to be the cost of trip in travel time and operating costs. A formal explanation of this approach is provided in Appendix D: Trip Loss Valuation. If transit-specific network metrics are recorded as specified by the user, lost transit trips are additionally monetized using a user-defined transit fare parameter in the configuration file.

Trip demand elasticity is an economic concept that relates the cost in travel time for a trip made on an asset to the number of trips that will be made on that asset in an equilibrium. A value of 0 implies that the trip will continue to be made if possible, even if the travel time becomes much longer. A value of -1 indicates that if the travel time between two zones doubles, the number of trips will be cut in half. Although there is little if any published literature on the elasticity of travel in response to a disruption, there is literature on steady state elasticities, assembled in VTPI’s 2019 report on elasticities.⁷⁷ This report suggested short term elasticities in the range of -0.5 (more-or-less) and long-term elasticities in the range of -1 .

Currently, the RDRM approach includes the following simplifying assumptions:

- treats all vehicle travel as passenger travel, and
- converts tolls and similar use fees to travel time equivalents using the value of time.

The current version of the tool incorporates basic consideration of transit and trips by zero-car households, as described in the Routable Network Preparation section of the User Guide. To summarize, transit links incorporate wait time, travel time, and fare for all households. Road links have low costs for 1+ car households (the travel time and the occasional toll). For 0-car households, non-transit road links have higher costs (the wait time for a taxi/TNC, the travel time, and the fare).

6.2.2.2 Damage Repair and Cleanup Cost Savings

The tool uses default repair cost and time based on the asset type and also depending on the percent of asset damage as described in Section 5.3.2.2.2. Due to the limitations of the model, the user cannot make trade-offs between asset repair costs and time. In practice, post-hazard decisions about repair time and cost would be made using optimization that is inappropriate to apply within BCA.

The repair cost is only calculated for the assets on which the project is deployed. The ROI analysis only considers the marginal impact of the project as the damage to other assets occurs in both the project

⁷⁷ Litman, Todd. July 28, 2026. “Understanding Transport Demands and Elasticities: How Prices and Other Factors Affect Travel Behavior.” Victoria Transport Policy Institute. vtpi.org/elasticities.pdf, starting on page 48)

scenario and the baseline and therefore cancel each other out. The damage repair and cleanup costs are provided by the user and should align with the hazard event severity. If there is uncertainty about the damage repair and cleanup costs or these costs are naturally stochastic depending on factors beyond hazard severity, users can explore cost uncertainty by developing separate projects that reflect the range of potential damage repair and cleanup mitigation anticipated. In the BCA context, scenarios are defined as the “best guess” about future conditions and outcomes so that users should use the average damage repair and cleanup costs anticipated from the full distribution of potential damage repair and cleanup cost outcomes.

Once network performance benefits (for both the exposure period and the repair period) and damage repair and cleanup cost savings have been monetized for the marginal impact of the resilience project compared to the baseline, benefits across the period of analysis are computed and compared to resilience project costs. Total benefits are the sum of network performance benefits and repair and cleanup cost savings, weighted by risk of each uncertainty scenario and discounted. Net benefits subtract out project cost. Regret metrics are computed by ranking net benefits and assigning an ordinal value within (a) the entire scenario space, (b) each uncertainty scenario, (c) all resilience investments on the same asset.

The results of the ROI analysis provide economic performance metrics for each project over the range of scenarios and conditions considered in the analysis, and over the economic analysis period. The outputs are exported from the RDR ROI Analysis Tool in a comma-separated values file for the user to be able to review, use for analysis, and explore in the visualization module.

6.2.2.3 *External Highway Costs*

The impact of additional or circuitous travel by highway vehicles introduces additional safety, noise, and emissions costs that are not internalized by drivers, which are called external highway costs. Vehicles driving additional miles than they would have otherwise will increase noise on the additional routes, increase crash risk for other drivers, and contribute additional emissions which indirectly impact drivers and non-drivers alike. In the case of safety, the drivers do internalize a portion of the safety risk for driving the additional miles. The model does not have a mechanism for drivers to respond to the additional internalized safety risk, e.g., to incorporate the safety cost into the decision to forego the trip, as that is solely based on the travel time, but the internalized costs of travel are incorporated as well.

The resilience projects provide benefits by avoiding the additional vehicle miles traveled and their associated per-mile external costs, which provides positive benefits. In some cases, however, these benefit could be negative (or “disbenefits”) in the project scenario if the project introduces unforeseen travel circuitry during the hazard event and repair/clean up period. In either case these benefits or disbenefits are in the numerator of the benefit-cost ratio (BCR) and added to the net benefits (whether the sign is positive or negative).

Safety, noise, and emissions benefits are calculated for roadway vehicles (both passenger and transit) based on the per mile monetary value by vehicle type from the US DOT BCA guidance table from Table A-14 External Highway Costs.⁷⁸ The model assumes that all travel is on urban roadways and uses the urban per mile costs. The model also applies the reciprocal of the external share of the safety cost to the

⁷⁸ This approach significantly simplifies the per-mile cost calculation approaches used in the previous version of the tool for both Safety and Emissions benefits.

safety costs per mile to account for both the internal and external safety costs of additional miles. The `bca_metrics.csv` output file has safety, noise, and emissions outputs for each scenario. The output includes variables for safety, noise, and emissions benefits in the damage, exposure, and initial periods of the hazard and provides discounted values for these as well, labeled:

- 'damSafetyvsBase', 'expSafetyvsBase', 'initSafetyvsBase', 'damSafety_Discounted', 'expSafety_Discounted', and 'initSafety_Discounted'
- 'damNoisevsBase', 'expNoisevsBase', 'initNoisevsBase', 'damNoise_Discounted', 'expNoise_Discounted', and 'initNoise_Discounted'
- 'damEmissionsvsBase', 'expEmissionsvsBase', 'initEmissionsvsBase', 'damEmissions_Discounted', 'expEmissions_Discounted', and 'initEmissions_Discounted'.

7 RDR Reporting and Visualization Module

7.1 Purpose: Communicating the Results

The RDR Reporting and Visualization Module provides the user with a streamlined approach for reviewing the ROI analysis within the analysis team as well as decision-makers external to the analysis team. This section describes data outputs and the visualization module that can be used to explore the analysis results. It describes how to use the analysis output and visualization to rank the performance of a given resilience investment in relation to other asset project options and/or in relation to resilience investments for other assets.

7.2 Structure, Functions, and Parameters

7.2.1 Reporting

The RDR ROI Analysis Tool outputs are intended to be used in project prioritization based on the BCA, BCA-U/Regret, or Breakeven Analysis approaches. Agencies can use these outputs to rank the evaluated projects in order of their economic ROI. This ranking can contribute to overall project prioritization evaluations when combined with other prioritization factors considered by agencies.

Given that the RDR Tool Suite is used when the potential future conditions are very large in number (the scenario space is very large), reviewing and understanding the outputs of the analysis can be challenging in text form. Therefore, the RDR ROI Analysis Tool imports final outputs into visualization dashboards that can be accessed using Tableau (full licensed Desktop version or the free Reader version).

The RDR ROI Analysis Tool output file addresses three categories of variables: scenario factors (i.e., the parameters that define the scenario including uncertainty parameters, the project alternative, and the recovery process parameters), scenario outcomes (PHT, PMT, VMT, and trips), and the monetization and analysis variables such as the monetization of PHT, VMT, and trips and the projection prioritization metrics. The variables of the RDRM output data include:

- Scenario Accounting
 - IDResiliencyScenario: unique ID for the resilience project scenario
 - IDScenario: ID for the unique combinations of uncertainty values.
 - IDScenarioNoHazard: unique ID for unique combinations of uncertainty variables not including the hazard dimensions. This variable defines the baseline scenario, namely the scenarios in which there are no project alternatives.
- Policy Levers
 - ResiliencyProject: name of the resilience project as a string.
 - Asset: asset on which a resilience project is deployed.
 - ResiliencyProjectAsset: resilience project and the asset defined jointly, as a string.
 - Project Group: an optional field that defines the project group in which a given resilience investment is tested. In some cases, analysts may have grouped potential projects in the TDM analyses.
- Uncertainties
 - Year: expresses the period of analysis as a string, e.g., “2020-2045”
 - Hazard Characteristics:
 - HazardDim1: hazard severity in numeric (e.g., flooding depth/flood event frequency, earthquake intensity).

- HazardDim2: a second hazard dimension as a numeric. For example, in the Hampton Roads case, this is defined as the sea level rise.
 - Event Probability: annual probability of the hazard event (e.g., a 100-year flood has an annual probability of .01).
 - DurationofEntireEventdays: duration of the hazard event (including hazard event recession) in days.
 - Exposurerecoverypath: hazard event exposure recovery path describing each stage of the hazard by day before the hazard event cases. The exposure recovery paths are defined by the progression of network states (corresponding to hazard levels) for every day of the hazard event, e.g., a hazard with a duration of three days may have an exposure recovery path of '2,2,1' which denotes that the first stage at hazard level 2 lasts two days and the second stage at hazard level 1 lasts one day.
 - DamageRecoveryPath: damage recovery path which describes the level of damage at each stage of the repair duration period.
- EconomicScenario: economic scenarios in string (e.g., future economic conditions used to estimate trip changes for TDM analyses).
- TripElasticity: trip elasticity in numeric (reduces the number of trips based on the increase in travel time).
- FutureEventFrequency: annual growth value of the probability of the hazard events (if probability is anticipated to change over time).
- Metrics
 - ProjectCosts_Discounted: resilience project costs discounted over the period of analysis, assuming any redeployment and subtracting the residual cost.
 - TotalResidual_Discounted: resilience project residual values discounted over the period of analysis, given project cost, redeployment cost if enabled, and lifespan.
 - TotalMaintenanceCosts_Discounted: resilience project annual maintenance costs accumulated and discounted over the period of analysis if enabled.
 - TotalNetBenefits_Discounted: risk-weighted discounted net benefits summed across each hazard event for a given unique uncertainty profile.
 - Benefits_Discounted: discounted benefits which is the sum of the monetized performance metrics during the exposure and damage periods as well as the repair and cleanup cost savings.
 - ExpBenefits_Discounted: discounted benefits of the system performance metrics during the exposure event.
 - RepairCleanupCostSavings_Discounted: discounted repair and cleanup costs relative to those occurring in the base scenario.
 - DamBenefits_Discounted: discounted benefits of the system performance metrics for the period during which the asset is under repair after the exposure.
 - BCR_Discounted: benefit cost ratio for each scenario.
 - initSafety_Discounted: discounted safety costs during the initial hazard event derived from VMT.
 - expSafety_Discounted: discounted safety costs during the exposure event derived from VMT.

- damSafety_Discounted: discounted safety costs for the period when the asset is under repair after the exposure.
 - initNoise_Discounted: discounted noise costs during the initial hazard event derived from VMT.
 - expNoise_Discounted: discounted noise costs during the exposure event derived from VMT.
 - damNoise_Discounted: discounted noise costs for the period when the asset is under repair after the exposure.
 - initEmissions_Discounted: discounted emissions costs during the initial hazard event derived from VMT.
 - expEmissions_Discounted: discounted emissions costs during the exposure event derived from VMT.
 - damEmissions_Discounted: discounted emissions costs for the period when the asset is under repair after the exposure.
 - RegretAll: resilience project regret ranking across computed across all scenarios.
 - RegretScenario: resilience project regret ranking for individual scenarios.
 - RegretAsset: resilience project regret calculated only for projects on the same asset, by scenario.
- System Performance: There are 36 system performance variables defined by the period of performance, the system performance category, and the output category. The field names are constructed using the same convention, e.g., 'expPHTlevels' where 'exp' is the period of performance, 'PHT' is the system performance category, and where 'levels' is the output category.
 - Period of Performance: particular period over which the benefits are calculated.
 - "dam" refers to the system performance of the network when the asset is damaged and under repair and after the exposure has ended.
 - "exp" refers to the system performance of the network when the exposure event is occurring.
 - "init" refers to the system performance of the network for a single day at the highest level of exposure event.
 - System performance categories: includes PHT, VMT, and trips.
 - Output Category
 - Levels: absolute level of the system performance category value.
 - Tripslevel_dollar: monetized value of the Levels.
 - vsBase: expresses the difference between the baseline scenario and the resilience scenario for a given system performance category.
 - vsBase_dollar: monetized value of vsBase.
- Asset Damage Performance:
 - AssetDamagelevels: damage function value for the scenario in numeric.
 - DamageDuration: number of days of repair required to return the asset to its pre-hazard condition.
 - AssetDamagevsBase: damage value in the project scenario against the damage value against the base.
 - RepairCleanupCosts: annual repair and cleanup costs for the scenario.

- RepairCleanupCostSavings: value of the cleanup and repair costs against those costs in the base.
- RepairCostSavings: value of repair cost only against those costs in the baseline.

7.2.2 Tableau Dashboards

The Tableau dashboards are intended to provide summary information for each of the main RDR analyses and to provide exploratory tools to review the breadth of the analytical outputs. The workbook includes six dashboards that provide summary information.

The Regret and Exploratory Dashboards feature a filter panel, shown in Figure 7-1, which allow the analyst to select a subset of the scenarios in the scenario space. The Asset-Project filter list allows users to select which project, projects, or assets are shown in the dashboard, and the Uncertainties filter list allows users to select the scenario parameters for filtering. Filtering on one variable automatically adjusts the available range of values for other filters. A reset button reverts all filters to the full range. The Overview tab provides a high-level navigation. Each dashboard has button on the top right for a dictionary and help. The dictionary provides explanation of the different terms used throughout the dashboards. The help button provides additional information about how to use the tool and where to find particular information.

REVERT FILTERS

ASSET-PROJECTS

Asset: (All) ▼

Project Group: (All) ▼

Asset-Project Cost \$: \$0 to \$3,285,876.4

Resiliency Project: (All) ▼

Resiliency Project-Asset: (All) ▼

UNCERTAINTIES

Hazard Severity: 30 to 45

Exp. Duration (Days): 8 to 12

Exp. Recovery Path: (All) ▼

Dam. Duration (Days): 0 to 0

Dam. Recovery Path: (All) ▼

Event Probability: 0.01000 to 0.10000

Future Event Frequency: 1.00 to 1.00

Economic Scenario: (All) ▼

Trip Elasticity: -1.000 to -1.000

Figure 7-1: Tableau Workbook Dashboard Filter Panel.

The Benefit Cost Analysis Dashboard provides the high-level BCA and Breakeven information for all of the resilience projects considered in the analysis.



The example Benefit Cost Analysis dashboard shown in Figure 7-2 shows the results from RDR Quick Start 1.

- The number of projects shown in the table and figure can be adjusted using the “Enter number of projects to show” entry on the top left.
- The Benefits & Cost by Resiliency Project table contains a list of the projects in the analysis with fields for Project ID, Project Name, Benefit Cost Ratio (BCR) Rank, BCR, Total Benefits, Total Project Costs, and Total Net Benefits. Users can click on any element in the table to show only that project in Benefits by Hazard Scenario & Costs by Resiliency Project graphic below.
- Benefits by Hazard Scenario & Costs by Resiliency Project: this stacked bar chart shows for each project the benefits that accrue independently for each hazard scenario modeled, and the project costs (as negative). This in effect shows the benefits contribution from each independent scenario tested for each project.

Figure 7-2: The BCA Dashboard in the RDR Tableau Visualization Workbook (Example from RDR Quick Start 1)

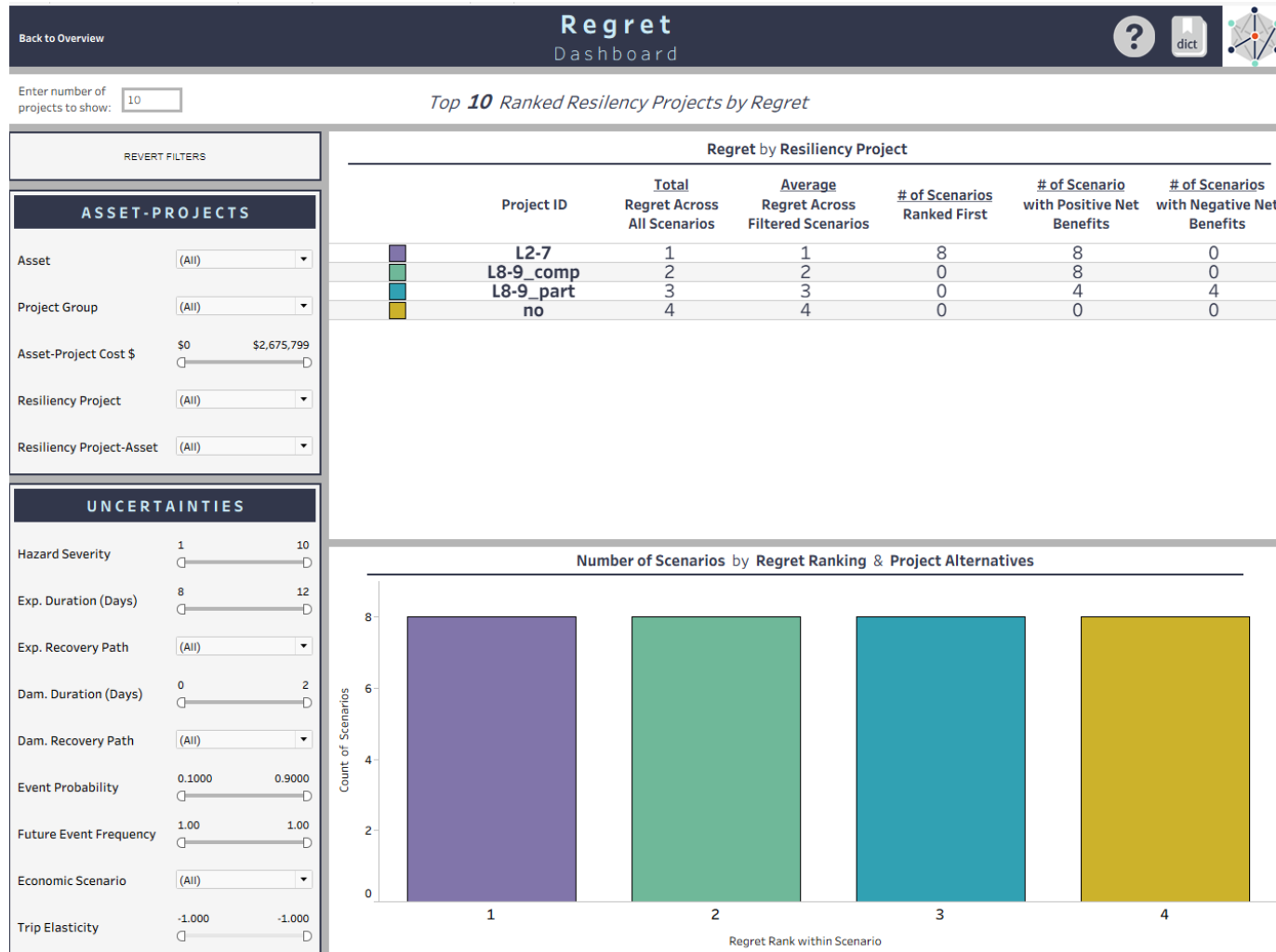


Figure 7-3: The Regret Dashboard in the RDR Tableau Visualization Workbook (Example from RDR Quick Start 1)

The Regret Dashboard provides the overview Regret information for all of the resilience projects considered in the analysis.

The example Regret dashboard shown in Figure 7-3 shows the results from RDR Quick Start 1.

- The number of projects shown in the table and figure can be adjusted using the “Enter number of projects to show” entry on the top left.
- The Regret by Resiliency Project table contains a list of the projects in the analysis with fields for color used in graphic, Project ID, Total Regret Across All Scenarios, Average Regret Across Filtered Scenarios, # of Scenarios Ranked First, # of Scenarios with Positive Net Benefits, # of Scenarios with Negative Net Benefits. Users can click on any element in the table to show only that project in Benefits by Hazard Scenario & Costs by Resiliency Project graphic below. Benefits by Hazard Scenario & Costs by Resiliency Project: this stacked bar chart shows the number of scenarios in which projects achieved a given ranking by project. Selecting a value in the graphic will highlight all of the results for that project.

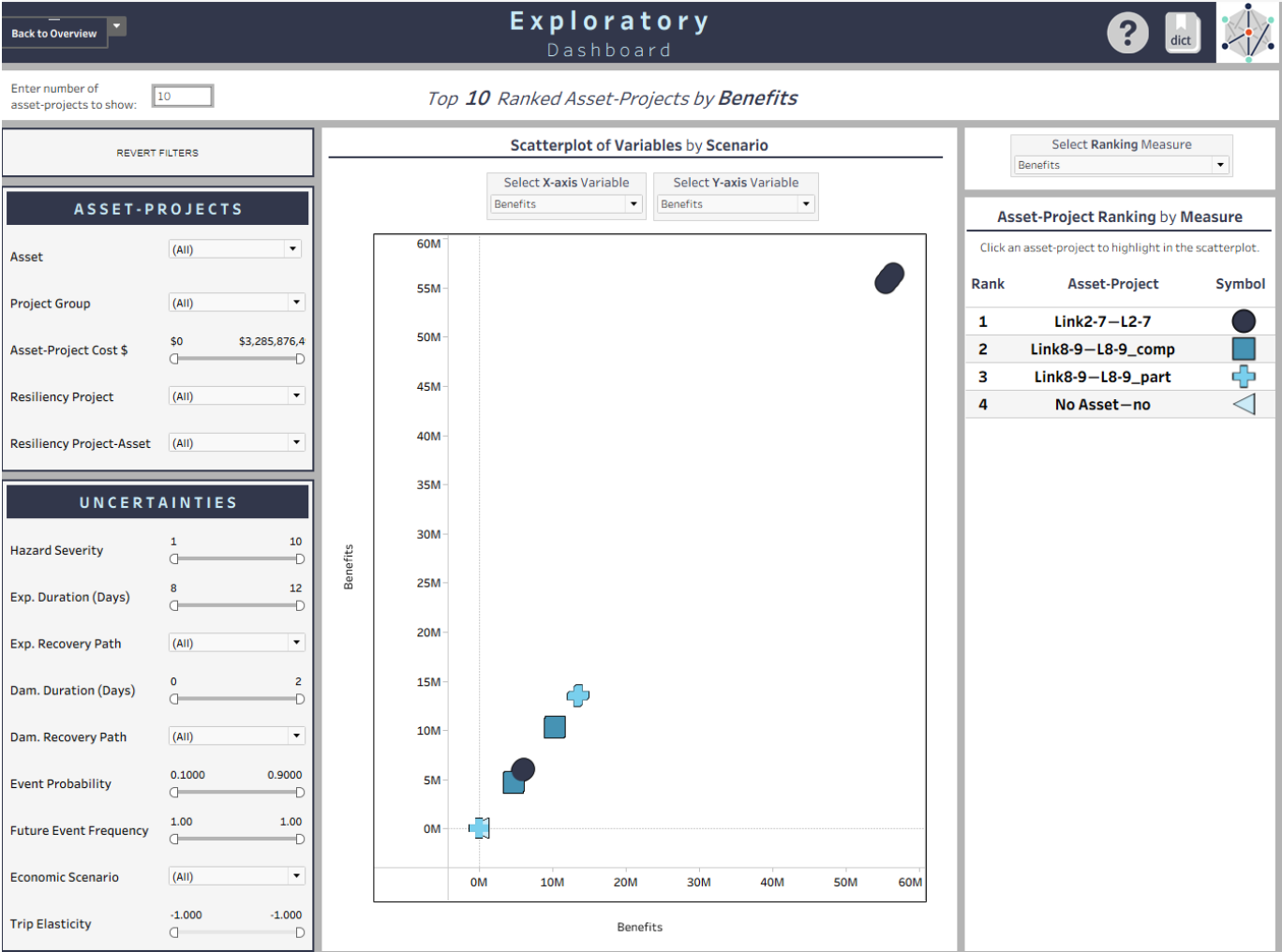
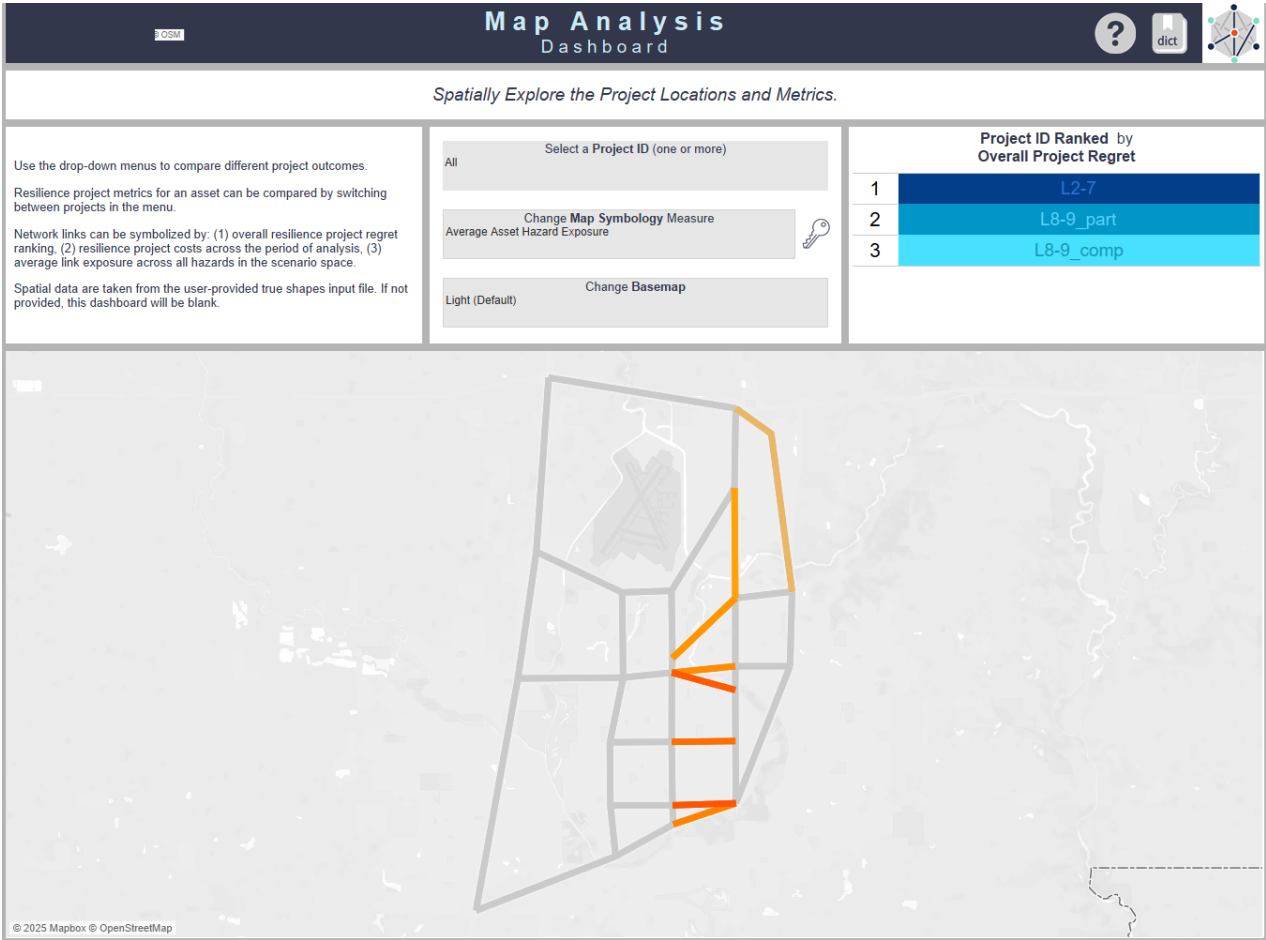


Figure 7-4: The Exploratory Dashboard in the RDR Tableau Visualization Workbook (Example from RDR Quick Start 1)

The Exploratory dashboard allows the analyst to explore detailed results for all of the projects and scenarios. There are two main components of this dashboard.

- **Scatterplot of Variables by Scenario:** Users can select the results outputs used in the X- and Y-axes variables independently using the dropdowns. The projects are identified by unique color and shape values, each instance of which represents a different scenario, which is reflected in the Asset-Project Ranking by Measures table.
- **The Asset-Project Ranking by Measures:** this feature shows the ranking of projects across all of the scenarios selected. Select an alternative ranking metric using the dropdown. The ranking measures include Regret All Scenarios, Mean Regret Scenarios, Net Benefits, and BCR.



The Map dashboard allows the analyst to explore all of the projects in a geospatial context. This dashboard will only populate if the optional true shapes input file is provided. There are three main components of this dashboard.

- **Dropdown Menus:** There are three options for navigating the map of project locations: filter by project, color by metric, and select basemap. Each menu can be used to configure the visualization. In particular, filtering by project will dynamically update the zoom extent on the map to narrow in on particular projects of interest.
- **Ranking Column:** This feature shows the ranking of projects across all of the scenarios considered. Selecting a project will update the map.
- **Map:** The map panel visualizes the true shapes network on a basemap (default, light, or streets). Network links are drawn based on the true shapes specifications; links are further symbolized based on the metric selected. A tooltip for each network link summarizes outputs from the run, including overall regret ranking, average hazard exposure, and project cost.

Figure 7-5: The Map Dashboard in the RDR Tableau Visualization Workbook (Example from RDR Quick Start 1)

Run Parameters Dashboard		
The detailed run parameters for this analysis.		
Category	Parameter	Value (hover over value for description)
AequilibraE Settings	aeq_max_iter	100
	aeq_rgap_target	0.01
	aeq_run_type	SP
	blocked_centroid_flows	FALSE
	calc_transit_metrics	FALSE
	run_minleq	0
Disruption Module Parameters	alpha	
	beta	
	beta_method	
	exposure_field	Value
	exposure_unit	Feet
	link_availability_approach	default_flood_exposure_function
	link_availability_csv	
	lower_bound	
	resil_mitigation_approach	manual
	upper_bound	
Model Run Configuration	zone_conn	0
	base_year	2017
	elasticity	-1.0
	end_year	2045
	event_freq_factors	1.0; 1.001
	future_year	2045
	hazard	haz1; haz2
	input_dir	C:\GitHub\RDR\scenarios\qs1_sioux_falls\Data\inputs
	lhs_sample_target	8
	metamodel_type	multitarget
	num_recov_stages	0; 1
	output_dir	C:\GitHub\RDR\scenarios\qs1_sioux_falls\Data\generated_files
	projgroup	02; None
	resil	1.0; 1.07; 1.0

The Run Parameters dashboard reports all of the different parameters that were used in the model run for all scenarios.

Figure 7-6: The Run Parameters Dashboard in the RDR Tableau Visualization Workbook (Example from RDR Quick Start 1)

8 RDR Benefits Analysis Tool

8.1 Purpose: Disaggregating Network Performance Statistics

The RDR Benefits Analysis Tool can be used to explore how impacts and benefits are related to any user-supplied variable that can be associated to TAZs. For this purpose, trip-level benefits, measured as reduction in miles and/or time spent on trips and maintenance of trips under hazard conditions with a resilience investment, are assigned to user-chosen TAZs or portions of the community within specific TAZs. The RDR Benefits Analysis Tool enables the user to examine the impact of one resilience project during one hazard event and show how impacts and benefits vary in relation to the user-supplied variable assigned to the TAZs in the scenario, allowing comparison.

8.2 Structure, Functions, and Parameters

The standalone RDR Benefits Analysis Tool incorporates high-level analysis of disaggregated TAZ impacts and performance benefits for an individual resilience investment during a specific hazard event through post-processing of core model run outputs.

- The tool allows users to provide any variable (e.g., categorization or quantity) at the TAZ level. For users that do not already have data assigned at the TAZ level, the tool also includes an optional helper tool to apply TAZ attributes of interest to the user's custom TAZ shapefile through an overlay method. For example, the user could identify TAZs based on urban/suburban/rural designations.
- The TAZ metrics component of the tool associates output trips from AequilibraE with the TAZ of origin and associates those trips with the user-supplied variable value of that TAZ of origin. It creates summary outputs for an individual resilience investment and hazard event in an HTML report. It also provides CSV file outputs with the underlying data results from the analysis so that users can perform further analyses in their analytical platform of choice.
- If the user-supplied variable is not numeric (or if it is numeric with fewer than 20 unique values), the tool treats it as a categorical variable, and runs the categorical version of the notebook (MetricsByTAZ_categorical.ipynb). If, instead, the user-supplied variable is numeric and has 20 or more unique values, the tool treats it as a continuous variable, and runs the continuous version of the notebook (MetricsByTAZ_continuous.ipynb). These two different versions of the notebook follow a parallel process, but due to the different assumptions about the nature of the user-supplied variable, the outputs are slightly different. Figure 8-1 below shows an example graph from the categorical version of the notebook, and Figure 8-2 shows an example graph from the continuous version of the notebook.
- Examples of questions that the tool helps address:
 - How does the baseline magnitude of trips/minutes per trip/miles per trip vary with respect to the value of the user-supplied variable?
 - How does the relevance of the disruption vary with respect to the value of the user-supplied variable?
 - How do the projected benefits of the resilience investment vary with respect to the user-supplied variable?

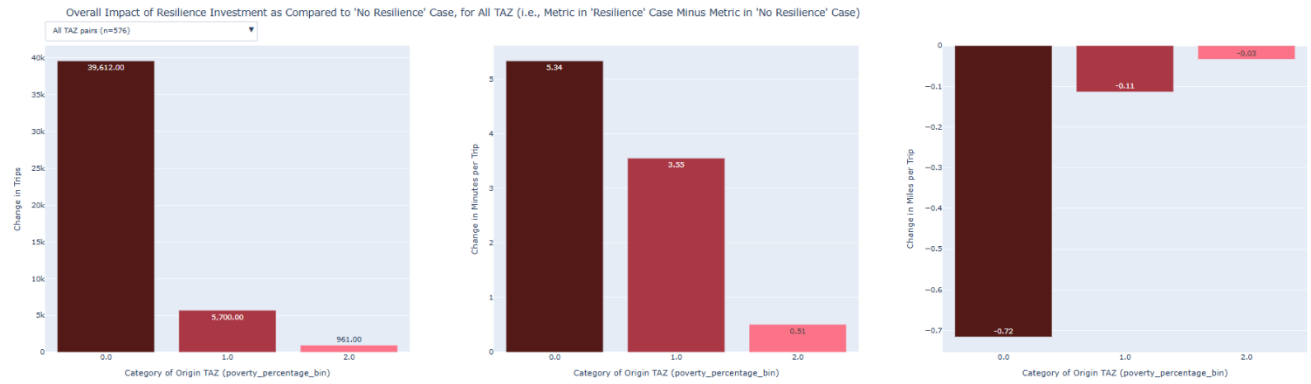


Figure 8-1 Example categorical output from benefits analysis

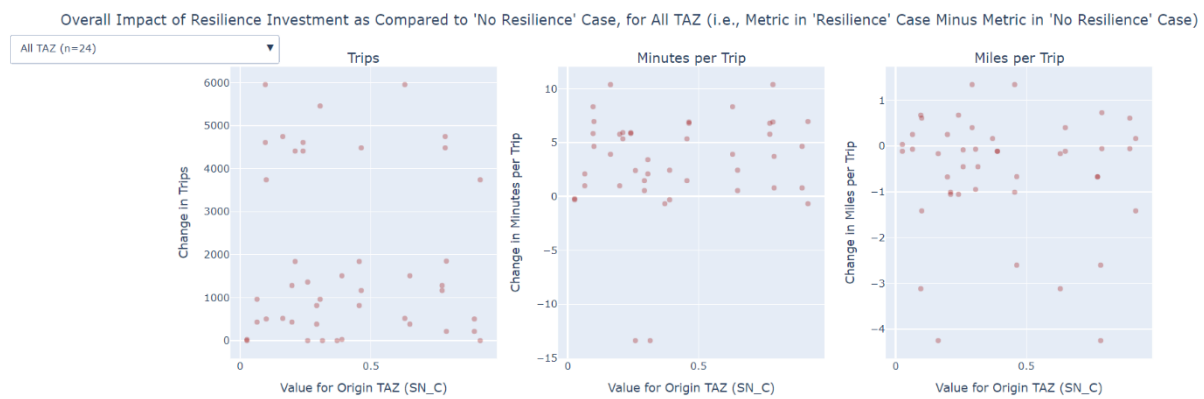


Figure 8-2 Example continuous output from benefits analysis

9 RDR Graphical User Interface

9.1 Purpose: Provide Guided User Interactions for RDR Run Setup

The RDR Graphical User Interface (RDR GUI) provides users with a guided, step-by-step set of interactions through the various steps involved in setting up a run of the RDR Metamodel and ROI Analysis Tool.

The RDR UI provides users with a browser-based interface (CLI) to:

- 1) set RDR configuration parameters,
- 2) set RDR parameters not contained in the configuration file, such as the Model Parameters, User Inputs, etc.,
- 3) save and load RDR UI configuration profiles (saves) in JSON format which can be used in place of a config file and non-config spreadsheets and matrices, and
- 4) build an RDR batch file that is automatically configured to run using a user-specified save. These features are designed to make it easier to set up an RDR run and/or multiple similar RDR runs.

The RDR UI also features a built-in help menu to enhance the user-friendliness of setting up RDR.

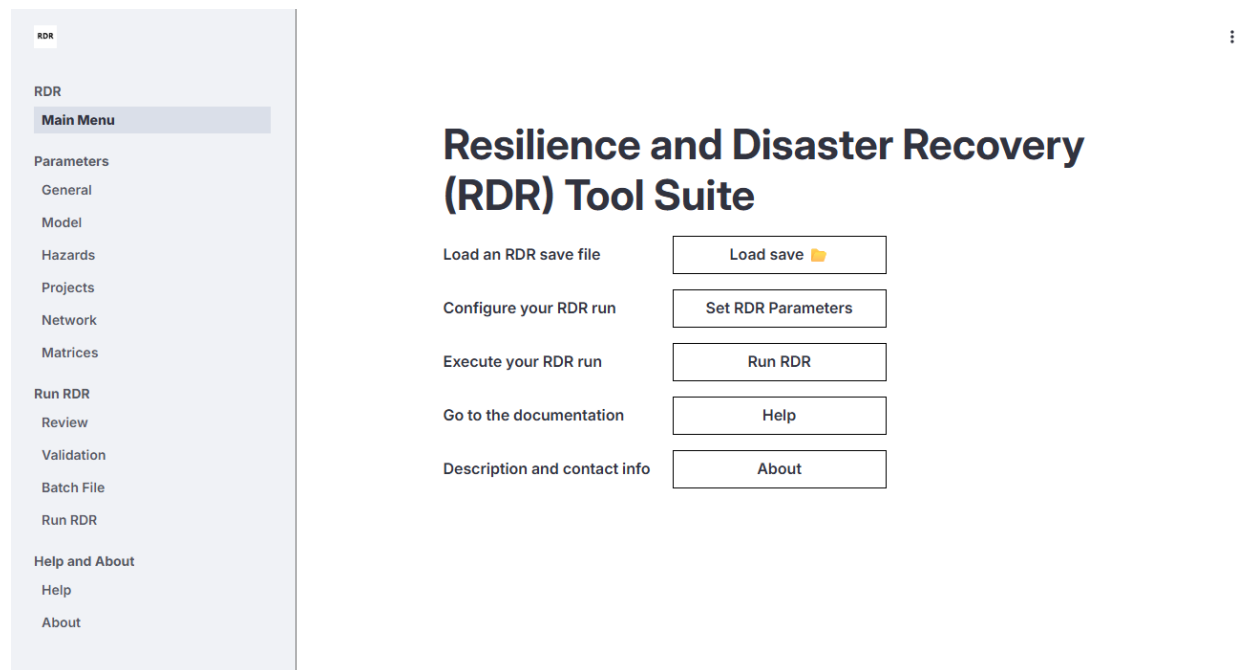


Figure 9-1 RDR GUI main menu

9.2 Structure, Functions, and Parameters

The RDR GUI structure can be represented graphically as a decision tree. See **Error! Reference source not found.** Figure 9-2 for a high-level look at the structure of RDR GUI.

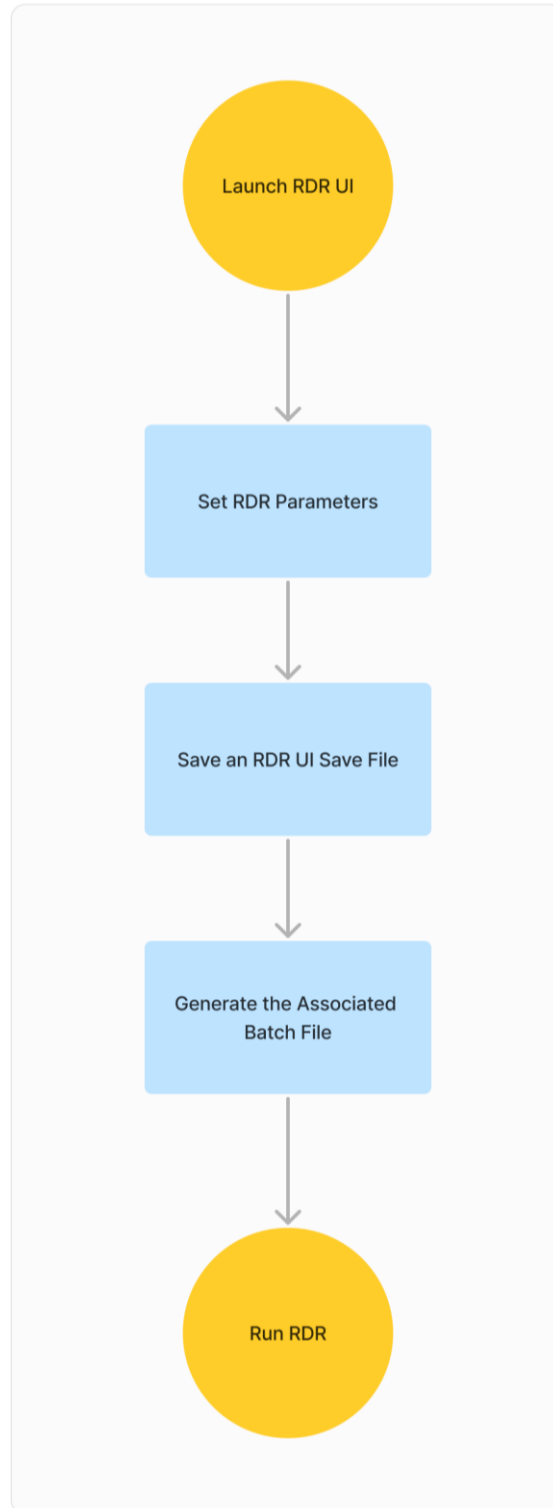
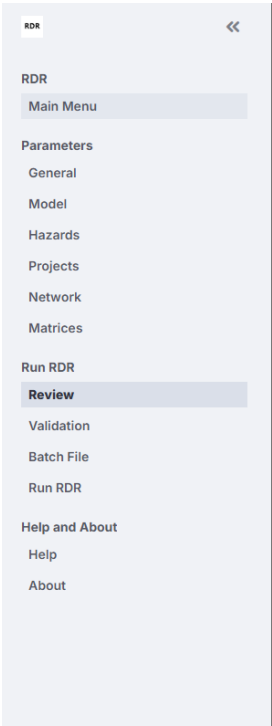


Figure 9-2 High-level RDR GUI user flow

- Users launch the GUI by running a downloaded batch file in their command line, using the Streamlit package in their environment.
- The RDR GUI is comprised of the following components:
 - Pages: pages are the interactive windows of RDR UI displayed one at a time in the command line, forming the building blocks of RDR UI through which the user navigates by entering **commands** into the **prompts**. See **Error! Reference source not found.**Figure 9-4 for details on the individual pages of RDR UI and their relationships.
 - Tabs: The GUI is structured as a set of tabs, visible on the left side of the page. The user works through these within four sections: Main Menu, Parameters, Run RDR, and Help and About.
 - Prompts: prompts are the place(s) on a **page** in which the user is allowed to enter **inputs**
 - Inputs: inputs are the pre-defined controls for RDR UI, consisting of commands and assignments.
 - Commands: commands are inputs which cause RDR UI to execute an action, such as page navigation, creating, overwriting, or loading a save, or exiting RDR UI.
 - Assignments: assignments are inputs which cause RDR UI to write data to a **parameter**.
 - Inputs: Within the Parameters tabs, the user fills in input boxes, whose content is then written to a parameter in the config file.
 - Parameters: parameters are the variables used in an RDR run, which RDR UI generates, stores in a JSON, and with which RDR UI associates a batch file.
 - Config parameters: variables previously assigned in the RDR config file, having a one-to-one value per parameter ratio.
 - Non-config parameters: variables previously assigned in the various RDR lookup tables and spreadsheets, some of which have a many-to-one value per parameter ratio.
 -
 - Data validation: backend validation checks to confirm that user-inputted assignments match the required parameter data type, such as integers, strings, and file paths.
 - See **Error! Reference source not found.**Figure 9-5 for a detailed view of what checks occur on each user input.
 - Review and Validation: The “Run RDR” tab group provides a visual overview of the user’s inputs and input files. After checking the inputs manually, the Validation tab allows the user to run backend validation checks to confirm that user-inputted assignments match the required parameter data type, such as integers, strings, and file paths. Finally, the “Batch File” and “Run RDR” tabs provide additional information on the generated batch file path and allow the user to run the batch file by

pressing a button.



Run RDR

Load save  Save  Save as  Clear all 

Review the current configuration

General RDR Configuration

	Parameter Name	Parameter Value
0	1.1 Input Directory	None
1	1.2 Output Directory	None
2	1.3 Run ID	None
3	1.4 Start Year of Analysis Period	None
4	1.6 End Year of Analysis Period	None
5	1.5 Year of Base Year Runs	None
6	1.7 Year of Future Year Runs	None
7	2.1 Metamodel Type	multitarget
8	2.6 Latin Hypercube Sample Size	None
9	2.3 AequilibraE Model Run Type	RT

Figure 9-3 Scenario review view in Run RDR tab group

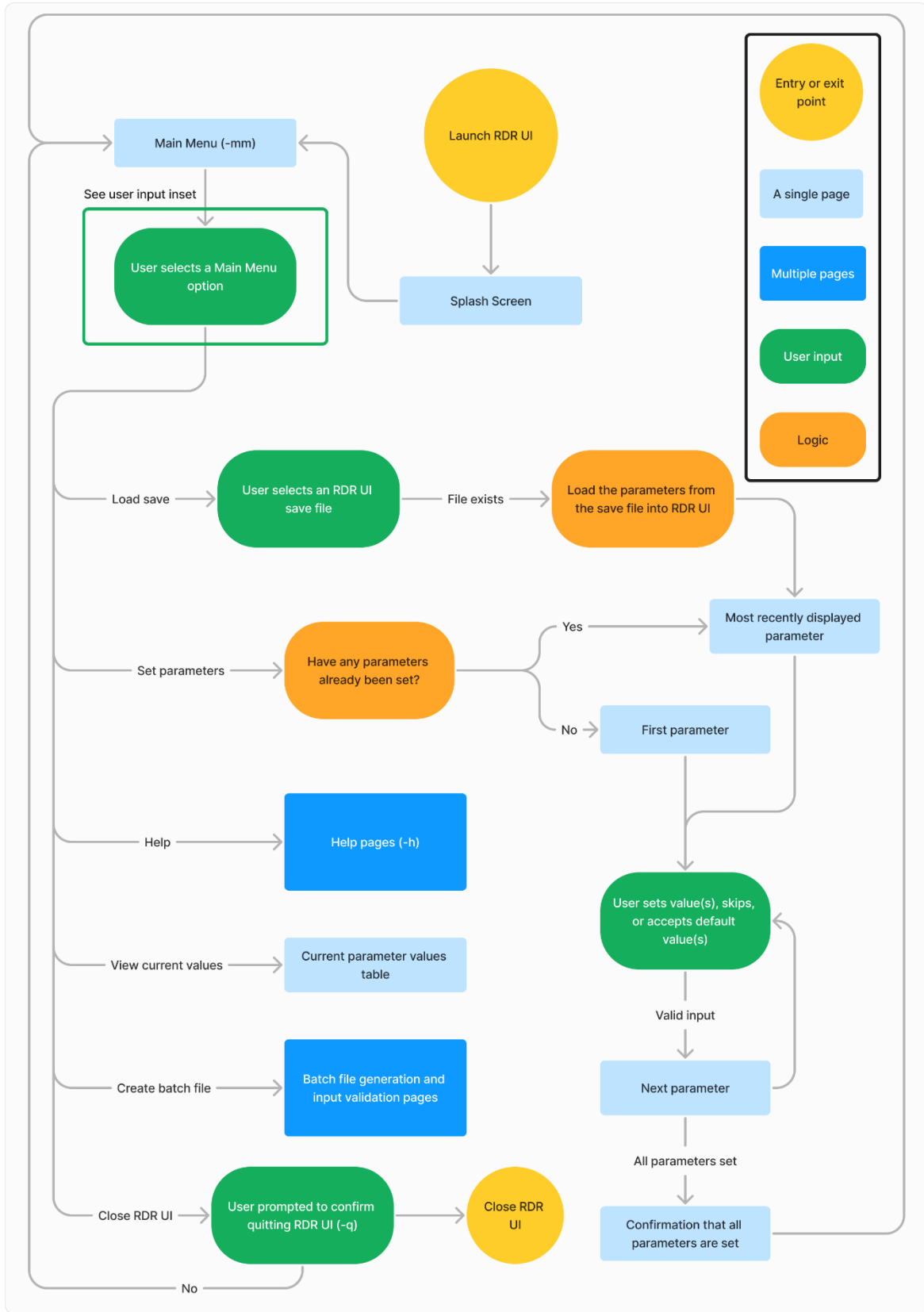


Figure 9-4 Detailed RDR GUI user flow

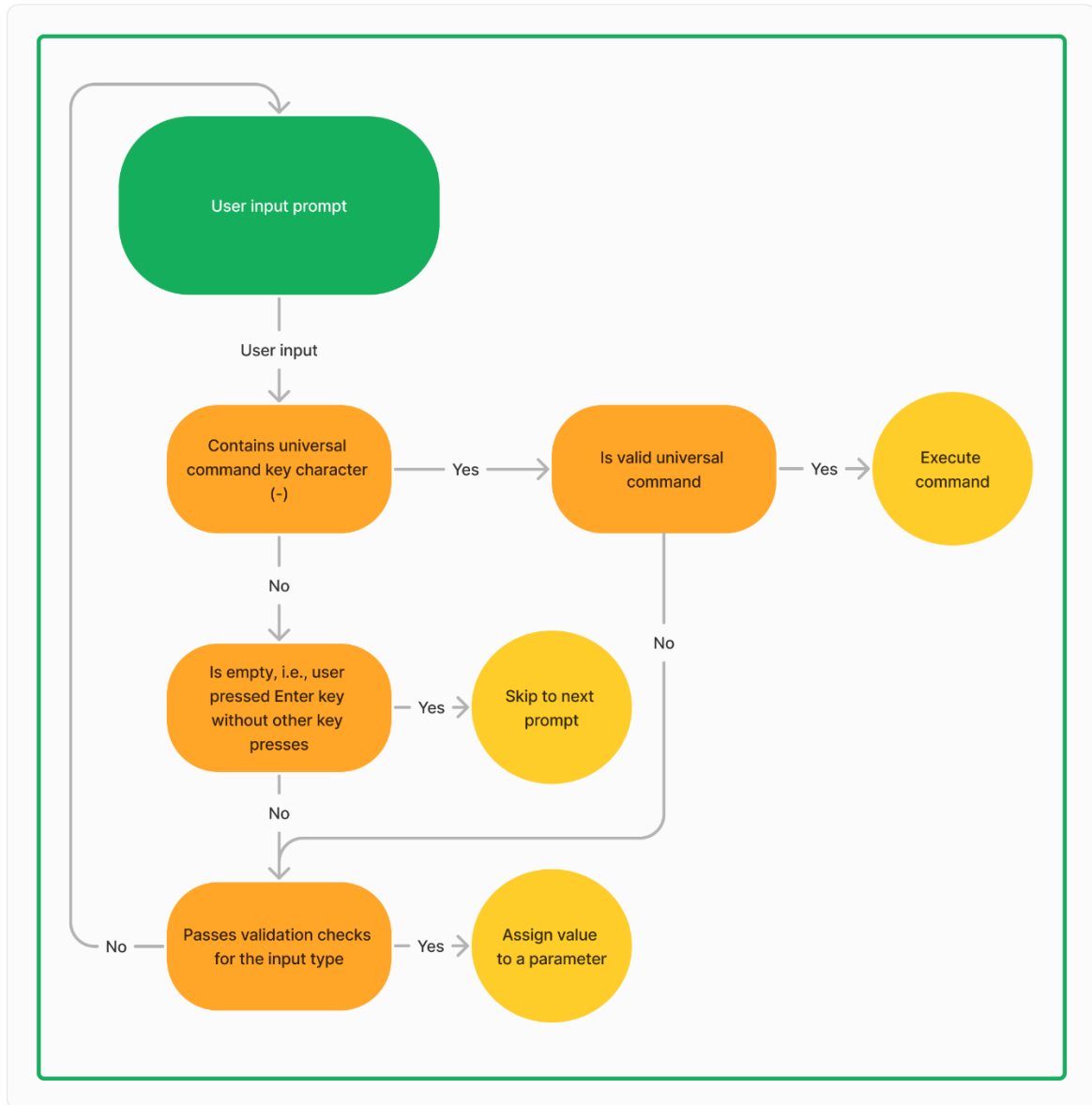


Figure 9-5 User input inset

10 RDR Tool Suite Pilot Testing

As part of the development for public release, the RDR Tool Suite was pilot tested with agency partners from:

- 1) The Hampton Roads Transportation Planning Organization (HRTPO), Hampton Roads Planning District Commission (HRPDC)
- 2) The Hillsborough Transportation Planning Organization
- 3) Houston-Galveston Area Commission (H-GAC)

The Hampton Roads partnership occurred in two phases: first, execution of preliminary analyses by the RDR Tool Suite developers to test functionality and validate the performance of the AequilibraE core model; and second, technology transfer/deployment to the HRTPO and HRPDC teams for in-house testing. The Hampton Roads partners tested the Exposure Analysis Tool as well as the main RDR Tool Suite.

The Hillsborough and H-GAC pilots focused on technology transfer and deployment of an initial version of the RDR Tool Suite for agency in-house testing and focused on the main RDR Tool Suite.

The pilot partners provided input on documentation, technology transfer, and tool usability. In addition, the RDR Tool Suite development team worked closely with the pilot partners to troubleshoot analyses. Subsequent testing with pilot partners focused on analyses employing the RDR Benefits Analysis Tool and transit network functionality. The input gained from the pilot partners during the testing led to a greatly improved initial public version of the RDR Tool Suite.

11 Limitations and Caveats

The following are known limitations and caveats to the current version of the RDR Tool Suite described in this documentation. These limitations also inform the following section on future enhancements.

- Scenario Definition
 - The RDR Tool Suite was developed to be hazard agnostic. Flooding inundation has been extensively tested in detail in the tool suite (along with hypothetical earthquake scenarios in the Reference Scenario Library document). The analyst is advised to exercise care in applying the tool to other hazards.
 - The RDR Tool Suite is focused on temporary hazard events, with the assumption that network performance returns completely to the baseline after the exposure recovery and repair recovery periods. Although the decision could be made to not rebuild a damaged facility, analyzing the permanent impacts of such a decision is out of scope for the tool suite.
 - Because only one resilience investment can be specified in a scenario, the effects of resilience investments are considered additively. Interactions between multiple resilience investments and their combined effect on the network are not considered. To model these interactions with the tool suite, the user can create a new resilience project combining the multiple resilience investments.
- Exposure
 - The automated RDR Exposure Analysis Tool provides a simple screening level method for assessing the intersection of a hazard with transportation assets. However, the assessment of potentially exposed assets should be reviewed carefully to determine if asset data (e.g., elevation for flooding) are accurately represented in the network dataset. For flood exposure, particular care should be exercised to review bridge, tunnel, and transit asset elevation data.
 - Projects that will add links to the transportation network will not be represented in the exposure analysis unless manually added, with appropriate attributes, to the network.
 - See the User Guide for additional limitations to the RDR Exposure Analysis Tool.
- Disruption
 - The current RDR Tool Suite allows multiple functions to be used to define disruption based on exposure extent. The default relationship in RDRM is specific to flooding and should not be used for other hazards.
- Damage Repair Cost and Time
 - The current RDR Tool Suite allows for repair costs and repair times to be provided by the analyst, or for the analyst to rely on the default look-up tables. Repair costs and times are based on asset type (e.g., roadway, bridge, transit, etc.) and further subcategories (e.g., functional class for roadways, length for bridges, etc.). Default values in the RDR Tool Suite for the potential time of repair were provided by VDOT or sourced from case studies in the literature and may not be applicable to all agencies. The analyst should use agency-specific estimates whenever possible.
 - The damage module for the RDR Tool Suite considers repair costs and repair times independently. Agency choice of the trade-off between repair cost and repair time can be adjusted through use of analyst-provided look-up tables, but evaluation of the trade-off between repair cost and repair time is not within the scope of the tool.

- The RDR Tool Suite currently does not have a mechanism for addressing asset hardening that does not change disruption but does change the need for repair afterwards.
- Metamodel
 - The metamodel attempts to predict the effect of resilience investments on performance metrics (e.g., trips, PMT, PHT) for the entire transportation network being modeled. In cases where the resilience investment spans only a few links in a large regional network, the magnitude of the effect can be very small, which can affect the accuracy of the metamodel. In particular, when compared to the effects of other variables (e.g., hazard severity, economic scenario), the effect of a resilience investment may be statistically insignificant and lead to unreliable predictions being propagated through the ROI analysis.
 - Relatedly, the extrapolation of metamodel outputs in the ROI analysis to calculate economic benefits across the full analysis period can lead to large levels of uncertainty in the data reported. This is particularly true at the extremes (high performing and low performing projects) for the linear regression models, as they are unbounded. Model evaluation of the RDRM regression model should be done in conjunction with analysis using the Tableau dashboards.
 - The metamodel coefficients and outputs are highly sensitive to both the number of samples used to fit the model and the model structure of the regression. Sensitivity analysis around these metamodel parameters should be conducted.
- Recovery modeling
 - The model for building out exposure recovery paths is currently limited to recession stages of equal length, given by the parameter setting “stage_length_method” = ‘Equal’ in the configuration file.
 - The RDR Tool Suite uses a fixed approach to approximate network impacts of partial asset repair during the repair recovery period, which cannot be set by the analyst.
 - Hazard recovery and damage recovery models are independent. There may be a relationship between these sections that could be important to model (e.g., longer storms require longer repair).
- Risk in Underestimating Damage
 - When conducting this type of analysis there is a risk of underestimating damage because it is not possible to capture all of the complexities of synergistic interactions, such as between sea level rise, storm surge, and increased precipitation or high heat and wildfire. The combined impact of scenarios that create multiple hazard factors would potentially be far greater than any one factor alone. However, the variability and uncertainty of those interactions make it very challenging to analyze those interactions. Therefore, most agencies focus on tractable exposure analyses that are accurate, but they should recognize the potential to underestimate damage.
 - The tool currently allows a user to prioritize projects based on the potential impact of a single hazard type. However, resilience investments are typically designed to mitigating the risk of a single hazard type, rather than providing some general kind of hazard mitigation. The tool does not yet allow a user to compare the benefits of mitigating different kinds of hazards.
- Economic valuation
 - There are benefits that have not been included in the tool that are proper for the ROI approaches, including travel time reliability. Future development of the tool may be able to address these shortcomings.

- The current version of the tool does not consider the pre-hazard condition of the asset and the repair does not impact the underlying lifespan of the asset. Future development of the tool may be able to address these shortcomings.

12 Conclusion

The existing approach for transportation investment analysis using standard TDMs does not integrate the potential for hazard events to disrupt and damage transportation assets. Nor does the current method evaluate the benefit of mitigating those impacts by investing in hazard resilient projects. This technical document describes both the conceptual background of robust decision making as a way to make decisions under highly uncertain conditions (e.g., future projections) and describes the technical details of the RDR Tool Suite and methodology that applies this approach to evaluating infrastructure resilience investment ROI.

The RDR Tool Suite described in this document enables MPOs and state DOTs to evaluate resilience ROI across a range of future scenarios and hazard conditions as part of their project prioritization process, an approach that builds on previously available resources and methodology.

The RDR Tool Suite complements existing federal and other resources⁷⁹ in providing additional background, considerations, and guidance in preparation for changes in the condition of transportation assets due to external factors. The approach is intended to be location and hazard agnostic, such that any state DOT or MPO can utilize it.

The RDR Tool Suite can be accessed at: <https://volpeusdot.github.io/RDR-Public>.

⁷⁹ Including the NCHRP 20-101: Extreme Weather and Climate Change: Guidelines to Incorporate Costs and Benefits of Adaptation Measures, Federal Highway Administration (FHWA) tools and resources including the Vulnerability Assessment and Adaptation Framework and U.S. Department of Transportation (USDOT) BUILD Grant guidance.

Appendix A: Flow Diagram

A SIPOC (Suppliers, Inputs, Processes, Outputs, Customers) diagram⁸⁰ is a table that describes the functions of a model or tool, focusing on source (“supplier”), inputs, processes, outputs, and end destinations (“customers”) of each function in the tool. It is a useful construct for mapping out the relationship among tools and modules of the RDR Tool Suite. The adapted SIPOC diagram below maps out the RDR Tool Suite component functions, inputs, outputs, and destinations for each function.

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
Exposure Analysis Tool	Tool setup ('exposure_grid_overlay' module in helper tools)	User	GIS-formatted exposure data and GIS-formatted transportation network.	Apply gridded exposure data to transportation network. Modify TDM network capacity based on exposure (asset level).	CSV link availability file (one per each unique hazard) quantifying exposure levels for each segment in the transportation network.	ROI Analysis Tool
Scenario Analysis Framework	Tool setup	User	Configuration file. Model parameters file. TDM futures (socioeconomic). TDM road network. (Optional) GTFS transit files. Future infrastructure projects. Future resilience projects.	Define matrix of hazard scenarios, TDM futures, resilience projects, recovery stages to be modeled. (Optional) Set up transit network.	Full list of scenarios output by uncertainty scenario builder (full_combos.csv). (Optional) Combined road and transit network files.	Latin Hypercube Scenario Sampling, Metamodel Regression.
ROI Analysis Tool – User Interface	Tool setup	User	RDR run parameters and analysis framework defining the scenario space. Prepared input files.	Create complete, validated configuration file to execute a run of the ROI Analysis Tool.	User interface JSON save file. Batch file to execute an ROI Analysis Tool run.	ROI Analysis Tool
ROI Analysis Tool – Disruption	'calc_link_availability' method in 'aeq_run' module	Exposure Analysis Tool	Exposure severity on true shapes network.	RDRM: Modify TDM network capacity based on exposure (asset level).	Modified TDM network with updated link availability (disruption) (asset/segment level).	GIS mapping, Core Models, RDRM.

⁸⁰ <https://www.isixsigma.com/tools-templates/sipoc-copis/sipoc-diagram/>

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
ROI Analysis Tool – Disruption	'calc_link_availability' method in 'aeq_run' module	Disruption analysis submodule	Modified TDM network. Resilience project locations and mitigation extent.	RDRM: Modify TDM network capacity based on resilience investment mitigation (asset-project level).	Modified TDM network with updated link availability (resilience) (asset/segment level).	GIS mapping, Core Models, RDRM.
ROI Analysis Tool – Damage	'recov_init' module	Exposure Analysis Tool and User (optional tables for damage, repair cost, and repair time)	Exposure severity on true shapes network. TDM network. Future resilience projects.	Damage and Repair analysis: Translate true shapes network exposure to repair cost and repair time (asset-project level).	Cost of repair and minimum time to repair (asset/segment level).	Metamodel Recovery, ROI Analysis Tool.
ROI Analysis Tool – Damage	'recov_init' module	Exposure Analysis Tool and User (optional tables for damage, repair cost, and repair time)	Exposure severity on true shapes network. TDM network. Future resilience projects.	Damage and Repair analysis: Recalculate repair cost and repair time based on resilience investment mitigation (asset-project level).	Cost of repair and minimum time to repair (asset/segment level).	Metamodel Recovery, ROI Analysis Tool.
ROI Analysis Tool – Core Model	'lhs' module	Scenario Analysis Framework	Matrix of scenarios/futures to be analyzed.	Latin hypercube sampling to identify Core Model runs to be performed.	List of scenarios to be analyzed in Core Models by TDM or AequilibraE.	Core Models.
ROI Analysis Tool – Core Model	'create_network_link_csv' method in 'aeq_run' module	Scenario Analysis Framework, Exposure Analysis Tool, and User	Modified TDM network (disruption, resilience) (asset/segment level). SQLite database of network.	Network Prep for AequilibraE: Convert baseline, disruption, and resilience mitigation scenarios to AequilibraE input files.	Modified SQLite database of network, directory for AequilibraE run.	Core Models.

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
ROI Analysis Tool – Core Model	'aeq_run' module	Latin hypercube sampling, and User (selection of TDM vs. AequilibraE runs), Scenario Analysis Framework	Modified SQLite database of network. Demand OMX or CSV files.	AequilibraE: Perform runs for disruption and resilience scenarios (project level).	PHT, PMT, trips for AequilibraE scenarios (overall and disaggregated by mode if specified).	Metamodel Regression.
ROI Analysis Tool – Metamodel Regression	'rr' module	TDM, AequilibraE, Scenario Analysis Framework	TDM Trips, Hours, Miles. AequilibraE Trips, Hours, Miles (overall and disaggregated by mode if specified). Full list of scenarios.	Fit and evaluate regression model for all scenarios.	PHT, PMT, trips for full matrix of scenarios (overall and disaggregate by mode if specified).	ROI Analysis Tool.
ROI Analysis Tool – Metamodel Recovery	'recov_init' module	User, and Damage and Repair analysis	Configuration file. Model parameters file. Time to repair (asset/segment level).	RDRM: Build out recovery modeling for uncertainty scenarios based on hazard event and repair time.	Full list of scenarios under different recovery paths (extended_scenarios.csv).	ROI Analysis Tool.
ROI Analysis Tool – ROI Analysis	'recov_calc' module	Scenario Analysis Framework, Metamodel Regression, Metamodel Recovery, Damage and Repair analysis	Regression PHT, PMT, Trips (full matrix of scenarios). Cost of repair and minimum time to repair (asset/segment level). Recovery paths (extended_scenarios.csv).	RDRM: Combine metamodel outputs with recovery module outputs to generate results for scenarios on VMT, PHT, trips. Interpolate across entire period of analysis.	Tableau report xlsx file of economic performance metrics (BCA, BCA-U/Regret) for scenarios under different recovery regimes.	Data Visualization.
ROI Analysis Tool – Data Visualization	'recov_calc' module	RDR ROI Analysis Tool	Tableau report xlsx file.	Tableau Dashboard: Rank and filter results. Project prioritization. Map visualization.	Dashboard outputs.	User.

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
Benefits Analysis Tool – Overlay	'TAZ_attribute_overlay' module	User	TAZ shapefile. Data layer with categories to assign to TAZs. TAZ_metrics configuration file.	Overlay and aggregation of category data onto TAZ polygons.	CSV file of TAZ categories.	Benefits Analysis Tool – TAZ Metrics Disaggregation.
Benefits Analysis Tool – Core Model	'aeq_run' module	User (selection of TDM vs. AequilibraE runs), Scenario Analysis Framework	Modified SQLite database of network. Demand OMX or CSV files.	AequilibraE: Perform runs for disruption and resilience scenarios (project level).	Disaggregate trip skims for AequilibraE scenarios.	Benefits Analysis Tool – TAZ Metrics Disaggregation.
Benefits Analysis Tool – TAZ Metrics Disaggregation	'TAZ_metrics' module	TAZ Attribute Overlay, Core Model	CSV file of TAZ categories. Trip-level skims from core model. TAZ_metrics configuration file.	Disaggregation and summarization of network performance metrics by TAZ categories.	HTML report of disaggregate impact by TAZ category on hazard scenario and resilience project.	User.
RDR Helper Tools – Baseline Network Run	'baseline_network_run' module	User	TDM base network (links and nodes), trip tables.	AequilibraE: Routing on the base network and trip tables.	Network with flows and volume/capacity ratios. Travel time skims between origins and destinations.	User.
RDR Helper Tools – Initial Year Core Model Run	'initial_year_run' module	User	Configuration file. Model parameters file. TDM initial year (socioeconomic). TDM road network. (Optional) GTFS transit files.	AequilibraE: Generate core model outputs for the initial year trip table for each hazard scenario.	PHT, PMT, trips for AequilibraE scenarios (overall and disaggregated by mode if specified).	RDR ROI Analysis Tool.

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
RDR Helper Tools – OSM to GMNS Conversion	'create_network_from_osm' module	User.	OpenStreetMap (OSM) relation ID.	Download OSM raw data. Convert to GMNS format. Generate node and link CSV files.	RDR-compatible road node and road link CSV files.	RDR Tool Suite (after User preparation of centroid connectors).
RDR Helper Tools – GTFS to GMNS Conversion	'GTFS2GMNS' module (initially developed by a third party)	User.	Transit agency GTFS files. User-supplied parameters.	Generate boarding, deboarding, transfer and enroute links in GMNS format.	GMNS network with transit links.	Transit Link Preparation.
RDR Helper Tools – Transit Link Preparation	'prepare_rdr_transit_network' module	GTFS2GMNS module.	Transit network. Centroid nodes.	Prepare transit network (synthesize centroid connectors to the transit stops).	GMNS network with transit links and transit centroid connectors.	Network Integration.
RDR Helper Tools – Network Integration	'prepare_rdr_transit_network' and 'calculate_transit_network_metrics' modules	User, Transit Link Preparation module.	Transit network with centroid connectors. Road network.	Combine transit and road networks. Calculate transit metrics (add appropriate costs and travel times to the transit links).	Combined network of transit links, transit centroid connectors, road links, road centroid connectors with appropriate costs.	RDR ROI Analysis Tool. RDR Exposure Analysis Tool.
RDR Helper Tools – Gravity Model	'GravityModel' module (initially developed by a third party)	User.	Production-attraction zones and trips.	Generate trip tables for both production-attractions and origin-destinations using a distance-based gravity model.	Trip table CSV files.	Trip Table Conversion.

RDR Tool Suite component	Process or step name	Who supplies the process inputs?	What inputs are required?	What are the major steps in the process?	What are the process outputs?	Who receives the outputs?
RDR Helper Tools – Trip Table Conversion	Format demand helper tools	User, Gravity Model module.	Network nodes (with centroids). Trip table in plain text CSV format.	Convert trip table to OMX binary format.	Trip table in OMX binary format.	RDR ROI Analysis Tool.
RDR Helper Tools – Scenario Input Validation	'rdr_input_validation' tool	User.	Input files for an RDR ROI Analysis Tool run.	Confirm necessary fields and data in input files to execute an RDR ROI Analysis Tool run.	Error list of input validation issues and a CSV file with summary statistics on certain parameters (for checking they are reasonable)	User.

Appendix B: Verification and Validation of the RDR Tool Suite

Core Model: AequilibraE model validation

This section briefly describes the validation performed during the Phase 1 pilot with Hampton Roads. Before the RDR Tool Suite analysis was conducted, the AequilibraE model was validated for the Hampton Roads, Virginia network. Using 2017 trip tables and network, the results of the AequilibraE routing run, and the all day network, flows for the core model were compared to average daily traffic counts on approximately 4,000 one-way (2000 two-way) links in the region. The results matched the counts nearly as well as those from the Hampton Roads TDM. One would not expect the results to be the same, due to the following simplifications employed in the fast AequilibraE model:

1. The AequilibraE model uses person trips for a generic peak hour. To compare these to vehicle trips, two adjustments needed to be made:
 - Generic peak hour to daily trips, using a factor of 13.4 (from the HRTPO model documentation).⁸¹ With this adjustment, the total number of trips match the total number of daily trips from the person trip tables.
 - Daily person trips to daily vehicle trips, dividing by a vehicle occupancy factor. The factor is chosen so that the sum of the daily vehicle volumes outbound from the centroids (on centroid connectors) is equal in the HTRPO output and the AequilibraE output. The occupancy factor thus calculated ranges from 1.13 (2017) to 1.3 (2045).
1. The trip tables also do not include truck trips and external trips. The breakdown of trip types from the HRTPO model, on outbound links from internal and external centroids, is as follows:

Trip Type	Daily Trips	Comment
TRUCK24	83,195	Truck trips, not in the AequilibraE model
HBW24	1,361,038	Trips that are in the AequilibraE model
HBO24	2,620,950	
NHB24	1,537,838	
VISITOR24	0	No visitor trips were included in model
EIIE24	228,760	External trips, not in the AequilibraE model
EE24	10,108	

Table B-0-1 Trip Table from HRTPO model

2. For this pilot, AequilibraE is not using a mode choice model, and is thus not explicitly considering transit trips. Transit trips are a small proportion of overall trips in the region (58,612 daily transit trips as reported in table 2.19 of the HRTPO version 2 methodology report).
3. The routing uses a generic peak hour for the entire day, rather than the 4 time periods used in the HRTPO model.
4. Finally, the version of AequilibraE used for this validation used Bureau of Public Roads (BPR) volume/delay functions, while the HRTPO model uses a conical function. For the pilot, the BPR

⁸¹ The factor 13.4 is the ratio of peak hour trips to all day trips in the HRTPO model.

parameters used in AequilibraE were adjusted to produce volume/delay results close to those from the conical function as seen in Table B-0-2).

Table B-0-2 Volume Delay Function Parameters Used in Hampton Roads Pilot

Functional Class	Conical Alpha	Conical Beta	BPR Alpha	BPR Beta
Freeway	9	1.06	0.83	5.5
Minor freeway/principal arterial	7	1.08	1	4
Major/Minor Arterials, Major Collectors	4.5	1.14	1	3
Minor Collectors/Locals	2	1.5	1	2

To check the reasonableness of model results the following comparisons of daily link flows were performed (Table B-0-3).

Table B-0-3 Checks for Reasonableness of Model Results

AequilibraE model	HRTPO core model	Traffic counts
Group 8 – 2045, person trips	Group 8 – 2045, allday, vehicle trips	N/A
NoBuild Network 2017, person trips	NoBuild Network – 2017, allday, vehicle trips	Recent counts from several thousand links, as reported in the Loaded network tables

The evaluation assessed differences between the RDRM model results and the expected outcomes. The validation identified tolling as one factor that had not been addressed sufficiently and was driving differences between the AequilibraE runs and the HRTPO data. The tolling function was added to the AequilibraE analysis and resulted in improved matching between this alternative core model and the real-world results.

The validation of the model with the Hampton Roads pilot demonstrated that the analyst should investigate the effects of the resilience projects at the network level in the model evaluation. The network-wide aggregate effect of each project is small. Consider, for example, the number of trips made with or without resilience investment 2045-122. Figure B-0-1 shows the numbers of trips at each of 7 stages of recovery (numbered 0 through 6) for three hazards:

- The pair of lines for the 100-year flood show a difference for the earlier recovery stages: more trips are possible with the resilience investment.
- The other two pairs of lines, for the 10-year and 0-year floods, overlay each other: they show little system-wide difference in the number of trips possible.

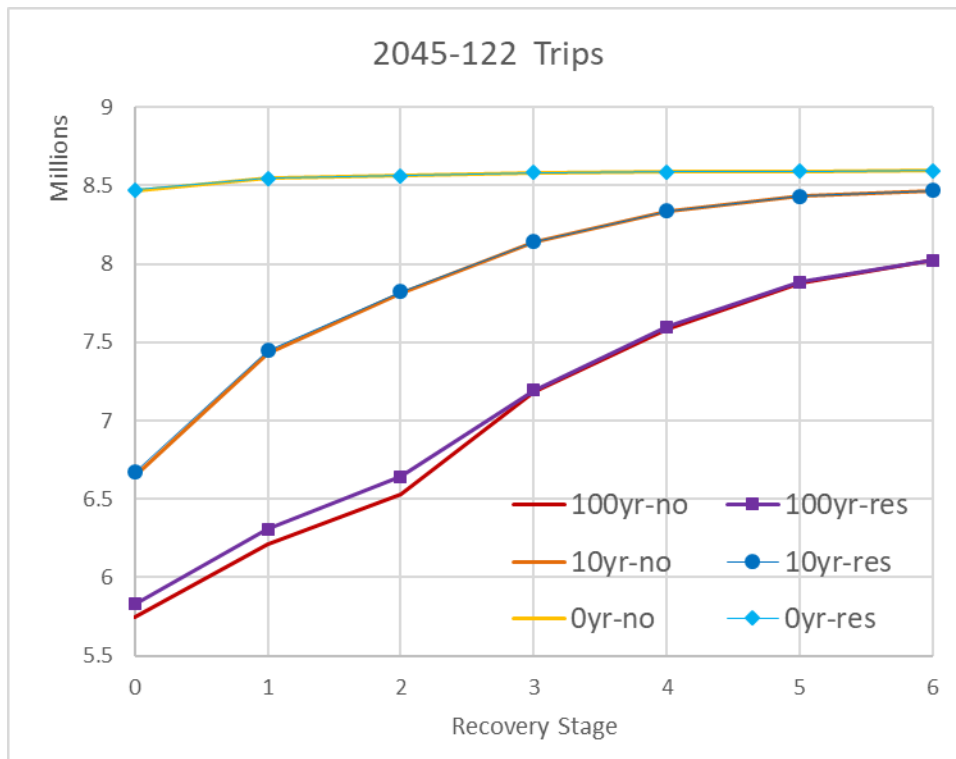


Figure B-0-1 Trips with or without resilience project 2045-122

In the future, to address the issue of small effects, it may be best to focus on origin-destination (O-D) skims, to show that while many O-D pairs will not be affected by the investment, some O-D pairs will see a significant difference in trips, PMT, and PHT. Breaking the analysis out by O-D pair will also aid with benefits analysis, as each travel zone has known demographic characteristics.

A linear regression model will have difficulty fitting statistically significant coefficients for the resilience projects if their individual effects on trips, PMT, and PHT are of the same order of magnitude as the variation observed in the AequilibraE modeling.

The analyst should consider conducting sensitivity analysis around the number of samples used to fit the regression model. The RDR team observed significant shifts in the regret rankings and total benefits as the regression model was fit on more AequilibraE core model runs. This variability in model outputs led to the decision to run all possible combinations with AequilibraE for the final Hampton Roads pilot analysis.

Furthermore, the analyst should consider the structure of their scenario space in determining if a regression model will be adequate in predicting trips, PMT, and PHT. For the Hampton Roads analysis, the large number of project groups and small number of resilience projects in each project group led to many no action baselines. The corresponding regression models were then overparameterized which led to overfitting on the small number of samples used for each regression model. In these cases of overfitting, the analyst should also consider implementing alternative specifications of the regression model but may need to consult economists or others with expertise in econometrics to determine the appropriate specification for their analysis.

These factors should be weighed in the analyst's decision for whether to rely on the regression model or to rely entirely on core model runs. The RDR team plans to further improve the regression model in the

next phase of development to improve performance in a range of conditions. The tool suite still provides a useful ROI function if the analyst chooses to forgo the time savings of the regression model and instead runs a core model run for every scenario in the space.

Validating Results

This section describes the verification and validation of the RDR Tool Suite that has been undertaken thus far to ensure that the tool works as intended and provides appropriate results.

Verification tests whether a model executes the functions intended and whether it works correctly as written. Verification tests check that calculations are done correctly, that outputs match inputs appropriately, and that as tool functions are added these calculations are maintained and are consistent across multiple sample analyses.

Validation tests whether the model matches expected results overall. In the case of the RDR Tool Suite, validation includes assessing whether the results of the scenarios evaluated by the RDR Tool Suite are similar enough to TDM runs to be useful and can provide results for a multitude of scenarios more quickly than could be accomplished with TDM runs alone.

Verification Testing

The following table shows a set of unit and branch test logs that ensured the proper calculation of results by various components of the RDR Tool Suite.

Table B-0-4. Log of verification tests performed during development of RDR Tool Suite.

Date completed	Module(s) tested	Code version used	Requirement Tested	Outcome
10/26/2020	lhs (rdr_LHS.py)	2020_10_20_rdr_full_metamodel_run branch	Latin Hypercube Sampling correctly generates a full list of scenarios for multiple resilience projects	PASS
10/27/2020	lhs (rdr_LHS.R)	2020_10_20_rdr_full_metamodel_run branch	Latin Hypercube Sampling correctly selects a sample of 10 to 100 runs as specified by the user.	PASS
10/27/2020	aeq_run	2020_10_20_rdr_full_metamodel_run branch	AEquilibraE module correctly generates user equilibrium routing results for base network and disrupted network (with mini-equilibrium) for sample of 10 LHS runs. Execute multiple (3+) times with different samples to ensure consistent results.	PASS
10/29/2020	Aeq manual	2020_10_20_rdr_full_metamodel_run branch	Automated aeq_run module correctly produces the same results as manually running the Aeq module for each run specified in the LHS sample.	PASS
10/29-30/2020	regression_setup	2020_10_20_rdr_full_metamodel_run branch	Regression setup step correctly prepares outputs from aeq_run as needed to parameterize the regression model.	PASS
10/30/2020	recov_init	2020_08_11_recovery_modeling branch	Creates a set of scenarios for analysis for the projects identified among HR's megaprojects.	PASS
10/30/2020	recov_calc	2020_08_11_recovery_modeling branch	Pulls together results generated by prior modules and produces Tableau dashboard input file for the projects identified among HR's megaprojects.	PASS
11/4/2020	all modules	2020_08_11_recovery_modeling branch	RDR tool can be initiated by running a batch file and executes as expected from start to finish.	PASS
11/4/2020	analysis modules (recov_init and recov_calc)	2020_08_11_recovery_modeling branch	Confirm that entire analysis section of RDR Tool Suite can be run from start to finish using batch file while keeping regression fit fixed and produces the same outputs as the initial analysis run.	PASS
11/4/2020	lhs	2020_08_11_recovery_modeling branch	Unit test for basic functionality of Latin Hypercube Sampling R code on a second machine in RDR environment.	PASS

Date completed	Module(s) tested	Code version used	Requirement Tested	Outcome
11/4/2020	analysis modules (recov_init and recov_calc)	2020_08_11_recovery_modeling branch	Analysis section of RDR Tool Suite runs correctly using batch file with a subset of uncertainty parameters used for regression fit selected in UserInputs file.	PASS
11/4/2020	analysis modules (recov_init and recov_calc)	2020_08_11_recovery_modeling branch	Analysis section of RDR Tool Suite runs correctly using batch file with uncertainty parameters in UserInputs file not used to fit the regression.	PASS
11/5/2020	analysis modules (aeq_run)	design_test_aeq	Multiple scenario runs executed to confirm that results can be used to provide pairwise comparisons of resilience investments, recovery steps, and hazards.	PASS
1/14/2021	all modules	2021_01_08_quick_starts branch	Entire RDR Tool Suite runs from start to finish using batch file for Quick Start 1 on a scenario comprised of 2 resilience projects.	PASS
1/14/2021	analysis modules (recov_init and recov_calc)	2021_01_08_quick_starts branch	Entire analysis section of RDR Tool Suite runs using batch file for: a subset of uncertainty parameters (Quick Start 2 Example A), a different set of recovery parameters (Quick Start 2 Example B), and a different method for disruption-damage calculation (Quick Start 2 Example C). Same scenario as Quick Start 1 of 2 resilience projects.	PASS
1/14/2021	all modules	2021_01_08_quick_starts branch	Entire RDR Tool Suite runs from start to finish using batch file for Quick Start 3 on a scenario comprised of 3 resilience projects across 2 project groups.	PASS

Validation Testing

Two types of validation testing were performed. First, AequilibraE-modeled link flows were compared to link flows from the HRTPO core model. Where traffic counts were available, the 2017 modelled link flows were also compared to the traffic counts. The validation identified tolling as one factor that had not been addressed sufficiently and was driving differences between the AequilibraE runs and the HRTPO data. The tolling function was added to the AequilibraE analysis and resulted in good matching between this alternative core model and the real-world results. Note that due to the differences between the AequilibraE model and the HRTPO core model, discussed in section 5.2.2.3, one would not expect the results to be exactly the same.

Second, the regression module was validated by assessing the sign and magnitude of the resulting regression models against the expected values for hazards, recovery steps, and other inputs.

For example, Figure B-0-2 shows validation tests for the trips, miles, and hours extrapolated from the regression fit using HRTPO example data. For the example scenario shown, which was not used in the fitting of the regression model, the estimated trips, miles, and hours all increase with increasing recovery steps, in ways which are expected and reasonable. Additional validation tests included similar reasonableness test for resilience investments and hazard levels.

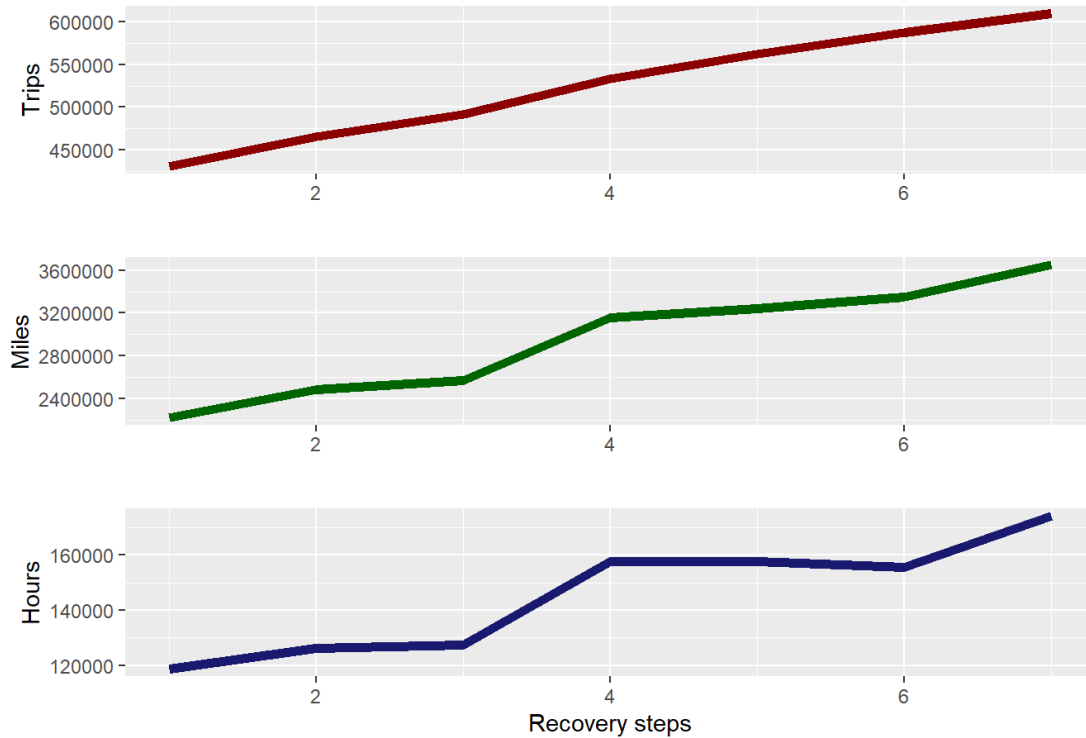


Figure B-0-2. Validation test for reasonable coefficients from the regression metamodel using HRTPO example data, for extrapolated trips, miles, and hours under a scenario not used in the fitting of the regression model across recovery.

AequilbraE model validation with transit network

Using the link flows produced by AequilbraE, the user can summarize transit usage, both for the trips by 1+ car households ('matrix' trip table), and for trips by 0-car households ('nocar' trip table). This analysis is performed using the link flow CSV files and calls for basic Excel or database skills. It relies on the following properties of the network:

1. All trips, road and transit, start at a centroid. The first link on a trip is a centroid connector, going from the centroid to another node in the network. If the travel time or cost is set high enough on the centroid connection to eliminate use of centroid connectors as intermediate links, we can summarize the trips by summarizing flows on these centroid connector links.
2. The topology of the network (i.e., the link_id, from_node_id, and to_node_id) is the same in the networks for the trips by 1+ car households and for the trips by 0-car households. Only the costs are different.
3. Node numbers follow a particular sequence:

- Lowest numbered nodes are the centroids
 - The next set of nodes pertains to the road network
 - The highest-numbered nodes pertain to the transit network
4. We can also summarize unlinked transit trips by looking at the flows on transit boarding links. Note that a transit trip may have more than one transit boarding.

Using Quick Start 4 (now Reference Scenario 4) as an example, the file names and locations are as follows. In the user's model, locations and filenames might be somewhat different:

Table B-0-5 Location of network files

Location	Filename	Description
C:\GitHub\RDRtransit\quick_starts\qs4_transit\Data\generated_files\aeq_runs\base\QS4\base01\matrix	Group01_baserun.csv	Network for the base run, with costs for trips by 1+ car households
C:\GitHub\RDRtransit\quick_starts\qs4_transit\Data\generated_files\aeq_runs\base\QS4\base01\matrix	Link_flow_base01.csv	Flows for the base run, using trips by 1+ car households
C:\GitHub\RDRtransit\quick_starts\qs4_transit\Data\generated_files\aeq_runs\base\QS4\base01\nocar	Group01_baserun.csv	Network for the base run, with costs for trips by 0-car households
C:\GitHub\RDRtransit\quick_starts\qs4_transit\Data\generated_files\aeq_runs\base\QS4\base01\nocar	Link_flow_base01.csv	Flows for the base run, using trips by 0-car households

The flows are contained in the matrix_ab or nocar_ab field of the link_flow_base01.csv files.

Steps taken by the user for summarizing the auto and transit trips are as follows:

1. Identify the sequences of node numbers for centroids, road network nodes, and transit network nodes. In the case of Quick Start 4, nodes 1, 2, and 3 are centroids, 10-15 are for the road network, and 20-27 are for the transit network.
2. Add the flows (from both link_flow_base01 files) to one of the network files (Group01_baserun). If property (2) above, holds, it should not matter which network file. Do this by joining on link_id, and creating two columns: one (matrix_ab) for the trips by 1+ car households and one (nocar_ab) for the trips by 0-car households.
3. Determine which links represent:
 - Outbound flows from centroids to road nodes. For Quick Start 4, the following function will identify these links: if(from_node < 4 AND to_node < 19), then YES, otherwise NO
 - Outbound flows from centroids to transit nodes. For Quick Start 4, the following function will identify these links: if(from_node < 4 AND to_node >= 20), then YES, otherwise NO
4. Add up the flows on these selected links. The user will end up with four values:
 - 1+ car trips by road (using the column matrix_ab)
 - 1+ car trips by transit (using the column matrix_ab)

- 0-car trips by road (using the column nocar_ab)
- 0-car trips by transit (using the column nocar_ab)

5. Check the results for reasonableness. You would expect to see relatively more 0-car trips using transit. The user can also check the overall transit mode share by adding up the transit trips and dividing them by the sum of all of the trips. Table B-0-5 shows the results for Quick Start 4. All 70 trips from the 1+ car households are on the road network; the remaining 30 trips from the 0-car households are on the transit network.

Components of this analysis and calculation of transit-specific metrics are incorporated into the RDRM for scenarios where the user specifies that transit metrics are calculated. These transit-specific metrics are then used in the RDR ROI Analysis Tool to calculate benefits and disbenefits for transit trips.

Table B-0-6 Quick Start 4 Results

link_id	from_node_id	to_node_id	Unused columns	centroid_road	centroid_transit	matrix_ab	nocar_ab	road_matrix	transit_matrix	road_nocar	transit_nocar
1	11	12	...	0	0	45	0	0	0	0	0
2	12	11	...	0	0	0	0	0	0	0	0
3	11	10	...	0	0	0	0	0	0	0	0
4	10	11	...	0	0	0	0	0	0	0	0
5	10	14	...	0	0	0	0	0	0	0	0
6	14	10	...	0	0	0	0	0	0	0	0
7	12	13	...	0	0	70	0	0	0	0	0
8	13	12	...	0	0	0	0	0	0	0	0
9	13	14	...	0	0	70	0	0	0	0	0
10	14	13	...	0	0	0	0	0	0	0	0
11	14	15	...	0	0	70	0	0	0	0	0
12	15	14	...	0	0	0	0	0	0	0	0
20	23	27	...	0	0	0	30	0	0	0	0
21	27	23	...	0	0	0	0	0	0	0	0
22	22	26	...	0	0	0	0	0	0	0	0
23	26	22	...	0	0	0	0	0	0	0	0
24	24	25	...	0	0	0	30	0	0	0	0
25	25	24	...	0	0	0	0	0	0	0	0
100	1	11	...	1	0	45	0	45	0	0	0
101	2	12	...	1	0	25	0	25	0	0	0
102	15	3	...	0	0	70	0	0	0	0	0
120	1	20	...	0	1	0	5	0	0	0	5
121	2	20	...	0	1	0	25	0	0	0	25
122	21	3	...	0	0	0	30	0	0	0	0
130	20	23	...	0	0	0	30	0	0	0	0

link_id	from_node_id	to_node_id	Unused columns	centroid_road	centroid_transit	matrix_ab	nocar_ab	road_matrix	transit_matrix	road_nocar	transit_nocar
131	20	22	::	0	0	0	0	0	0	0	0
132	26	24	::	0	0	0	0	0	0	0	0
133	27	24	::	0	0	0	30	0	0	0	0
134	25	21	::	0	0	0	30	0	0	0	0
							Sum	70	0	0	30

Appendix C: Generalized Modeling Network Specification (GMNS)

Network to AequilibraE Network Conversion

This appendix describes the node and link network attributes used by AequilibraE. RDR takes a CSV file network (in GMNS format) and converts it to the SQLite tables used by AequilibraE. This material should not be needed by the end user but is documented to aid future RDR development.

Table C-0-1 Node specifications for GMNS and AequilibraE networks

GMNS Network	How to convert from GMNS to AequilibraE	AequilibraE network
node_id	Copied directly	ogc_fid and node_id
x_coord	Used only to show network in a GIS	
y_coord	Used only to show network in a GIS	
node_type	If node_type == "centroid", then is_centroid = 1, else is_centroid = 0	is_centroid

Table C-0-2 Link specifications for GMNS and AequilibraE networks

GMNS Network	How to convert from GMNS to AequilibraE	AequilibraE network
link_id	Copied directly	ogc_fid and link_id
from_node_id	Copied directly	a_node
to_node_id	Copied directly	b_node
directed	Copied directly	direction = 1
geometry	Used only to show network in a GIS	
length (miles)	Copied directly	distance (miles)
facility_type	Copied directly	link_type ⁸²
capacity (veh / day / lane) ⁸³	Converted to a daily link capacity in persons per day	capacity_ab (veh / day)
free_speed (mph)	Copied directly	speed_ab (mph)
lanes	Used to calculate capacity	
allowed_uses	For now, always set to AUTO	modes = 'c'
toll (cents)	See RDR User Guide	toll (cents)
toll_nocar (cents) ⁸⁴	Used for the network that runs with the "nocar" trip tables	toll (cents)
travel_time (minutes)	See RDR User Guide	Free_flow_time (minutes)
travel_time_nocar (minutes)	Used for the network that runs with the "nocar" trip tables	Free_flow_time (minutes)
alpha	See RDR User Guide	alpha
beta	See RDR User Guide	beta

⁸² Two [link types](#) are reserved in AequilibraE: centroid_connector and default.

⁸³ Standard GMNS has capacity in veh / hour / lane. However, RDR works with daily, and not hourly flows.

⁸⁴ GMNS allows user defined fields. The fields toll_nocar, travel_time, travel_time_nocar, alpha and beta are not part of the default GMNS specification.

Appendix D: Trip Loss Valuation

US DOT does not have a standard method for monetizing lost trips, e.g., trips that are foregone because the path is not traversable or if the trip time makes the trip infeasible. Instead of a fixed value for lost trips, the tool uses the economic theory concept of consumer surplus to value lost trips. The trip values are computed using the implied value of trips based on the trip elasticity value and the network equilibrium generated in the core model. Loss of consumer surplus represents an economic cost over and above the economic value of additional travel time and vehicle operating costs that occurs when trip that continue to be made become more circuitous or time-consuming.

This concept is demonstrated in Figure D-0-1 below. The supply curves are upward sloping and exhibit steep slopes, demonstrating the non-linear impact of congestion on trip price. The initial supply curve is that of the baseline scenario in which a hazard event occurs, but no resilience alternative has been deployed, S_0 , while the supply curve S_1 exists when the resilience alternative has been deployed. The demand curve, D_0 , is downward sloping and does not change between a baseline scenario and its associated alternative scenarios. The change in supply from S_0 to S_1 measures the impact of the resilience alternative relative to the baseline scenario, which enables an increase in the number of trips that can be taken in the event of a disruption from Q_0 to Q_1 , while also reducing the cost (including the economic value of the time they require) of trips that would continue to have been taken during the disruption from P_0 to P_1 . The economic value of the additional trips enabled by the investment in improved resilience is measured by the shaded area under the demand curve D_0 and above the supply curve S_1 between Q_0 and Q_1 , which is a gain in consumer surplus.

Functionally, the trip valuation in the model is defined by the following equation:

$$\text{Trip Valuation} := \frac{(P_0 - P_1) * (Q_1 - Q_0)}{2}$$

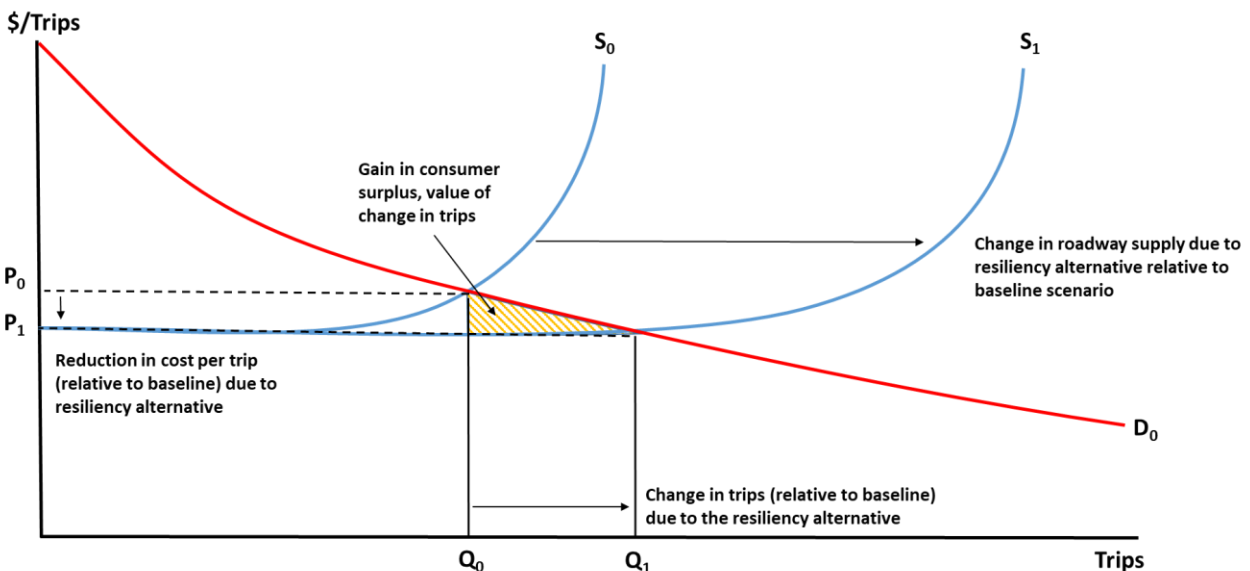


Figure D-0-1: Trip Valuation based on change in Consumer Surplus due to a given Resiliency Investment

The consumer surplus valuation of lost trips is applied to both trips lost due to drivers choosing to not make the trip due to the increased travel time and trips lost on routes made impossible by the loss of links in the network. For impossible routes, the value of the trip is P_1 as P_0 is equal to 0 in the baseline representing that the consumer would be unwilling to pay any price for a trip that could not be made. Essentially, this approach values trips at their travel time under the conservative assumption that all trips are at least as valuable to make as the time it takes to make them.